Optimizing live aboveground carbon estimations: streamlining data acquisition techniques, improving modeling efficiency and accuracy, and quantifying modeling and product error

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ABSTRACT

Blah blah blah

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INTRODUCTION

The journey of a carbon model is a long process, beginning with data acquisition to model development to the final step of real-world implementation. Developing carbon models can be an arduous task, only to have it outperformed by another approach. However, all the trials and tribulations are not carried out in vain, for as climate change threatens the current balance on Earth, humans around the world have joined together in collaboration to alter our current trajectory of increasing global average temperatures by 2º C. Many solutions have been proposed to combat rising atmospheric temperatures over the past decades, including alternative energy sources, green technologies, and other ambitious (and sometimes controversial) plans. One proposal has gained traction in recent decades due to its potential to make meaningful changes in atmospheric carbon: using forests to offset carbon. Harnessing the power of trees to capture excess CO_2 in the atmosphere and convert it to stored carbon (defined as carbon bound within a plant), along with transitioning away from petroleum-based products towards wood-based products, seems to be an elegant solution to our ever-growing climate threats. But this is only possible if properly executed along every step, from accurately quantifying carbon capture and release, to appropriate land management practices that maximize carbon acquisition and storage, and to adoption of public policy to ensure these goals can be achieved.

While the process currently in place to make meaningful changes in our current climate trajectory is multifaceted, requiring thoughtful economic, political, and ecological planning, this dissertation aims to improve one small facet of this global effort: modeling live above ground carbon. Before carbon estimation models can be developed, training data must be acquired. While ground based forest inventories are fundamental in modeling forests, they may lack non-tree and stand-level information like environmental variables, which influence forest growth and yield (Yang *et al.* 2022). In addition, multispectral bands have also been helpful to monitor changes in forests. Environmental and multispectral data, often provided by national and international organizations like the European Space Agency (ESA) or the U.S.'s National Aeronautics and Space Administration's (NASA), are stored as geospatial data. In addition, many existing carbon products are accessible through geospatial data, providing various estimates of carbon or biomass. One example is the ESA's Climate Change Initiative Biomass data (https://climate.esa.int/en/projects/biomass/data/), which is the product of international collaboration, highly studied, and utilized globally across fields of study (Wang *et al.* 2026).

Improving geospatial data accessibility

Research gaps and needs

While geospatial datasets can be incredibly helpful in providing additional variables and even carbon estimates, accessing geospatial data can be cumbersome enough to either be inefficient or abandoned altogether, resulting in geospatial products being underutilized. Issues with accessing geospatial data can be partially attributed to high technical barriers, including specialized data handling and programming skills. Even though geospatial programs and packages already exist that make geospatial data easier to work with, they still have severe

limitations, like requiring expensive licenses or technical barriers that are still too high to overcome.

Objectives and approach

Chapter 1 outlines a Python application design and accompanying web application that can be used as a framework to reduce the technical skills required for simple geospatial processes. This framework is demonstrated in the created *forestspot* Python application, a widget-oriented approach to conducting pixel extractions of user-provided coordinates from Google Earth Engine datasets. Designed to complement plot-based forest inventories through the application's streamlined point extraction functions, the application also accounts for the sensitive nature of some forest inventory plots and provides buffering approaches to conceal true plot coordinates from Google servers. While the *forestspot* application is able to help make the process of conducting point extractions from Google Earth Engine datasets easier than previously possible, the widget-based framework can be replicated to accomplish other geospatial tasks with minimal skills.

Live aboveground carbon prediction models

Research gaps and needs

Carbon estimates can vary as much as the modeling approaches used. Previous studies have compared estimates of the various carbon models and products available (Hudak *et al.* 2026). A commonality between all models is that they are often validated using forest inventory data with species-specific biomass equations, which produces the most accurate values without destructive sampling and is accepted as the "true" value. However, this method is time intensive, from conducting full inventories to processing data. Therefore, there is always a need to develop models that aim to be as accurate and precise as the full inventory and analysis methodology but at a reduced cost. Particularly, models that are able to produce accurate estimates quickly at large scales allow for better and faster real-time forest monitoring and management, which is advantageous to forest managers and practitioners. One way to accomplish this is through producing stand-level carbon estimates that remove the need to conduct intensive field inventories and calculate individual tree biomass (Yang, Brandeis, and Skiba 2025). Successful stand-level carbon models alleviate the processing time needed to generate carbon estimates, compared to the intensive process of fitting individual tree biomass models and then aggregating to the stand level.

Objective and approach

Chapter 2 compares model performance between stand-only variables and combined stand and environmental variables using three modeling techniques to estimate stand-level live above ground carbon. The three models tested are linear mixed-effect models, mixed-LASSO regression, and generalized additive mixed models (GAMMs), all with a random effect on forest type. This approach allows for large scale estimations on the scale of federal lands, which are

typically large contiguous areas that can contain high species and forest diversity within. The study region for this chapter focuses on ten national parks and forests in southern Appalachia. As southern Appalachia is a hotspot of biodiversity and threatened by various invasives and changing climate, developing models for federal lands in southern Appalachia aid in the regions' carbon and forest monitoring efforts. Additionally, we tested two different datasets with each region and approach. One dataset consists solely of stand-level predictors calculated from the U.S. Forest Service Forest Inventory and Analysis permanent plots that are within the regions of interest. The second dataset uses the aforementioned stand-level variables in combination with WorldClim BIO climate variables, as previous research suggests that the inclusion of environmental variables increase prediction performance. Furthermore, findings from this study can be expanded to other regions to generate quick and accurate estimates.

Expanded use of carbon prediction models

Research gaps and needs

Finally, chapter 3 focuses on implementing carbon models and products to monitor carbon storage, which ultimately allows for successful forest monitoring and management. As other carbon models and products are available, understanding where the models tested in this study fit into the current landscape of carbon estimation products is crucial to determining when and how tested models and products can be used. One particular extension of these models can be used to estimate live aboveground carbon in nearby forests. As forest systems expand beyond land boundaries, while models can be trained from a specific region, they may still have high predictive power for similar forests outside the training region. The ability to expand these models to predict live aboveground carbon on nearby forests will be beneficial to forest managers and practitioners who can confidently use these pre-fitted models on their properties, and thereby increasing prediction efficiency.

Objectives and approach

In this study, we applied the Daniel Boone National Forest linear mixed-effects model with stand-only predictors from chapter 2 to the nearby Forest Recreation and Research Center (FRREC) at the University of Tennessee, Knoxville. This property serves as a working research forest, with overlapping management goals to other private landowners' objectives. Furthermore, the FRREC is already enrolled in a carbon inventory program with data available to serve as validation data. Two alternative carbon products, the European Space Agency (ESA) Carbon Climate Initiative and CTrees biomass maps, were selected to test against. The ESA and CTrees products were selected for their public availability and differentiation in methodology, input data, and resolution. As resolution size impacts the scale at which products can best be applied at, we evaluated the model and products are two aggregation levels: plot and stratum. Results from this study allows us to understand error present when extending a model beyond its trained region, rank this regional model's performance against other available products, and investigate overall performance of all models at the scale of small landowner properties like those found in FRREC.

The three topics contained within this dissertation (improving data acquisition, testing novel carbon estimation approaches, and extending the use of these approaches to nearby forests) aim to make carbon estimations easier and more accurate in order to meet the need of today's forest managers seeking to quantify live aboveground carbon easily and efficiently without sacrificing accuracy. Developing a Python application that reduces technical barriers expands accessibility to forest managers who were previously barred from utilizing geospatial databases due to high technical barriers. The development and testing of novel carbon prediction models allows for fast and accurate carbon estimation with reduced efforts needed. Again, this allows forest managers to quickly compute carbon estimates at a fraction of the cost needed to conduct efficient forest monitoring and management. Lastly, we tested the best prediction model to quantify its accuracy in extended applications, which again allows for efficient carbon estimations by using existing models.

Chapter 1

forestspot: a forester's app to perform point extractions from Google Earth Engine

Abstract

Technical challenges regarding handling geospatial data remain a major barrier that has prevented the uptake of geospatial data into forestry fields, despite the advantages that geospatial data can provide. Google Earth Engine, which is a large, publicly available database of popular geospatial data, can be leveraged along with existing geospatial technologies to remove many of the existing barriers to accessing and handling geospatial datasets, which will allow geospatial data to be more easily integrated into forest management workflows. To fill this gap, the *forestspot* Python and Streamlit web applications were developed to perform simple pixel extractions from Google Earth Engine's geospatial datasets with minimized technical knowledge needed. By making the retrieval process easier, geospatial data available on Google Earth Engine becomes more accessible and easier to incorporate into forest studies. A case study was conducted using querying environmental data with a national forest inventory with an existing plot buffering method. Results showed that the *forestspot* application returned similar or more accurate results to the true plot values compared to the inventory's existing buffered methodology.

Introduction

Forests are complex, highly variable systems whose structure and dynamics are shaped by environmental conditions. These environmental conditions, like climate, temperature, and precipitation metrics, are often difficult to measure at meaningful spatial scales via the common plot-visit based forest inventories (Toledo *et al.* 2011, Itter *et al.* 2017, Wei *et al.* 2025). Operational forest inventories, such as national forest inventories and those used for landscape-scale forest management, are designed to capture tree and stand characteristics across large scales and at high resolution. However, they typically lack detailed environmental data, despite known relationships between environmental factors and forest growth.

As a result, forest modelers utilize geospatial data to provide environmental data for their research studies and management plans. Geospatial data provide continuous, long-term climate records, short-term environmental metrics, and other remotely sensed metrics that are best measured from satellite imagery. One case is the Normalized Difference Vegetation Index (NDVI), which is calculated from Landsat's multi-spectral bands (Chi *et al.* 2016, Chinembiri, Mutanga, and Dube 2022, Yang *et al.* 2022a, Chinembiri, Mutanga, and Dube 2024, Jha *et al.* 2024). As long-term climate change disrupts normal weather patterns, understanding how forests may adapt can be achieved by utilizing large operational forest inventories in tandem with geospatial imagery. Together, inventories and geospatial data can provide better forest monitoring at a macro scale. In turn, land managers can better observe and respond to changes in the forests.

Environmental data can be found in many publicly available databases, like government websites, and stored in a variety of data formats, like tabular, array, or geospatial data. Geospatial data can be stored as vector or raster datasets, and they combine location and attribute

information into multi-dimensional spatial data that can be easily mapped. To handle these geospatial datasets are various geospatial technologies, like ArcGIS or R. However, there are many barriers that can arise when trying to utilize geospatial data due to the specialized technical knowledge needed. Existing technological barriers are significant enough to deter forest managers from utilizing geospatial data to its full potential (Ziegler and Chasins 2023, Fassnacht *et al.* 2024).

Existing geospatial processing technologies of the modern age are powerful but still highly technical, requiring geospatial literacy and possibly programming skills. Geospatial software like ArcGIS eliminates the majority of the programming knowledge needed through its program interface which makes it a popular software within natural resources, but users must still have geospatial data knowledge to perform many operations (Morgenroth and Visser 2013). In addition, licenses may have to be purchased, which can be expensive and deter potential users. Open-source programs like Python, QGIS, and R provide the most user flexibility with geospatial data handling, but the programming required is more challenging. Even if users have the required technical knowledge and software available, there are still more challenges.

One major obstacle is data decentralization, as geospatial datasets are collected and maintained by numerous domestic and international government agencies and private corporations, like Google. Datasets hosted on various public and private servers make data retrieval less efficient, both in locating and downloading the data. When datasets are especially large, handling them can be cumbersome. Such inefficiencies with geospatial data retrieval, handling, and processing can also deter potential data users.

A solution to the data decentralization problem can be found when using Google Earth Engine (GEE), which is a free, public repository of many popular remotely sensed and other geospatial datasets (Mutanga and Kumar 2019). GEE datasets are stored in a cloud server, which allows users to avoid having to download large datasets locally, which reduces data handling and processing time. GEE provides a Code Editor where users can work completely on a cloud server using JavaScript language (Amani *et al.* 2020, Ermida *et al.* 2020). For non-JavaScript users, GEE has been integrated into existing geospatial analysis software. The GEE Toolbox provides GEE integration in ArcGIS products (Google Earth Engine Community [2024] 2026). The R package *rgee* provides GEE integration into R environments by translating R code to Python and passing it to the Google Earth Engine API (Aybar *et al.* 2020). Meanwhile, earthengine-*api* (ee) in Python provides users the full processing flexibility similar to GEE's Code Editor, and Python packages like *geemap* provides additional functionality to streamline workflows (Wu 2020, Google Earth Engine, n.d.). Such tools provide easier access to GEE and reduce set up and run time needed to perform geospatial analysis, but the technical threshold for manipulating geospatial data remains.

One way for geospatial data to be paired with operational forest inventories is to perform simple point extractions of the pixel value that contains an inventory plot's coordinates. To

accomplish this, users upload their coordinates, access GEE and select a dataset, perform any necessary data manipulations to ensure file compatibility, extract desired information, and return results. While point extractions are possible to produce with existing software, there is a need to make this process more efficient and less technically complicated. The entire workflow can be distilled down to only require users to provide the necessary input information and perform all other steps in the background. By integrating the streamlined workflow into a user-friendly interface with minimal coding, not only will foresters benefit from the increased accessibility, but experienced geospatial users are provided an efficient tool to perform simple point extractions.

Forest inventories often keep plot coordinates confidential in order to protect detailed information of public and private landowner information, as well as preserve plot and inventory integrity. Confidentiality agreements to protect plot coordinates may not allow data users to run actual plot coordinates through GEE without being modified. The U.S.'s national forest inventory, maintained by the USDA U.S. Forest Service's Forest Inventory and Analysis program, protects their permanent plot coordinates for the reason listed above. To compensate for the loss of coordinate data, the FIA provides publicly available coordinates that are buffered using the methodology described in Burrill *et al.* (2024). However, proprietary or other public forest inventories may not have established buffering methodologies developed to conceal plot coordinates to use on unauthorized external servers like Google's (Amani *et al.* 2020, Zhao *et al.* 2021, Alshehri, Zhang, and Liu 2025). Users with concerns with running protected coordinates through Google Earth Engine can benefit from a quick buffering approach that randomizes private coordinates prior to using GEE.

Currently, a streamlined workflow designed for forest practitioners to more efficiently perform point-extractions from GEE-derived raster data without advanced coding or software does not exist. In this study, the *forestspot* Python application was developed to build off *geemap*'s *extract_values_to_points* and incorporate other Python packages into a system that lowers the technical threshold otherwise needed. This project alleviates setup and performance time to conduct point extractions with the goal of helping integrate geospatial data more efficiently into forest studies and inventories.

Methods

Software design

The *forestspot* Python application was designed to reduce technical barriers for accessing GEE and producing point extractions. As privacy concerns can arise with using sensitive plot coordinates in GEE, the application provides two coordinate buffering methodologies to conceal protected coordinates. Point-and-click widgets are used to outline the few user inputs needed, while all additional processing is performed behind the user interface. While these workflows were designed to prioritize reducing technical barriers, the workflows can also be efficient

alternatives to existing approaches due to quick processing times for buffering and extracting data.

The application is divided into two task groups: point extractions and coordinate buffering, which can be visualized in Figure 1. The point extraction approaches access and retrieve information from GEE, while the coordinate buffering approaches provide the option to safeguard confidential plot coordinates prior to point extraction. Each task group contains two modules, for a total of four created modules. This application relies on a number of dependencies. The *ipywidgets* package serves as the foundation for all modules to construct the front-end user interface to provide a point-and-click workflow without extensive data handling and coding (Martindale *et al.* 2024). *Pandas* and *geopandas* are leveraged in each module to

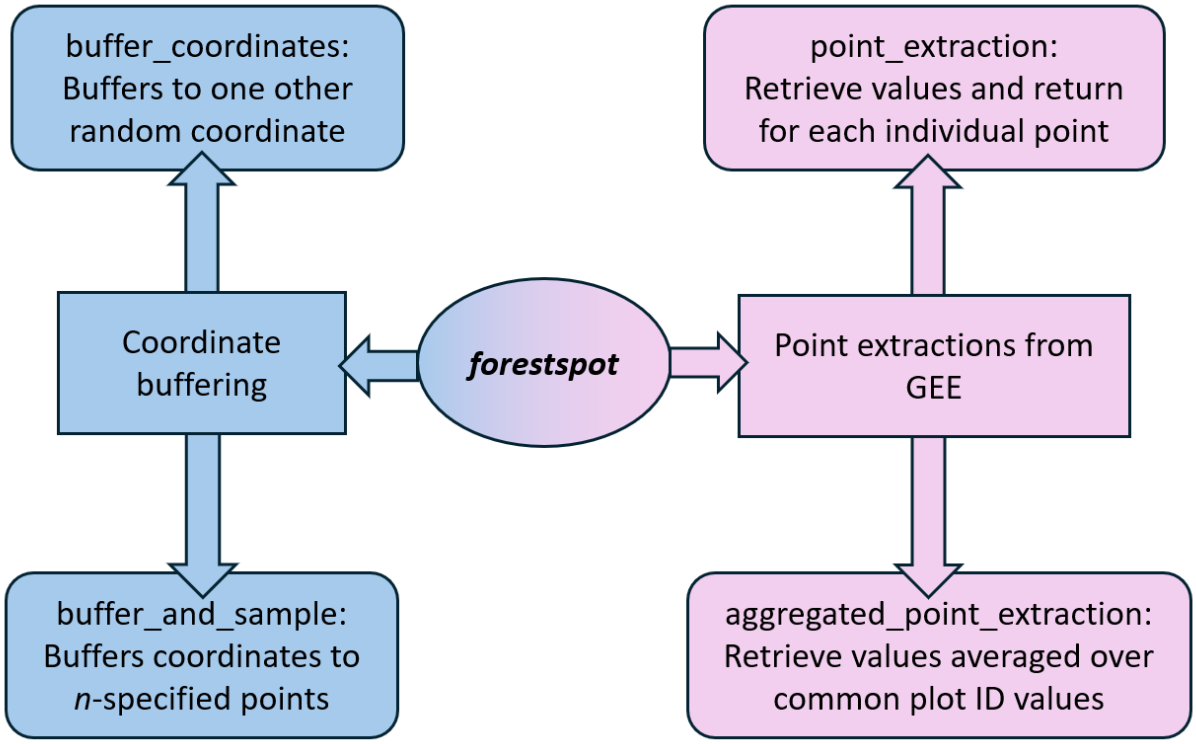


Figure 1: Diagram of package structure, broken into its two functions (coordinate buffering and point extractions). Each function has two modules, which differ in aggregation methods applied.

handle geospatial file handling throughout the algorithm (McKinney 2010, Jordahl *et al.* 2020). The *ee* and *geemap* Python packages are exclusively called within the point extraction modules to handle Google Earth Engine objects and retrieve pixel values (Gorelick *et al.* 2017, Wu 2020). In the buffering approaches, *shapely*, *pyproj*, and *numpy* help generate random coordinate points (Gillies *et al.* 2022, Snow *et al.* 2025, Harris *et al.* 2020). As the buffering modules' packages do not access GEE, coordinate points run through these buffering modules are not run through Google servers.

Formatting for Plot Coordinates File

Users begin all created modules by providing a file of coordinate points to perform operations on. The user-provided file must be properly formatted to prevent data handling errors. This file must be saved as a CSV file and contain the three following columns: a plot identifier column, a latitude column, and a longitude column. Accepted naming conventions for these columns are shown in Table 1 (Additional columns may be present but will be ignored). The plot identifier column is used as a reference column that is grouped over in the point creation and averaging processes conducted in the *buffer_and_sample* and *aggregated_point_extraction* modules. Latitude and longitude columns should be formatted in decimal degrees. Coordinates will be internally set to coordinate reference system WGS64.

Point Extraction Modules

Table 1: Accepted column names for user-provided list of coordinates. File must be saved as a CSV.

CSV Columns	***Accepted Names***
latitude	lat, latitude, y, LAT, Latitude, Y
longitude	log, long, longitude, x, LON, Longitude, Long, X
plot ID	id, ID, plot_ID, plot_id, plotID, plotId

The point extraction modules contain the processes to perform pixel extractions from GEE datasets. The modules are built around *geemap*'s *extract_values_to_points* function, simplifying the process to perform extractions. As point extraction modules access Google Earth Engine internally, GEE must be initialized and authenticated as described in the GEE python documentation prior to running any query. The dropdown menu contains all available GEE datasets, updated daily from Open Geospatial Solutions' master GEE catalog (https://github.com/opengeos). Two date picker widgets allow users the option to filter GEE datasets if desired or allowed.

The modules' inputs have certain conditions. The file containing coordinate points must follow the formatting criteria described in the Data Formatting section above. Accepted GEE datasets must be classified as Images or ImageCollections but not FeatureCollections. Due to the design of the app's data filtering process, the additional data filtering and handling capabilities needed to filter ImageCollections with additional properties or FeatureCollections are absent. Acceptable ImageCollections require start and end dates, which can be either the dataset's entire availability or a specified time frame. Images do not require time filters due to their single image property. Information about a dataset's classification as an Image, ImageCollection, or FeatureCollection, its temporal availability, and additional image properties can be found on the dataset's GEE documentation page. The link to the documentation is conveniently provided once a dataset is selected from the drop-down menu.

Two point-extraction modules that follow this general framework were created: *point_extraction* and *aggregated_point_extraction.* Both modules perform the same extraction process but differ in aggregation technique. *point_extraction* returns the extracted data for each coordinate uploaded, whereas *aggregated_point_extraction* extracts the pixel values then averages over shared plot IDs. The *aggregated_point_extraction* can only be used when returned values are numerical, not categorical data (i.e., land cover). Differences between returned results are demonstrated in Figure 2. The following Python script is used to initialize GEE and load the point extraction modules:

```
import ee
# Initialize Earth Engine
ee.Authenticate()
ee.Initialize(project="[your-project-name]")

# For non-aggregated point extraction
import skiba.point_extraction as spe
point = spe.PointExtraction().vbox
point

# For aggregated point extraction
import skiba.aggregated_point_extraction as sape
agg = sape.AggregatedPointExtraction().vbox
agg
```

Coordinate Buffering Modules

Users who have concerns about operating GEE using sensitive plot coordinates can first run their coordinates through one of two buffering approaches created to quickly generate randomized coordinates within a defined radius. Randomized coordinates can then be applied in the point extraction functions. The buffering modules are called separately from the point extraction modules, and as they do not initialize GEE, coordinates are hidden from GEE. Like the point extraction modules, these buffering modules only require two steps: users provide a correctly formatted CSV file of raw coordinates, then set the buffer radius and optionally, the number of randomized coordinates to generate. The returned dataset is returned correctly formatted for the point extraction modules with the original coordinates removed.

Two modules were created to buffer sensitive coordinates. Both modules follow the same randomization process: buffered coordinates are randomly generated to achieve uniform spatial distribution within a circular area with radius *r*. The difference between the two modules lies in how many randomized coordinates are generated. The module *buffer_coordinates* produces one randomized coordinate for each user-provided coordinate; *buffer_and_sample* returns *n* random coordinates for each provided coordinate, where *n* is an integer provided by the user. A visualization comparing buffering approach is provided in Figure 2 and Figure 3 .

Setting buffer radius *r* for randomized points to be generated within can be determined by consulting the data's user guide, the pixel size for the GEE dataset of interest, and user

discretion. Additionally, if the plot radius is considerably large, it may need to be considered when determining an appropriate buffer radius. Note that pixel values extracted using buffered coordinates could return biased results as the raw and buffered coordinate may lie in different pixels of a raster dataset. Some visual examples of how buffered coordinates may lie outside of the original coordinate's pixel are shown in. This is demonstrated in Figure 4 and Figure 5, where some buffered coordinates lie in a neighboring pixel. Users may desire to repeat the buffering process and point extractions to compare differences in returned values and determine if the values are acceptable. The demonstration conducted later in the Case Study section quantifies the bias present for one example. The following script is used to load the point buffering modules:

```
# For single point buffering
import skiba.buffer_coordinates as sbc
single = sbc.buffer_coordinates().vbox
single

# For multiple point buffering
import skiba.buffer_and_sample as sbs
multiple = sbs.buffer().vbox
multiple
```

Workflow

User workflows for the *forestspot* application depend on whether returned data is returned aggregated or not. The overall simple workflow is shown in Figure 6. The first step is whether or not to buffer the coordinates. While both buffering approaches protect the uploaded coordinates, the *buffer_and_sample* module can also be used to create random plot samples within a circular area with radius *r*. After the optional buffering is conducted, the point extraction method selected will depend on how the user wants the data returned. While the *buffer_coordinates* module was designed to complement the *point_extraction* module, and likewise, *buffer_and_sample* complements *aggregated_point_extraction*, users may be interested in examining the individual values returned by multi-point buffered coordinates. Therefore,

Plot IDs	GEE Extracted Values
1	1
1	2
2	3
2	4
3	5

point_extraction

Plot IDs	[Variable]
1	1
1	2
2	3
2	4
3	5

Plot IDs	GEE Extracted Values
1	1
1	2
2	3
2	4
3	5

aggregated_point_extraction

Plot IDs	[Variable]
1	1.5
2	3.5
3	5

Figure 2: Demonstration of how data aggregations differ for the two point-extraction modules.

Plot IDs	Lat	Long
1	35.13840	-84.1800
2	35.05636	-83.6507
3	35.79585	-82.2298
4	35.80452	-82.3504
5	35.89833	-81.8107

buffer_coordinates

Plot IDs	Lat	Long
1	35.13414	-84.1669
2	35.05425	-83.6422
3	35.78975	-82.2444
2	35.79726	-82.3424
4	35.89902	-81.8106

Plot IDs	Lat	Long
1	35.1384	-84.18
2	35.05636	-83.6507
3	35.79585	-82.2298
4	35.80452	-82.3504
5	35.89833	-81.8107

buffer_and_sample
(n = 2, r = 5280 ft)

Plot IDs	Lat	Long
1	35.13414	-84.1669
1	35.13019	-84.172
2	35.13935	-84.1802
2	35.05425	-83.6422
3	35.05923	-83.6538

Figure 3: Demonstration of point creation differences for the two coordinate buffering modules.

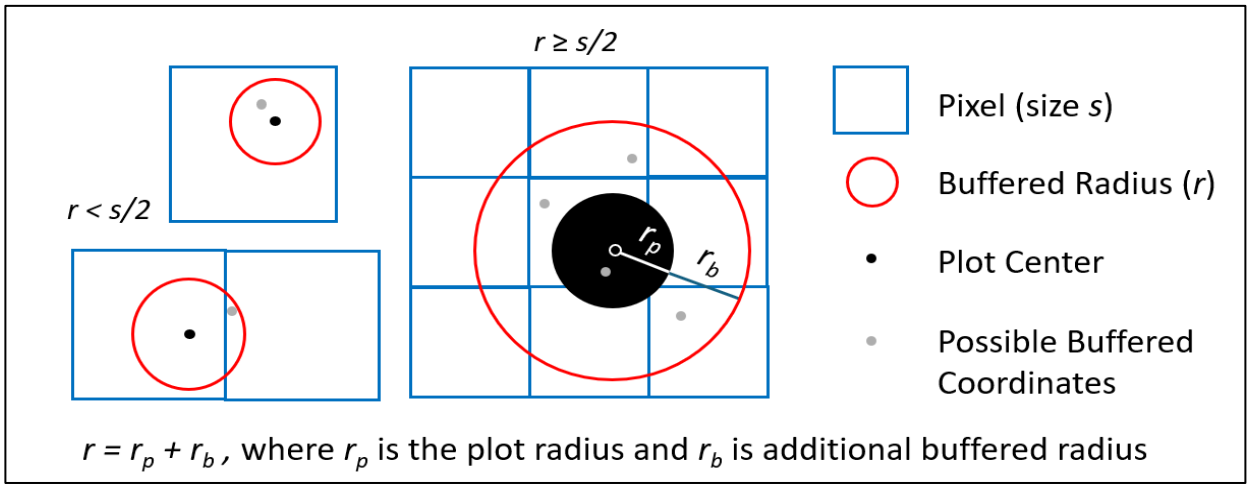


Figure 4: Examples of where randomized pixels may be created in pixels different from the original coordinate

coordinates buffered through *buffer_and_sample* can be used in *point_extraction*, and user-provided datasets that contain shared plot ID values can be run through *aggregated_point_extraction* to quickly aggregate results over the shared values. A diagram of all potential workflows is shown in Figure 7.

Methodological limitations

Limitations exist regarding dataset compatibility. For example, WorldClim Climatology V1 (WORLDCLIM/V1/MONTHLY) and NEON Canopy Height Model (projects/neon-prod-earthengine/assets/CHM/001) are both ImageCollections, which are able to be handled by the created modules. However, both of these datasets contain metadata fields attached to the collection objects themselves called Image Properties, which are used to filter, sort, and select images. This changes the process for filtering GEE datasets and was not considered in the design process of the *forestspot* application. Therefore, GEE ImageCollections that contain Image Properties are incompatible with this application.

Streamlit web application

While the *forestspot* Python application removes many technical barriers and streamlines coordinate buffering and point extraction processes, the required Python skills needed to operate this application are enough to still deter users from utilizing this application. To remove any remaining coding requirements, a web application hosted on Streamlit (https://streamlit.io/) was created that contains the same modules as the Python application but with some differences in the back-end development.

While each of the four modules in the Python application described above are recreated in the web application, there are structural differences due to *ipywidgets* not being compatible with Streamlit. Streamlit offers its own collection of widgets, which was used to recreate the Python package in a Streamlit environment. This approach means that the *forestspot* Python package is not called within the web application. Limitations with this approach result in different functionalities between the Python package and web application. For example, if a mapping feature is added to the Python package, the same functionality will have to be developed specifically for the web application.

In addition, GEE is initialized through a pre-designated Google account held by the application maintainer, and web application users are not able to use their own GEE Project. This differs from the Python application, where GEE is manually initialized with a user-owned account. This is notable particularly when users with sensitive plot coordinates are only permitted to run raw coordinates through an approved Google account. Debugging error messages is also difficult since the detailed error codes are only visible to maintainers.

While limitations exist with the web application that can be improved in the future, the major advantages are that all coding requirements are removed. This, in turn, provides maximized accessibility to users with limited coding knowledge from any computer with internet access, sans Python environment.

Case study

Operational Forest Inventory

We demonstrate the utility of the *forestspot* application by quantifying differences in returned values produced via the different buffering approaches developed. The U.S. Forest

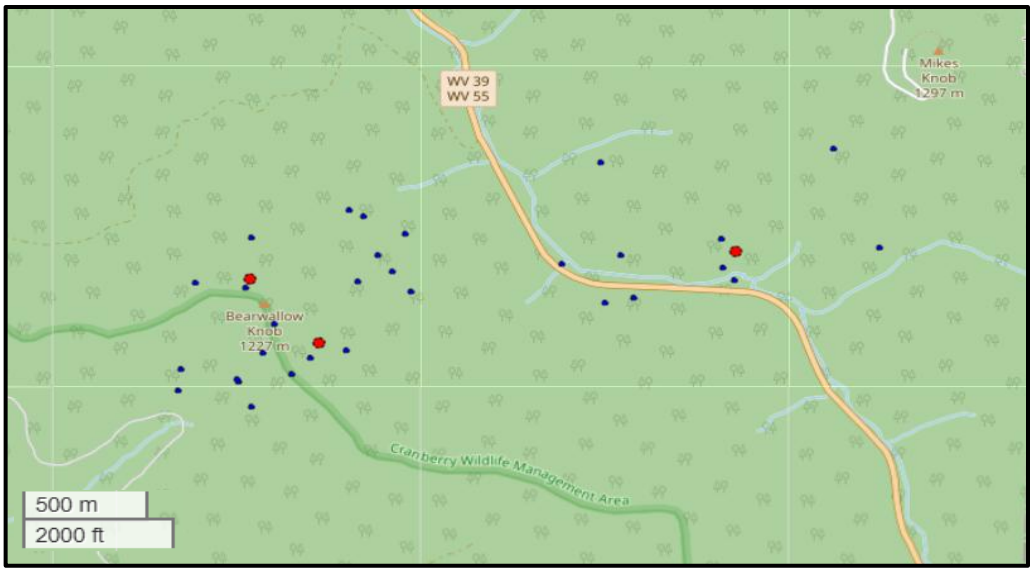


Figure 5: Example of buffer_and_sample approach with r = 914.4 m and n = 10. Uploaded coordinates are marked red, and the 10 buffered coordinates are blue.

Service Forest Inventory and Analysis's (FIA) long-term forest monitoring inventory provided plot coordinates and an existing buffering methodology that serves as our baseline performance.

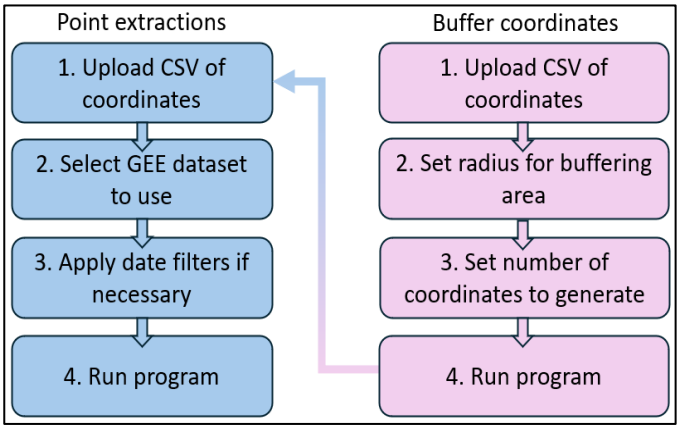


Figure 6: General framework for application use. Point extraction modules can be used alone, or coordinates can first be buffered using the buffer coordinates modules and fed into the point extraction modules.

The FIA's forest monitoring program serves as the United States' national forest inventory and has been used to provide stand-level metrics and as validation data (Cankaya and Radtke 2025, Jiang *et al.* 2025). The FIA's long-term forest monitoring program consists of repeated permanent plot measurements, sampling approximately 0.41 ha and recording comprehensive tree and stand-level measurements. To preserve the integrity of the permanent plot data and protect private landowner information, exact plot coordinates are protected. Instead, coordinates that have been buffered following Westfall *et al*. (2022) are publicly available. Public FIA coordinates are located approximately half to one mile away from the actual plot location. Plots located on private land receive additional fuzzing by swapping plot identifiers. This case study used FIA plots located on federal lands in southern Appalachia which include the following lands: Chattahoochee-Oconee, Cherokee, Nantahala, Pisgah, George Washington-Jefferson, Daniel Boone, Monongahela, and Wayne National Forests; and Great Smoky Mountains and Shenandoah National Parks). As these regions are on federal lands, public FIA coordinates were only buffered within a mile and did not swap identifiers. For this case study, we obtained both the public and protected plot coordinates. The private plot coordinates serve as the "true" value, and the public coordinates error with respect to the true value is the baseline. A total of 2,694 observations were used in this study.

While FIA provides public plot coordinates, running these coordinates through the point extraction modules may contain some amount of error depending on factors like pixel size compared to buffered radius. To determine how the buffering methods perform against the public coordinates, we use the various buffering approaches in tandem with the point extraction module to simulate supplementing FIA forest inventories with WorldClim BIO variables. The WorldClim BIO variables V1 dataset in GEE is commonly used to provide environmental data in existing forest studies, including supplementing FIA data (Marchi *et al.* 2019, Wertz and Wilczyński 2022, Yang *et al.* 2022a, Gou *et al.* 2025). WorldClim BIO has 19 variables containing various climate metrics relating to precipitation and temperature (Table 2). Summary statistics of WorldClim BIO variables used in this case study are found in **Error! Reference source not found.**. It is gridded at a spatial resolution of 927.7 m (3043.5 feet), with time availability from Jan. 1, 1960 – Jan.1, 1991. As the distances between public and private FIA plot coordinates

typically range between 804.5 to 1609.3 m (0.5 to 1 mi), the public and private plot coordinates may lie in adjacent pixels. When this happens, the returned values may not accurately reflect the true conditions at the private plot location.

In this study, we compared the differences in returned values against the private FIA coordinates for three datasets to identify which dataset produced the least error. The three datasets are as follows: one dataset of the public FIA plot coordinates and one dataset from each of the two buffering approaches, buffered using the private FIA coordinates. Buffered coordinates from private FIA coordinates were generated within a 1609.3 m (1-mi) radius, similar to how the FIA's public coordinates are generated. For this analysis, version 2.6.1 was used, with the seed set to 42. At this distance, the probability of buffered points landing within one pixel with 927.5 m (3,043 ft) is approximately 33%, but the probability of the buffered points being within the same pixel is even lower. One buffered dataset contains the coordinates from a single point buffer, and the other contains multi-point buffering using buffer_and_sample module with n=10. Using the raw dataset as the baseline, the average percent differences and average absolute differences were calculated from the 2,694 plot coordinates for all 19

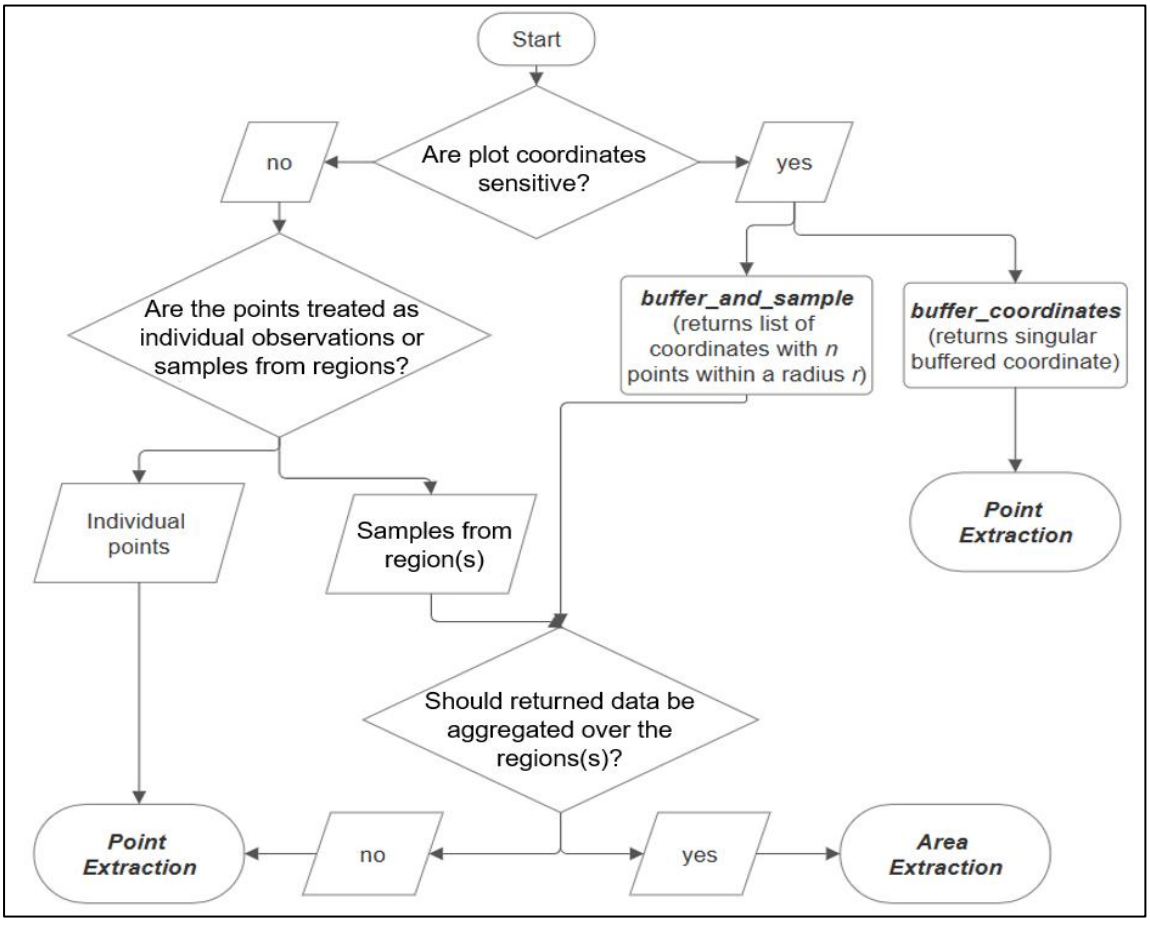


Figure 7: Complete workflow options for module use. The workflow shows how the different modules can be combined in various ways to produce the desired results. For example, the buffer_and_sample module can be used in either point extraction module based off if returned data should be returned aggregated over common plot identifiers or not.

Table 2: WorldClim BIO variables and description

Category	WorldClim BIO var	Label	Description
Temp.	bio01	Ann_t	Annual mean temperature (°C, 0.1 scale)
	bio02	Diurnal_range	Mean diurnal range (mean of monthly (max temp - min temp)) (°C, 0.1 scale)
	bio03	Iso_therm	Isothermality (bio02/bio07 * 100) (%)
	bio04	Season_t	Temperature seasonality (Standard deviation * 100) (°C, 0.01 scale)
	bio05	Warm_mon_tmax	Max temperature of warmest month (°C, 0.1 scale)
	bio06	Warm_mon_tmin	Min temperature of coldest month (°C, 0.1 scale)
	bio07	Ann_t_range	Temperature annual range (bio05-bio06) (°C, 0.1 scale)
	bio08	Wet_qtr_t	Mean temperature of wettest quarter (°C, 0.1 scale)
	bio09	Dry_qtr_t	Mean temperature of driest quarter (°C, 0.1 scale)
	bio10	Warm_qtr_t	Mean temperature of warmest quarter (°C, 0.1 scale)
	bio11	Cold_qtr_t	Mean temperature of coldest quarter (°C, 0.1 scale)
Precip.	bio12	Ann_ppt	Annual precipitation (mm)
	bio13	Went_mon_ppt	Precipitation of wettest month (mm)
	bio14	Dry_mon_ppt	Precipitation of driest month (mm)
	bio15	Season_ppt	Precipitation seasonality (Coefficient of variation)
	bio16	Wet_qtr_ppt	Precipitation of wettest quarter (mm)
	bio17	Dry_qrt_ppt	Precipitation of driest quarter (mm)
	bio18	Warm_qtr_ppt	Precipitation of warmest quarter (mm)
	bio19	Cold_qtr_ppt	Precipitation of coldest quarter (mm)

WorldClim bio-variables. We calculated the relative and absolute differences for each: the public, single buffered, and 10-point buffered datasets compared to the raw dataset. In addition, the coefficients of variation calculated quantify the variance within the multi-point buffered points for each plot and is used to understand the range of values within each plot's multi-point buffered coordinates.

Results

Percent and absolute differences were calculated between the various buffered datasets, shown in

seen in Table 4. Therefore, despite biovariables 04 and 12 appearing to be less accurate than other variables, the percent differences are still small.

Discussion

The buffering methods returned results similar to public FIA coordinates, which is as expected, as the buffered coordinates are generated within the same radius as public coordinates. The reduced inaccuracies may be explained by two things. One, public coordinates are typically buffered within 804.5 to 1609.3 m (0.5 to 1 mi), while the created buffering methods produce

coordinates within the full range from 0 to 1609.3 m. Two, as public coordinates may lie in a pixel adjacent to the private coordinates, the multi-point buffering approach can return values closer to the truth by sampling plots containing the true value.

Table 4. Both buffered datasets produced values similar to the baseline when *r = 1609.3 m (5280 ft)*, as seen with the small percent differences and absolute differences. In some cases, the buffering methods on average produce values closer to true value than the public coordinates produced. Some instances are seen in the percent difference from single point buffer for bio01, bio08 and bio13, as well as the absolute difference for the multi-point buffer for bio01, bio07 and bio14. The single point buffer, however, did not produce average absolute differences smaller than the baseline. The 10-point buffer mostly produced percent differences equal to the baseline, with only one variable where the multi-point buffer performed better (bio14).

Percent differences for all variables across the three comparisons were all small, with most being less than 1%. Biovars 08 and 09 had the largest percent differences (mean temperatures of wettest and driest quarters, respectively). Absolute differences were fairly equal across the comparisons. The differences lie in the range of values across the biovariables. While most absolute differences were under 5, biovariables 04 and 12 (temperature seasonality and annual precipitation) had average absolute differences around 20. However, biovariables 04 and 12 have unscaled values ranging from 6788 to 8784 and 917 to 2067, respectively. These value ranges are between one and two orders of magnitude larger than the values of other biovariables,

Table 3: Summary table of WorldClim BIO variables using protected FIA coordinates. Minimum, 1st quartile, median, mean, 3rd quartile, and maximums are calculated.

bio01	**bio02**	**bio03**	**bio04**	**bio05**
Min. : 60	Min. : 97.0	Min. :31.00	Min. :6788	Min. :206.0
1st Qu.: 95	1st Qu.:124.0	1st Qu.:36.00	1st Qu.:7204	1st Qu.:260.0
Median :108	Median :129.0	Median :38.00	Median :7645	Median :275.0
Mean :108	Mean :127.7	Mean :37.46	Mean :7611	Mean :273.7
3rd Qu.:122	3rd Qu.:133.0	3rd Qu.:39.00	3rd Qu.:7922	3rd Qu.:290.0
Max. :166	Max. :146.0	Max. :43.00	Max. :8784	Max. :327.0
bio06	**bio07**	**bio08**	**bio09**	**bio10**
Min. :-110.00	Min. :292.0	Min. : 5.0	Min. :-48.00	Min. :154.0
1st Qu.: -78.00	1st Qu.:326.0	1st Qu.: 53.0	1st Qu.: -6.00	1st Qu.:190.0
Median : -63.00	Median :337.0	Median :166.0	Median : 62.00	Median :203.0
Mean : -63.09	Mean :336.7	Mean :133.3	Mean : 72.09	Mean :202.8
3rd Qu.: -49.00	3rd Qu.:347.0	3rd Qu.:186.0	3rd Qu.:160.00	3rd Qu.:216.0
Max. : 0.00	Max. :374.0	Max. :220.0	Max. :207.00	Max. :256.0
bio11	**bio12**	**bio13**	**bio14**	**bio15**
Min. :-48.000	Min. : 917	Min. : 94	Min. : 56.00	Min. : 6.00
1st Qu.:-10.000	1st Qu.:1114	1st Qu.:113	1st Qu.: 75.00	1st Qu.:10.00
Median : 6.000	Median :1300	Median :130	Median : 89.00	Median :12.00
Mean : 6.039	Mean :1351	Mean :139	Mean : 90.39	Mean :11.96
3rd Qu.: 22.000	3rd Qu.:1543	3rd Qu.:165	3rd Qu.:104.00	3rd Qu.:13.00

Max. : 70.000	Max. :2067	Max. :207	Max. :155.00	Max. :20.00
bio16	**bio17**	**bio18**	**bio19**	
Min. :272.0	Min. :173.0	Min. :266.0	Min. :173.0	
1st Qu.:319.0	1st Qu.:237.0	1st Qu.:311.0	1st Qu.:239.0	
Median :366.0	Median :284.0	Median :349.0	Median :296.0	
Mean :383.7	Mean :291.4	Mean :353.7	Mean :323.1	
3rd Qu.:444.0	3rd Qu.:333.8	3rd Qu.:390.0	3rd Qu.:415.0	
Max. :557.0	Max. :483.0	Max. :519.0	Max. :530.0	

seen in Table 4. Therefore, despite biovariables 04 and 12 appearing to be less accurate than other variables, the percent differences are still small.

Discussion

The buffering methods returned results similar to public FIA coordinates, which is as expected, as the buffered coordinates are generated within the same radius as public coordinates. The reduced inaccuracies may be explained by two things. One, public coordinates are typically buffered within 804.5 to 1609.3 m (0.5 to 1 mi), while the created buffering methods produce coordinates within the full range from 0 to 1609.3 m. Two, as public coordinates may lie in a pixel adjacent to the private coordinates, the multi-point buffering approach can return values closer to the truth by sampling plots containing the true value.

Table 4: Average percent and absolute differences of public, single buffered point, and 10 buffered points against raw data for all WorldClim BIO variables (raw data is used as the baseline or observed values. The average coefficients of variation within the 10 buffered points for each plot are shown for each WorldClim BIO variable. Units reported in Celsius are unscaled (i.e., results are calculated with raw returned values). See Table 2 for conversion factors.

	Percent Differences (%)			***Absolute Difference***			***Coefficient of Variation***
BIO var	Public vs. Private (baseline)	Single Point Buffer on Private vs. Private	10-Point Buffer on Private vs. Private	Public vs. Private (baseline)	Single Point Buffer on Private vs. Private	10-Point Buffer on Private vs. Private	Within Each Plot's 10-Point Buffer
01	0.45 °C	0.43 °C	0.48 °C	2.51 °C	3.46 °C	2.40 °C	3.57 °C
02	0.18 °C	0.19 °C	0.21 °C	1.56 °C	2.14 °C	1.53 °C	1.84 °C
03	0.08	0.09	0.11	0.30	0.39	0.34	1.28
04	0.03 °C	0.04 °C	0.03 °C	20.12 °C	27.43 °C	18.89 °C	0.37 °C
05	0.21 °C	0.21 °C	0.23 °C	3.56 °C	4.92 °C	3.39 °C	1.94 °C
06	-0.37 °C	-0.40 °C	-0.38 °C	1.88 °C	2.56 °C	1.77 °C	-4.87 °C
07	0.08 °C	0.07 °C	0.08 °C	1.76 °C	2.48 °C	1.74 °C	0.78 °C
08	1.80 °C	1.64 °C	1.89 °C	3.46 °C	4.67 °C	3.51 °C	5.88 °C
09	-0.93 °C	-2.67 °C	-1.32 °C	4.10 °C	5.91 °C	5.10 °C	-15.62 °C
10	0.23 °C	0.23 °C	0.25 °C	2.79 °C	3.86 °C	2.64 °C	2.05 °C
11	-0.36 °C	-2.23 °C	-0.97 °C	2.34 °C	3.23 °C	2.24 °C	-5.33 °C
12	-0.17	-0.17	-0.17	22.02	29.63	20.76	2.36

13	-0.13	-0.12	-0.12	1.68	2.26	1.61	1.78
14	-0.19	-0.21	-0.18	1.91	2.54	1.80	3.02
15	0.34	0.45	0.48	0.27	0.35	0.30	3.67
16	-0.14	-0.14	-0.14	5.11	6.85	4.84	1.94
17	-0.17	-0.20	-0.18	5.66	7.58	5.33	2.81
18	-0.15	-0.15	-0.15	4.91	6.70	4.74	2.00
19	-0.14	-0.17	-0.15	5.24	6.91	4.87	2.45

While there are performance differences between the 3 testing datasets, overall, the differences are incredibly small. Percent differences may only differ between buffered datasets by 0.05%. The small differences in performance between all testing datasets indicate public FIA coordinates capture values as accurately as buffering methods developed in the *forestspot* application. Therefore, it is unnecessary to run raw FIA coordinates through *forestspot* buffering methods when querying from WorldClim BIO variables; public coordinates are sufficient.

As the analysis performed in this study was conducted using a specific radius and dataset, repeating this analysis with other radiuses and dataset resolutions can return different results and performance metrics. Reducing the radius may further reduce error, and users should minimize the buffering radius to as small as possible. In addition, the pixel size of the GEE dataset with respect to the set buffering radius should be considered to understand how likely the buffered coordinates will contain the true value. The ratio of buffer radius to pixel size can also help inform the number of buffered coordinates to produce in the multi-point buffering approach. This can ensure the samples have a high enough point density to capture the true value. Future studies can investigate these remaining uncertainties with using this application.

The *forestspot* application reduces the number of steps to query data from GEE compared to what was previously required and for alternative querying approaches, like ArcGIS or *rgee.* Through the development of the *forestspot* application, the number of steps required to query data from GEE are reduced, increasing overall efficiency.

Conclusions

As forest management and research studies continue to cover larger spatial and temporal scales, geospatial techniques will be required to accomplish large-scale forest monitoring. Therefore, it is imperative to increase user-friendliness with accessing and utilizing the plethora of geospatial data available. The *forestspot* Python application was developed to remove technical challenges required to perform point extractions from GEE and buffer sensitive coordinates to run through Google servers. By developing a widget-oriented application, many of the technical skills needed to access and handle GEE raster datasets are removed. The supplemental web application on Streamlit removes the remaining coding skills needed in order to further increase user accessibility. This app has an open-source repository on GitHub and is downloadable from PyPI.

As this app is hosted on GitHub, users have the option to increase functionality to fit additional needs, such as expanded GEE dataset handling or complementary mapping features. Despite existing limitations, a framework has been established to demonstrate how to remove technical complexities with using geospatial data. This application can be further built on to increase user accessibility, adapting the framework and design to other languages like R, which will in turn, increase the uptake of geospatial data among foresters and other natural resource professionals.

Chapter 2

Comparing live aboveground carbon modeling techniques utilizing stand-level forest inventory and environmental datasets: a case study on national parks and forests of southern Appalachia

Abstract

Accurately estimating live aboveground carbon through species-specific biomass and carbon equations can be rigorous in data collection and processing. Developing alternative carbon estimation models that are efficient but still accurate can allow for faster processing and better forest monitoring. This study compared three parametric and semi-parametric models with 2 input datasets to investigate alternative stand-level estimation techniques to producing estimates through species-specific, individual tree equations. The three parametric models were linear mixed-effects models, LASSO, and generalized additive mixed models, with a random effect added on assigned forest type to allow flexibility in modeling larges areas with different forest compositions. Models were fit on national forests and parks of southern Appalachia using either stand-level data collected by the USDA Forest Service, Forest Inventory and Analysis (FIA) program, or in combination with WorldClim BIO environmental variables. Results show that the three models produced accurate results, with an exception for Shenandoah National Park. Models fit with only forest inventory variables generally produced more accurate and less biased predictions compared to combined forest and environmental variables. Out of all modeling approaches, the linear mixed-effects model was the most efficient with the most accurate estimates. The accuracy and efficiency of methods tested can allow for easier large scale forest monitoring through reduced inventory needs and fast processing.

Introduction

Forest growth and yield models are primarily developed as single tree species, as tree growth and forms can vary greatly between species. Single-species models can be easily expanded to produce stand-level models in single-species softwood stands, like loblolly pine plantations in the southern United States or spruce/pine forests in the Pacific Northwest, because of the species' largely uniform tree growth and form. Hardwood stands are more difficult to model due to greater ecological diversity and structural complexity, because tree form varies not only across species, but within a single species. Although single-species growth and yield models can be fit for individual hardwood species, combining the single-species models to produce stand-level estimates can be cumbersome. Despite challenges associated with modeling hardwood forests, the development of stand-level growth and yield equations in hardwood forests is necessary for forest health monitoring, particularly as existing and future threats jeopardize forest ecosystems.

The southern Appalachia mountains are an ancient mountain range with high biodiversity thanks to the range of climates present, with many endemic plant and animal species. Comprised primarily of mixed-species hardwood forests, southern Appalachia is threatened by both man-made and ecological disturbances, like logging, the introduction of invasive pests and species, and the effects of climate change. For example, Eastern hemlock (*Tsuga canadensis*) is a foundational species in southern Appalachia that has been in severe decline since the

introduction of the hemlock wooly adelgid (*Adelges tsugae*), despite decades of efforts to combat the pest (Ellison *et al.* 2018). Because some tree species are critical for the survival of flora and fauna within these forest communities, the loss of any tree species can have devastating impacts. Previous research has indicated ecological and climate threats have altered carbon storage and sequestration patterns (Keyser, Fearer, and Harper 2016). Such threats have necessitated increased attention on the southern Appalachia region to protect vulnerable forest ecosystems and the biodiversity they contain. Developing growth and yield models for mixed species and hardwood forests like southern Appalachian can allow land managers to better monitor the health or decline of the vital and sacred forests.

To study mixed species forests, mixed-effect models have been used as an alternative to previous data grouping techniques, which require the time-consuming process of using species-specific carbon equations and models. By deploying a mixed-effect model, which consists of fixed-effect predictors and a random-effect on forest type, we can make stand-level estimations over large spatial areas without developing species-specific models. Applying a random effect on forest type builds a global model that can capture differences in forest growth between forest types. This mixed-effect technique was applied in (Yang, Brandeis, and Skiba 2025) to estimate stand-level carbon using a linear mixed-effect model and stand-level predictors. However, alternative estimation approaches and the inclusion of environmental covariates were not evaluated. The inclusion of environmental variables has been shown in previous studies to improve prediction performance (Aguirre *et al.* 2021, Liepiņš *et al.* 2022, Yang *et al.* 2022b). Because environmental variables can help increase performance, predictors taken from the U.S. Forest Service, Forest Inventory and Analysis (FIA) forest monitoring program were combined with WorldClim BIO variables V1 dataset were fit to three models: linear regression mixed-effect model (LME), least absolute shrinkage and selection operator regression (LASSO), and generalized additive mixed model (GAMM). Results provide important information on modeling techniques and predictor variables to efficiently estimate stand-level carbon.

Despite complexities involved with modeling mixed species forests, there is a need to develop models to help monitor important yet vulnerable mixed-species hardwood forests, like in southern Appalachia. In particular, being able to quickly estimate carbon at the stand-level allows for better forest health monitoring and understanding of how forests respond to on-going and future threats. This study tests multiple viable methodologies for estimating stand-level carbon and provides information about model performance using two variable sets. Utilizing long-term national forest inventory data from the FIA long-term forest monitoring program and WorldClim BIO environmental variables, models tested can be implement monitor forest resources and vulnerable ecosystems of southern Appalachia and expanded for use in other regions to continuously.

Methods

Study Area

This study's regions of interest include 8 national forests and 2 national parks: Chattahoochee-Oconee, Cherokee, Nantahala, Pisgah, George Washington-Jefferson, Daniel Boone, Monongahela, and Wayne National Forests; and Great Smoky Mountains and Shenandoah National Parks. These regions are described in Yang *et al.* (2025).

Tree data

Tree and stand-level data for this study comes from the Forest Inventory and Analysis's long-term forest monitoring program, which serves as the United States' national forest inventory. This inventory contains plot measurements, remeasured every 5-7 years in the Southeast. Each plot contains four fixed-radius subplots (7.3-m/24-ft radius), with all trees with diameter at breast height (DBH) greater than or equal to 12.7 cm or 5 in. recorded. Four fixed-radius microplots (2.1-m/6.8-ft) are nested within each subplot, and trees with diameters between 2.54-cm (1-in.) and 12.7-cm (5-in.) are recorded. The entire plot is divided up based off site condition classifications, like land use, ownership, forest type, stand size, and more. A plot can fall entirely in one distinct condition or divided into multiple condition classes that run across the subplot and microplot boundaries. Because of the similarities within a condition class, the condition class was treated as the sampling unit and is subsequently referred to as "stand-level". Data was obtained following the data processing protocol described in Yang *et al.* (2025). Tree-level carbon estimates, calculated using Westfall *et al.* (2024). By-region summary statistics are presented in Table 6.

The distribution of FIA forest compositions varies across and within the regions of interest, similar to Yang *et al.* (2025). Although these forests are primarily classified as oak-hickory, each study region contains a variety of forest types with differing growth and yield patterns. Shenandoah National Park, the smallest region included, has only 14 distinct forest types present, while Cherokee and George-Washington National Forests (the two largest regions in this study) have 33 distinct forest types each.

Environmental data

Environmental variables like average annual precipitation and temperature play a strong role in determining how a forest ecosystem develops, both in structural complexity and growth rates (Fotis *et al.* 2018, Ali *et al.* 2020, Ehbrecht *et al.* 2021). The WorldClim BIO variable dataset was used to provide environmental data for this study due to its temporal compatibility with inventory data (January 1, 1960, to January 1, 1990) and 19 temperature and rainfall variables (listed in Table 5). The combination of time compatibility and abundance of variables provides a comprehensive picture of the environmental conditions that the forests of this study developed under. The spatial resolution is 927.67 m., which is larger than the FIA plot size.

The environmental data was queried using FIA private plot coordinates in the *forestspot* Python application, using the entire time availability which produces values from the median composite image. This approach captures how previous climate conditions influenced later carbon values. Table 6 shows the summary statistics (average and standard deviation) of environmental variables, by region. While the regions share many environmental characteristics, evidenced by the relatively small range within each variable, there are some differences that would result in different carbon stocking values. For example, the variation in average temperature between wettest and driest quarters (*bio08* and *bio09*, respectively) show inverted relationships between regions. For example, Chattahoochee-Oconee, Cherokee, and Nantahala have low average temperatures in the wettest quarter and high temperatures during the driest quarter. Conversely, Pisgah, Washington-Jefferson, and Daniel Boone experienced high temperatures during the wettest quarter and low temperatures during the driest quarter. The Great Smoky Mountains National Park has the highest average slope, whereas Wayne National Forest is the flattest of all regions. Additionally, annual precipitation varied greatly between regions. Under the theory that resource competition drives forest development, these environmental differences can significantly change forest characteristics that appear in each region.

Modeling Approaches

All analyses were performed in R statistical software (Version 4.5.3, https://www.r-project.org/). Three modeling approaches were chosen for their various strengths when working with high-dimensional datasets with complex relationships: linear mixed-effect models (LME), least absolute shrinkage and selection operator regression (LASSO), and generalized additive mixed model (GAMM). The three modeling are simplicity, feature selection, and non-linearity strengths, respectively. The LME model is a simple methodology that has proven performance (Yang, Brandeis, and Skiba 2025). The LASSO model is one regularization technique that performs variable selection to prevent overfitting and increase model interpretability (although model interpretation was not conducted in this study as we focused on prediction power) (Sohil, Sohali, and Shabbir 2022). GAMMs are a semi-parametric approach that produces non-linear models, which is particularly helpful to capture non-linear relationships.

Each model was fit with stand-level variables alone and stand and environment variables combined.

Linear mixed-effects model

The linear mixed-effects model (LME) was built as the following:

$$C = \sum x\beta + \mathrm{b} + \epsilon$$

$\sum x\beta$ is the linear combination of predictors x and their coefficients β, b is a forest-type-specific random effect, and error term ϵ. We assume that b follows a normal distribution with forest-type-specific variance σ^2_{forty} ($b \sim N(0, \sigma^2_{forty})$), and ϵ follows a normal distribution with

Table 5: List of stand and environmental variables compiled from U.S. Forest Service, Forest Inventory and Analysis database and WorldClim BIO variables.

Variable Group	Label	WorldClim BIO var	Description
Stand-level	*D*		Mean diameter at breast height (cm)
	H		Mean height (m)
	H-D		Mean height-diameter ratio
	QMD		Quadratic mean diameter (cm)
	BA		Basal area per hectare (m2/ha)
	TPH		Number of trees per hectare
	SDI		Site density index
	Aspect		Aspect (° from due north)
	Slope		Slope (°)
Environmental	*Ann_t*	*bio01*	Annual mean temperature (°C, 0.1 scale)
	Diurnal_range	*bio02*	Mean diurnal range (mean of monthly (max temp - min temp)) (°C, 0.1 scale)
	Iso_therm	*bio03*	Isothermality (bio02/bio07 * 100) (%)
	Season_t	*bio04*	Temperature seasonality (Standard deviation * 100) (°C, 0.01 scale)
	Warm_mon_tmax	*bio05*	Max temperature of warmest month (°C, 0.1 scale)
	Cold_mon_tmin	*bio06*	Min temperature of coldest month (°C, 0.1 scale)
	Ann_t_range	*bio07*	Temperature annual range (bio05-bio06) (°C, 0.1 scale)
	Wet_qtr_t	*bio08*	Mean temperature of wettest quarter (°C, 0.1 scale)
	Dry_qtr_t	*bio09*	Mean temperature of driest quarter (°C, 0.1 scale)
	Warm_qtr_t	*bio10*	Mean temperature of warmest quarter (°C, 0.1 scale)
	Cold_qtr_t	*bio11*	Mean temperature of coldest quarter (°C, 0.1 scale)
	Ann_ppt	*bio12*	Annual precipitation (mm)
	Went_mon_ppt	*bio13*	Precipitation of wettest month (mm)
	Dry_mon_ppt	*bio14*	Precipitation of driest month (mm)
	Season_ppt	*bio15*	Precipitation seasonality (Coefficient of variation)
	Wet_qtr_ppt	*bio16*	Precipitation of wettest quarter (mm)
	Dry_qrt_ppt	*bio17*	Precipitation of driest quarter (mm)
	Warm_qtr_ppt	*bio18*	Precipitation of warmest quarter (mm)
	Cold_qtr_ppt	*bio19*	Precipitation of coldest quarter (mm)

mean zero and variance σ^2 ($\epsilon \sim N(0, \sigma^2)$). The "lmer" function from the *lme4* R package was used to fit the linear mixed-effect models (Bates *et al.* 2015).

LASSO regression

The mixed-effect LASSO (least absolute shrinkage and selection operator) regression is similar to the linear mixed model in form, but with an additional penalty term that error is minimized on. This penalty term gives LASSO regression its most notable asset: feature selection (Faraway 2015). When using datasets with many correlated variables, as is the case for

the stand-level and environmental variables used in this study, this feature selection identifies which variables do not aide in explaining the response variable. This process is an efficient alternative to traditional model optimization processes like stepwise regression while still using traditional techniques to perform model selection. Compared to LME models which keep all input variables, LASSO models shrink non-influential variables and retain the best predictors.

LASSO's feature selection is possible due to the shrinkage penalty added to the error that is being minimized:

$$min(RSS + \lambda \sum |\beta_j|)$$

where RSS is the residual sum of squares, and the second term is where the additional penalty is included. λ (lambda) can be any positive value, with coefficients β from j equals 1 to p number of predictors.

As lambda can be adjusted, LASSO models have an additional level of flexibility to optimize fitness. When lambda is 0, the model reduces down to least-squares regression. Lambda can either be chosen via previous knowledge and research or optimized through iterative processes to choose the best. Using an optimized lambda value ensures that non-influential predictors' coefficients shrink to zero. For our analysis, a five-fold cross validation was used to tune lambda for each bootstrap iteration, with lambda being tested at increments of 5 between 0 and 500.

GAMM

GAMs (short for generalized additive models) are an extension of generalized linear models that allow for modeling non-linear relationships between dependent and independent variables, like those found in forest systems (Adhikari, Montes, and Peduzzi 2023). While linear models form relationships using linear functions, mixed GAMS (or GAMMs) create relationships using smooth functions. The generalized additive mixed model was built as the following:

$$g(\mu_i) = \boldsymbol{X}_i\beta + \boldsymbol{Z}_i\boldsymbol{b} + \boldsymbol{\epsilon}$$

where μ_i is the expected value, g is a smooth monotonic 'link function'. Fixed effects are contained in the first term, where $\boldsymbol{X}_i$ is the ith row of a model matrix $\boldsymbol{X}$, and β is a vector of unknown parameters. The forest-type random effect has model matrix $\boldsymbol{Z}$ and vector $\boldsymbol{b}$, the latter which follows a normal distribution with mean expected value of zero and covariance matrix ψ_θ (i.e., $\boldsymbol{b} \sim \boldsymbol{N}(\boldsymbol{0}, \boldsymbol{\psi_\theta})$, and unknown parameters in θ. Error ϵ follows a normal distribution with mean 0 and variance Λ_θ, where Λ_θ is a positive definite matrix used to model residual autocorrelation. The full theory and description are found in Wood (2017). The "gamm" function from the *mgcv* R package (Wood 2017) was used to fit these models. Default parameters (i.e. *k=10*) were used for the stand-only model. In the stand+environmental variable model,

Table 6: Summary table of predictor variables by region, with median values and standard deviations. WorldClim BIO variables are unscaled from raw queried data. Refer to Table 1 for conversion factors.

Variable	Chattahoochee-Oconee	Cherokee	Nantahala	Pisgah	George Washington-Jefferson	Daniel Boone	Monongahela	Wayne	Great Smoky Mountains	Shenandoah
C	73.5 (44.8)	67.2 (47.1)	73.6 (45.4)	76.3 (57.4)	69.7 (41.3)	63.7 (45.1)	88.7 (48.3)	50.9 (41.7)	90.2 (47.7)	76.9 (42.0)
D	23.2 (5.3)	22.1 (6.2)	23.2 (5.7)	23.8 (6.4)	22.4 (5.2)	22.3 (5.5)	25.8 (5.5)	24.7 (5.4)	24.9 (5.3)	23.7 (6.2)
H	18.1 (3.6)	17.3 (4.6)	17.6 (4.1)	17.6 (4.6)	16.0 (3.8)	18.2 (3.9)	18.1 (3.9)	18.7 (3.7)	18.6 (4.2)	15.2 (4.1)
H-D	88.6 (13.4)	89.9 (15.8)	88.1 (14.5)	83.9 (16.2)	81.0 (15.6)	95.6 (14.5)	77.9 (13.0)	83.3 (13.0)	87.3 (13.2)	76.2 (10.6)
QMD	25.9 (6.1)	24.9 (7.1)	26.3 (6.6)	26.6 (7.3)	25.1 (5.9)	25.6 (6.1)	28.5 (6.2)	27.1 (6.2)	28.4 (6.2)	27.4 (7.0)
BA	24.1 (11.7)	23.0 (12.1)	23.7 (10.9)	25.8 (12.9)	23.8 (10.4)	20.1 (10.6)	27.7 (11.7)	16.8 (9.4)	29.1 (12.7)	25.2 (9.4)
TPH	890.3 (757.3)	897.8 (904.7)	886.3 (890.6)	846.4 (901.5)	957.2 (801.0)	897.8 (985.0)	476.0 (752.2)	327.2 (566.4)	861.2 (709.6)	886.9 (619.9)
SDI	943.0 (570.5)	920.9 (593.1)	921.4 (570.2)	921.8 (625.3)	942.2 (561.0)	960.7 (635.6)	577.5 (653.8)	385.7 (472.3)	1061.0 (711.9)	976.3 (495.5)
Aspect	170.0 (112.9)	184.0 (106.8)	176.0 (99.6)	205.0 (95.1)	175.0 (104.3)	176.5 (102.7)	175.0 (107.3)	160.0 (109.9)	195.0 (107.4)	171.0 (113.9)
Slope	34.5 (20.2)	45.0 (20.8)	42.5 (19.5)	42.0 (18.6)	35.0 (18.2)	35.0 (17.9)	30.0 (18.5)	26.0 (17.2)	52.0 (19.0)	34.0 (17.3)
Ann_t	127.0 (11.5)	121.0 (14.3)	110.0 (11.5)	115.0 (13.8)	104.0 (11.0)	123.8 (3.8)	83.0 (9.1)	110.0 (5.7)	111.0 (16.2)	97.0 (10.1)
Diurnal_range	131.0 (4.1)	134.0 (6.0)	130.5 (8.0)	132.0 (7.5)	129.0 (6.0)	132.0 (3.1)	119.8 (7.3)	129.0 (2.0)	133.0 (9.1)	116.5 (7.4)
Iso_therm	40 (1.2)	39 (1.0)	40 (1.4)	39 (1.3)	37 (1.3)	37 (0.9)	35 (1.5)	35 (0.5)	40 (1.7)	33 (1.5)
Season_t	7093.0 (230.1)	7403.5 (149.8)	6984.0 (152.3)	7091.0 (185.3)	7871.0 (218.7)	8028.5 (158.9)	7912.0 (64.4)	8547.0 (102.0)	7133.0 (143.7)	8072.0 (97.8)
Warm_mon_tmax	285.5 (14.7)	286.0 (19.1)	267.0 (16.2)	274.0 (20.3)	276.0 (14.8)	298.0 (3.1)	247.0 (13.7)	293.0 (5.1)	270.0 (21.2)	266.0 (15.1)
Cold_mon_tmin	-41 (10.2)	-49 (11.2)	-56 (7.6)	-52 (10.1)	-68 (11.1)	-57 (5.5)	-92 (7.2)	-73 (6.6)	-59 (11.4)	-78 (7.7)
Ann_t_range	324.0 (6.7)	334.0 (8.2)	322.0 (9.8)	325.0 (10.9)	345.0 (10.1)	352.5 (5.7)	340.0 (6.9)	365.0 (3.4)	328.0 (10.4)	343.0 (8.6)
Wet_qtr_t	45 (10.9)	56 (70.5)	32 (18.5)	126 (42.4)	183 (17.8)	205 (3.0)	163 (9.3)	198 (5.1)	41 (42.4)	177 (40.8)
Dry_qtr_t	174 (17.2)	148 (65.8)	153 (18.4)	48 (39.5)	2 (26.6)	182 (24.7)	-17 (50.6)	-6 (33.4)	128 (22.2)	-12 (9.9)
Warm_qtr_t	215 (13.4)	214 (16.2)	198 (12.8)	204 (16.0)	202 (12.0)	224 (2.9)	179 (10.0)	217 (4.9)	200 (17.6)	197 (11.0)
Cold_qtr_t	31.0 (10.6)	23.0 (12.6)	16.0 (10.5)	19.0 (12.2)	0.0 (11.4)	15.0 (5.3)	-24.5 (9.0)	-6.0 (7.2)	14.0 (14.5)	-12.0 (9.9)
Ann_ppt	1678.0 (137.7)	1455.0 (167.8)	1778.0 (156.6)	1396.0 (215.7)	1058.0 (93.4)	1265.0 (46.0)	1320.8 (122.4)	1063.0 (26.6)	1542.0 (166.1)	1131.0 (69.1)

Went_mon_ppt	180 (11.2)	154 (19.9)	184 (13.2)	136 (21.5)	105 (8.5)	129 (2.8)	131 (12.1)	114 (4.1)	160 (14.9)	111 (5.9)
Dry_mon_ppt	109 (12.7)	91 (10.0)	117 (17.3)	100 (19.4)	72 (8.2)	79 (3.5)	93 (10.9)	69 (3.7)	94 (16.7)	74 (5.0)
Season_ppt	13 (1.9)	13 (1.5)	11 (2.0)	9 (1.6)	12 (1.8)	12 (1.7)	10 (1.6)	16 (1.4)	11 (2.1)	14 (0.7)
Wet_qtr_ppt	487 (31.4)	416 (49.3)	498 (33.3)	377 (52.7)	297 (23.7)	359 (5.9)	375 (31.5)	318 (4.9)	429 (40.0)	320 (17.4)
Dry_qrt_ppt	361 (38.8)	303 (34.8)	389 (47.0)	309 (60.0)	224 (26.3)	274 (11.3)	294 (33.9)	222 (11.6)	327 (48.6)	227 (15.9)
Warm_qtr_ppt	398.0 (37.6)	362.0 (26.2)	428.5 (38.8)	372.0 (48.3)	289.0 (21.0)	338.0 (5.9)	367.5 (30.2)	313.0 (6.5)	391.0 (39.1)	312.0 (12.3)
Cold_qtr_ppt	455.0 (32.5)	391.0 (69.1)	470.0 (32.9)	314.0 (60.4)	224.0 (28.5)	303.0 (20.1)	294.5 (34.8)	222.0 (11.7)	402.0 (38.7)	227.0 (15.9)

Table 7: Summary table of forest type groups and observation count by region.

	Aspen/ birch		Elm/ash/ cottonwood		Exotic HW		Loblolly/ shortleaf		Maple/beech/ birch		Oak/gum/ cypress		Oak/ hickory		Oak/pine		Other HW		Spruce/fir		White/red/ jack		Total	
	Obs	FT	Obs	FT	Obs	FT	Obs	FT	Obs	FT	Obs	FT	Obs	FT	Obs	FT	Obs	FT	Obs	FT	Obs	FT	Obs	FT
Chattahoochee-Oconee	0	0	7	2	0	0	154	5	1	1	0	0	1152	15	279	5	14	1	0	0	74	3	1681	32
Cherokee	0	0	0	0	0	0	143	5	76	3	0	0	887	15	145	5	17	1	1	1	66	3	1335	33
Nantahala	0	0	0	0	0	0	15	5	19	2	0	0	417	13	32	4	10	1	0	0	22	2	515	27
Pisgah	2	1	0	0	0	0	6	2	11	1	0	0	373	12	35	4	23	1	4	1	15	2	469	24
George Washington-Jefferson	0	0	0	0	1	1	106	3	81	4	2	1	1990	15	231	4	18	1	2	1	60	3	2491	33
Daniel Boone	0	0	3	1	0	0	7	3	34	3	0	0	508	14	16	4	0	0	0	0	8	1	576	26
Monongahela	2	1	2	1	0	0	4	3	607	4	0	0	510	13	18	3	28	1	51	1	22	3	1244	30
Wayne	10	1	25	3	0	0	6	1	31	2	0	0	275	13	17	2	1	1	0	0	19	1	384	24
Great Smoky Mountains	0	0	1	1	0	0	16	5	58	3	1	1	221	13	23	4	53	1	9	1	10	3	392	32
Shenandoah	0	0	1	1	0	0	0	0	0	0	0	0	164	11	1	1	3	1	0	0	0	0	169	14

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environmental variables' k parameters were set to 2 to constrain the flexibility of the smooth terms. This parameter was increased within the model to the minimum as needed. The maximum likelihood estimation method was used to prioritize unbiased fixed effects over unbiased random effects.

Model Building and Evaluation Statistics

The various combinations of modeling approaches and variable groups (stand-level and stand+environmental) were built under 1,000 cluster bootstrap samples to approximate the sampling distribution of evaluation statistics (Murphy 2012). Because of the repeated plot measurements, potential temporal autocorrelation was accounted for by splitting training and testing datasets (also referred to as in-bag and out-of-bag data) on the sampling unit in a 70-30 split between training and testing datasets. Using this approach, if a plot condition is sampled in the training dataset, all repeated measurements for that plot condition are grouped into the training dataset and will not be included in the testing dataset. Likewise, data included in the testing dataset will not have been used in the training dataset. As a specific example, if sampling unit A was selected for the in-bag data, its measurements in 2012, 2017, and 2022 are all in the in-bag data. Sampling unit B was selected for out-of-bag data, then its measurements in 2008, 2013, and 2018 are all contained in the out-of-bag data. For each bootstrap sample, R^2, mean bias (MB), and root mean square error (RMSE) were calculated for the in-bag and out-of-bag data using the following equations:

$$R^2 = 1 - \frac{\sum(e^2)}{\sum(C - \bar{C})^2}$$

$$MB = \frac{\sum(e)}{n}$$

$$RMSE = \sqrt{\frac{\sum e^2}{n}}$$

$\bar{C}$ is the mean of C, $e = C - \hat{C}_{final}$, and n is the total number of observations. Once the 1,000 bootstrap sample were trained, the median and 95% confidence intervals evaluation statistics were computed. In addition, for the LASSO models fit, coefficients of the fixed-effect variables were extracted from each bootstrap sample, and the median and 95% confidence intervals were calculated to determine significant predictors.

To visualize fit, one out-of-bag sample for each study area was used to plot predicted vs observed carbon values. A 1:1 diagonal line and LOESS smoothing curve with 95% confidence interval were added, using the *ggplot2* R package (Wickham 2016).

Results

Generally, all 3 modeling approaches fit using the nine stand-level predictors performed equally well, providing accurate predictions. Of the three models tested, the LME models produced the least biased predictions, followed by GAMMs, and lastly, LASSO. The differences between the LME model with GAMMs and LASSOs are large enough to identify the LME model as the best predictive model with the highest stability, as GAMM and LASSO models have greater performance variability. Shenandoah National Park is one example of the model instability; the lower 2.5% of R-squared values produced by the stand+environmental LASSO bootstrapped models are smaller than 0.06. The notable difference between Shenandoah National Park, which presented many challenges throughout this study, and other study areas is that the former is the smallest, with the least amount of acreage and fewest number of observations (169 observations). To compare, the next largest region is Wayne National Forest, with 384 observations and greater predictive power.

The inclusion of environmental predictors did not drastically improve model performance across the three methodologies tested and produced more biased results than the stand-only models. This is counter to the findings of other studies (Yang *et al.* 2022b), with caveats. The only approach that improved when environmental variables were added was LME. With the other modeling approaches, the inclusion of environmental variables could severely reduce performance, as observed in Shenandoah with the LASSO and GAMM models. Again, the same issues with small sample size could be insufficient to fit models with more complex computations. This, combined with the increased number of predictors, may cause models to be highly unstable or fail altogether (Figure 3).

WorldClim Data Conditions

As the WorldClim BIO dataset contains an assortment of temperature and precipitation variables, many which are related to each other, it is unsurprising that multicollinearity is present (Pradhan and Setyawan 2021). Although multicollinearity has been found in WorldClim BIO data, in order to produce a quick model that does not need intense data cleaning prior to use, we did not account specifically for multicollinearity for prediction models. Instead, variables were removed only as needed to allow models to be fit. Model failure was initially observed across all stand+environmental LASSO and GAMM models when WorldClim variable *bio07* (annual temperature range) was included, likely due to *bio07* being calculated as the difference of *bio05* and *bio06*. In addition, the model summary produced from the LME model dropped *bio07*.

With *bio07* removed, the LASSO model and almost all GAMMs were able to be fit, but the GAMM fit on Shenandoah continued to fail due to remaining multicollinearity. To identify problematic variables in the Shenandoah GAMM, a linear regression model was fit, and the model summary's coefficient estimates were not available for *bio07*, *11*, and *19* (annual temperate range, mean temperature of coldest quarter, and precipitation of coldest quarter, respectively). Upon further investigation, these variables had direct correlations with another

variables. For example, *bio09* and *bio11* (mean temperature of driest quarter and coldest quarter, respectively) have a 1:1 relationship. Keeping *bio09* within the model requires *bio11* to be removed, and vice versa. From this investigation, we removed *bio07*, *11*, and *19* from Shenandoah's GAMM model. Coefficients from LME and LASSO models are provided in Table 10 and Table 11.

LME

Both variable sets of LME models performed well across all regions, as evidenced by high R^2 values in Table 8. No clear pattern was observed among performance between regions, based on sample size or area. Compared to Yang *et al.* (2025), model performance with similar across all metrics. Out-of-bag MB was slightly larger for this study, but the median and range of RMSE values were smaller than in Yang *et al.* (2025). The inclusion of environmental variables in the LME models produced slightly more accurate predictions than stand-only models, with increases in median, upper and lower 95% quartiles of R^2 values (Table 9). However, potential bias increased compared to stand-only models, with larger ranges of MB and RMSE values. The increased bias may be due to overfitting the model with the extra variables, ultimately resulting in a poorer performing model.

LASSO

Across variables sets and regions, LASSO performed considerably worse than either of the other modeling techniques. While the median values of all summary statistics do not appear to perform much worse than LME or GAMM, the lower 2.5% quartiles show that some bootstrap samples produced poor models. The worst performing region was Shenandoah, where the lower 2.5% of bootstrapped samples had R^2 values lower than .06 (stand + environmental predictors used). The 95% range of MB was multiple times larger than those of LME models, and upper 97.5% of RMSE values up to twice as large as the LME models. Therefore, potential model bias can vary greatly between each model trained and is less reliable than the LME models.

Even in other models that performed better than Shenandoah, out-of-samples showed high enough variability, with average prediction performance lower than LME models and GAMMs. In both stand-only and stand+environmental predictor models, it was observed that performance between regions was related to regional sample size, as regions with more observations had higher prediction performance than those with fewer observations (Figure 8 and Figure 9). A region's number of forest types presents did not appear to have an effect on performance.

Furthermore, median values of the optimized lambda parameter were mostly 0, indicating that the majority of fitted models were reduced to least-squares regression, ultimately performing the same regression process as LME models (Table 11). This indicates that using the LASSO approach is altogether unnecessary.

GAMM

GAMMs fit with stand-level variables performed mostly equally as well as LME models for fit metrics across all regions. R-squared, MB and RSME values were approximately the same between the linear mixed-effect model and GAMMs. GAMMs were observed to have lower R^2 confidence intervals ranges (by around 1 to 4%). Average OOB MB values were mostly smaller, but with larger confidence intervals. Average OOB RMSE for GAMMs were larger than LME models for a few regions (Wayne National Forest and the Great Smoky Mountains and Shenandoah National Parks), with larger confidence interval ranges as well.

As for when environmental variables are added, the pattern changes. With environmental variables added, performance metrics of GAMMs dropped noticeably compared to LME models. While Chattahoochee-Oconee and Cherokee had similar performance compared to the LME models, all other models performed worse. The most notable decrease in performance is seen in Shenandoah National Park. While the average R-squared value was 0.82, the lower 2.5% of models had R-squared values less than 0.06. Shenandoah's MB and RMSE are noticeably higher than for Shenandoah's LME model.

Out-of-bag performance

While the metrics for these models can appear satisfactory, further investigation into models' out-of-bag predicted versus observed values highlight further issues. For both stand-only and stand and environmental predictors, LME models struggled to accurately predict higher carbon values (C > 150 MgC/ha), consistently underpredicting in this range to various degrees (Figure 10). LASSO models' predicted versus observed graphs further emphasize the modeling approach's poor performance, with particularly rounded loess curves, further exacerbated when environmental variables were included. These stand + environmental models' loess curves are so acute that predictions are overfit in the middle range of carbon values (50-100 MgC/ha) and severely underfit at higher carbon stocking levels.

Discussion

Evaluation of dataset combinations

This study tested whether including environmental variables in three modeling approaches increased predictive power, as other studies have previously found (Aguirre *et al.* 2021, Liepiņš *et al.* 2022, Yang *et al.* 2022b). With the three modeling approaches tested, the inclusion of environmental variables resulted in model performance approximately as well as models fit without environmental variables. For the linear mixed-effect model, the inclusion of environmental variables improved model performance compared to only-stand variables, providing a stable, highly accurate prediction model for stand-level live aboveground carbon. For LASSO models and GAMMs, the inclusion of environmental did not improve predictive power, and was even detrimental in some regions, most clearly observed in Shenandoah National Park. In addition, as previously mentioned, issues arose with including environmental variables in

GAMMs, and additional data cleaning was required to identify and subsequently remove variables that prevented models from being fit. The inclusion of environmental variables in LASSO models and GAMMs reduced performance to the point that models could be unusable.

If including environmental variables in datasets is unable to be included for various reasons, using stand-only predictors with a linear mixed-effects model still provides highly accurate estimate. However, if resources are available and WorldClim BIO variables can be added, more accurate results are returned, which could be desirable in some monitoring programs.

Evaluation of model approaches

Out of the three modeling approaches tested, LME models performed the highest and were the fastest due to its simple form Its accuracy is similar to those reported in Mohd Zaki *et al*. (2017), which used multiple linear regression techniques combined with field and remotely sensed forest metrics. It consistently produced accurate estimates across all regions, even with smaller sample sizes. GAMMs performed the second best; stand-only GAMMS produced as accurate estimates as stand-only LME models, but some regions had models that performed far worse and possibly unusable altogether. Sample size appeared to have an effect on performance, with larger samples having better estimates. GAMMs were able to produce accurate estimates, but their inconsistent performance in smaller regions and additional data cleaning needed is more cumbersome than the LME models. However, GAMMs should not be fully discounted, as predicted versus observed figures show that GAMMs can better estimate larger carbon values than LMEs, which demonstrates one advantage of GAMMs' flexibility. In addition, GAMMs did not take considerably longer to compile than LMEs, which keeps GAMMs as a model contender for carbon estimation. Caution should be taken to ensure that the model is well trained before implementation in small regions covering similar areas as Wayne National Forest and Shenandoah National Park.

LASSO models consistently performed the worst, along with having the longest processing time. We hypothesized that when adding 19 temperature and precipitation variables, some may be redundant. Using LASSO's feature selection to remove some variables should help with model overfitting and increase predictive power. However, we observed the opposite. Model performance diminished enough to determine that LASSO models are inadequate for estimating live aboveground carbon. In addition, many samples' optimized lambda parameter was zero, defeating the point of conducting LASSO regression. Due to the five-fold cross-validation approach used to optimize lambda, fitting the LASSO models expended a large amount of computational power. While the cross-validation approach can be removed to expedite processing time, the LASSO approach can be abandoned altogether for the more efficient LME approach.

While these LASSO models were not able to provide the most accurate estimates, the two regions when the median lambda estimate was not zero (stand-only variables in Great Smoky

Mountains National Park as well as stand-only and stand + environmental variables in Shenandoah National Park) can be later studied to understand what differentiates these cases from the other regions and models. Further exploring lambda optimization and coefficient interpretation to identify notable predictors can be conducted, which can provide deeper insight into the influence of stand and environmental conditions on live aboveground carbon.

Conclusion

When implementing the results from study, LME models are able to be trained the fastest and with great accuracy. However, if larger carbon values need to be estimated, GAMM models may be more favorable. However, when using GAMM models, a large enough dataset is required to train the models to have enough accuracy. Also, all models tested produced predicted carbon values less than 0, which does not have biological meaning and should be removed in post-modeling analysis. General mixed model approaches with the same inputs and parameters were used for all regions of interest (with some exceptions), and future work can be conducted to tailor model parameters and inputs specifically to a region. The inclusion of environmental variables improves accuracy for LME models but can severely worsen predictive power for LASSOs and GAMMs, even rendering some models completely unusable.

The development of these models offers alternative methodologies to estimate live aboveground carbon using a regionally trained model with forest inventories and optional environmental variables.

Table 8: Evaluation statistics for the three modeling approaches fitted with stand-level variables for each region. R-squared, root-mean-squared-error (RMSE), and mean bias (MB) were calculated for in-bag (IB) training data and out-of-bag (OOB) testing data with median estimates and 95% confidence intervals.

STAND-LEVEL VARIABLES

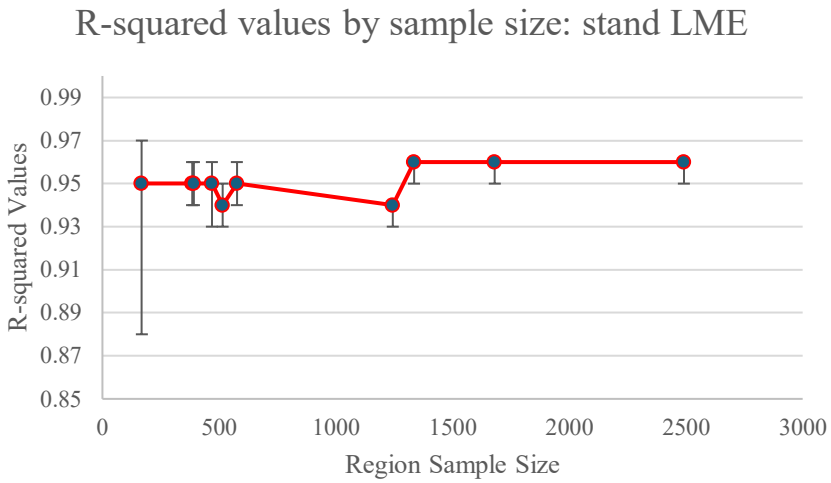
		R-squared						**MB**						**RMSE**					
		IB	CI		OOB	CI		IB	CI		OOB	CI		IB	CI		OOB	CI	
LME	*Chattahoochee-Oconee*	0.96	0.95	0.96	0.95	0.94	0.96	0.00	0.00	0.00	-0.03	-1.56	1.72	9.01	8.54	9.46	9.71	8.66	10.79
	Cherokee	0.96	0.95	0.96	0.95	0.94	0.96	0.00	0.00	0.00	-0.07	-2.20	2.13	9.41	8.79	9.94	10.46	9.06	11.83
	Nantahala	0.95	0.93	0.97	0.93	0.88	0.96	0.00	0.00	0.00	-0.06	-4.10	4.40	10.21	7.26	11.48	11.95	8.32	16.29
	Pisgah	0.96	0.95	0.97	0.94	0.90	0.97	0.00	0.00	0.00	-0.36	-4.58	3.90	10.83	8.60	11.94	13.18	9.83	17.24
	George Washington-Jefferson	0.96	0.95	0.96	0.95	0.94	0.96	0.00	0.00	0.00	0.03	-1.27	1.39	8.26	7.80	8.64	8.83	7.87	9.80
	Daniel Boone	0.95	0.95	0.96	0.93	0.90	0.95	0.00	0.00	0.00	0.05	-3.50	3.56	9.49	8.43	10.39	11.39	9.12	14.09
	Monongahela	0.94	0.93	0.95	0.93	0.91	0.95	0.00	0.00	0.00	-0.01	-2.34	2.24	11.73	11.09	12.25	12.71	11.26	14.22
	Wayne	0.93	0.91	0.95	0.90	0.85	0.93	0.00	0.00	0.00	-0.05	-4.12	4.54	10.45	9.13	11.46	12.74	10.19	15.44
	Great Smoky Mountains	0.95	0.94	0.96	0.93	0.89	0.95	0.00	0.00	0.00	-0.15	-4.52	4.29	10.07	8.67	11.16	12.53	10.09	15.34
	Shenandoah	0.98	0.97	0.99	0.94	0.88	0.97	0.00	0.00	0.00	-0.25	-5.10	4.37	4.84	3.45	6.48	9.94	5.95	13.82
LASSO	*Chattahoochee-Oconee*	0.96	0.96	0.97	0.95	0.90	0.96	0.00	0.00	0.00	-0.18	-8.62	7.82	8.69	8.20	9.13	10.22	8.62	13.95
	Cherokee	0.96	0.95	0.96	0.94	0.88	0.96	0.00	0.00	0.00	0.10	-9.94	9.67	9.75	9.16	10.21	11.58	9.85	15.87
	Nantahala	0.95	0.94	0.97	0.90	0.74	0.95	0.00	0.00	0.00	-0.19	-14.45	15.21	10.24	7.89	11.22	14.09	9.60	22.70
	Pisgah	0.96	0.95	0.97	0.89	0.63	0.94	0.00	0.00	0.01	0.17	-20.32	21.14	11.58	9.22	12.69	18.74	13.41	31.11
	George Washington-Jefferson	0.96	0.96	0.97	0.95	0.92	0.96	0.00	0.00	0.00	0.01	-6.31	6.73	7.93	7.55	8.25	9.01	7.91	11.94
	Daniel Boone	0.96	0.95	0.96	0.92	0.75	0.95	0.00	0.00	0.01	-0.33	-14.37	12.83	9.41	8.27	10.16	12.95	10.27	21.02
	Monongahela	0.94	0.94	0.95	0.92	0.87	0.94	0.00	0.00	0.01	0.28	-9.05	9.15	11.53	10.82	12.07	13.35	11.58	16.83
	Wayne	0.95	0.95	0.96	0.90	0.63	0.94	0.00	-0.01	0.01	-0.25	-15.16	14.64	8.94	7.78	9.72	13.05	9.80	21.44
	Great Smoky Mountains	0.97	0.96	0.97	0.92	0.72	0.96	0.00	0.00	0.01	0.07	-18.92	16.65	8.73	7.70	9.47	13.68	9.75	23.40
	Shenandoah	0.98	0.96	0.98	0.86	0.19	0.96	0.01	0.00	0.03	-0.63	-25.29	23.20	6.49	5.12	8.22	15.07	8.45	30.53
GAMM	*Chattahoochee-Oconee*	0.96	0.96	0.97	0.95	0.94	0.96	0.00	0.00	0.00	-0.13	-1.81	1.60	8.68	8.18	9.13	9.73	8.45	11.08
	Cherokee	0.96	0.95	0.96	0.95	0.93	0.96	0.00	0.00	0.00	-0.02	-2.09	2.02	9.60	8.97	10.07	10.60	9.30	12.04
	Nantahala	0.95	0.94	0.97	0.93	0.88	0.96	0.00	0.00	0.00	-0.12	-3.92	4.37	9.92	6.98	11.10	11.34	8.33	16.13
	Pisgah	0.97	0.96	0.98	0.95	0.90	0.97	0.00	0.00	0.00	-0.39	-4.29	4.13	10.09	8.10	11.26	12.85	9.63	17.36
	George Washington-Jefferson	0.96	0.96	0.96	0.95	0.94	0.96	0.00	0.00	0.00	-0.02	-1.29	1.24	8.07	7.65	8.42	8.75	7.90	9.73
	Daniel Boone	0.95	0.94	0.96	0.92	0.87	0.95	0.00	0.00	0.00	-0.20	-3.94	3.84	9.87	8.61	10.86	12.40	9.58	15.75
	Monongahela	0.95	0.94	0.95	0.93	0.91	0.95	0.00	0.00	0.00	-0.06	-2.35	2.11	11.05	10.44	11.60	12.37	10.83	14.23
	Wayne	0.93	0.91	0.95	0.89	0.82	0.93	0.00	0.00	0.00	0.05	-4.30	4.79	10.51	9.12	11.69	13.56	10.45	16.55
	Great Smoky Mountains	0.95	0.94	0.96	0.92	0.88	0.95	0.00	0.00	0.00	0.01	-4.82	4.38	10.31	8.75	11.42	12.98	10.49	16.12
	Shenandoah	0.98	0.98	0.99	0.93	0.85	0.97	0.00	0.00	0.00	0.16	-5.99	5.28	4.80	3.51	6.21	10.71	6.62	14.85

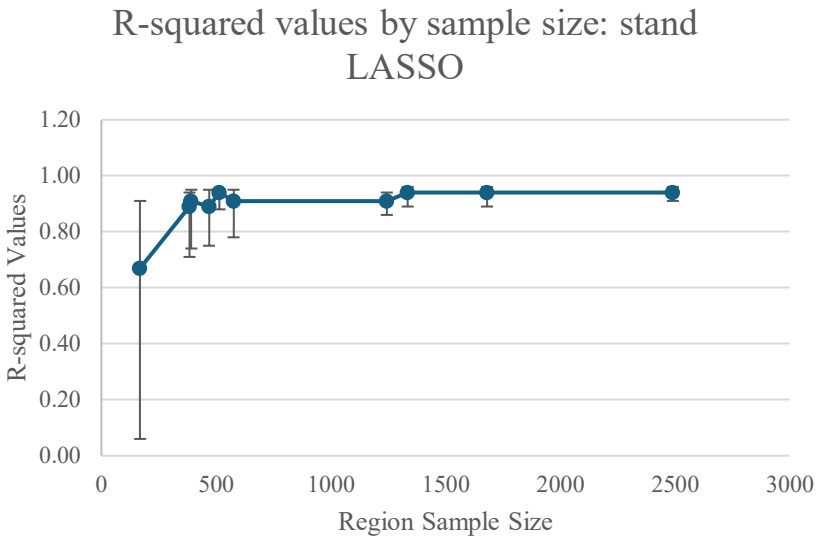
Table 9: Evaluation statistics for the three modeling approaches fitted with stand and environmental variables for each region. R-squared, root-mean-squared-error (RMSE), and mean bias (MB) were calculated for in-bag (IB) training data and out-of-bag (OOB) testing data with median estimates and 95% confidence intervals.

STAND-LEVEL & ENVIRONMENTAL VARIABLES

		R-squared						MB						RMSE					
		IB	CI		OOB	CI		IB	CI		OOB	CI		IB	CI		OOB	CI	
LME	*Chattahoochee-Oconee*	0.96	0.96	0.97	0.96	0.96	0.96	0.00	0.00	0.00	-0.01	-2.61	2.46	8.40	7.95	8.84	8.81	8.67	9.28
	Cherokee	0.96	0.96	0.96	0.96	0.95	0.96	0.00	0.00	0.00	0.04	-2.86	3.01	9.37	8.75	9.86	9.90	9.74	10.46
	Nantahala	0.96	0.95	0.97	0.95	0.93	0.95	0.00	0.00	0.00	0.06	-4.28	5.33	9.46	6.95	10.36	10.62	10.12	12.29
	Pisgah	0.97	0.96	0.98	0.95	0.93	0.96	0.00	0.00	0.00	-0.20	-6.34	6.50	10.42	8.13	11.46	12.65	11.75	15.18
	George Washington-Jefferson	0.97	0.96	0.97	0.96	0.96	0.96	0.00	0.00	0.00	0.01	-1.92	1.98	7.70	7.35	8.03	8.00	7.91	8.30
	Daniel Boone	0.96	0.96	0.97	0.95	0.94	0.96	0.00	0.00	0.00	-0.04	-4.48	4.49	8.58	7.46	9.34	9.96	9.42	11.26
	Monongahela	0.95	0.94	0.95	0.94	0.94	0.94	0.00	0.00	0.00	-0.15	-2.65	2.68	11.32	10.54	11.93	11.88	11.69	12.28
	Wayne	0.96	0.96	0.97	0.95	0.93	0.95	0.00	0.00	0.00	-0.20	-4.34	4.25	7.89	6.84	8.54	9.44	8.87	10.94
	Great Smoky Mountains	0.97	0.97	0.98	0.96	0.95	0.97	0.00	0.00	0.00	-0.16	-5.14	5.43	7.55	6.60	8.45	9.44	8.73	11.06
	Shenandoah	0.99	0.98	0.99	0.94	0.85	0.97	0.00	0.00	0.00	-0.15	-8.09	7.88	4.29	3.35	5.40	9.98	7.28	16.40
LASSO	*Chattahoochee-Oconee*	0.96	0.96	0.97	0.94	0.89	0.96	0.00	-0.01	0.01	-0.22	-8.53	8.79	8.77	8.30	9.20	10.55	8.96	14.49
	Cherokee	0.96	0.95	0.96	0.94	0.88	0.95	0.00	0.00	0.01	0.00	-9.56	9.42	9.73	9.12	10.22	11.86	10.15	15.83
	Nantahala	0.95	0.94	0.97	0.90	0.72	0.95	0.00	-0.01	0.01	-0.79	-15.91	16.35	10.19	7.74	11.17	14.34	9.96	23.10
	Pisgah	0.96	0.95	0.97	0.89	0.59	0.94	0.00	-0.01	0.02	-0.58	-20.21	21.78	11.52	9.16	12.72	19.04	13.76	32.81
	George Washington-Jefferson	0.96	0.96	0.96	0.95	0.92	0.96	0.00	-0.01	0.00	-0.02	-6.33	6.22	8.10	7.70	8.45	9.34	8.17	11.69
	Daniel Boone	0.96	0.95	0.97	0.91	0.70	0.95	0.00	-0.01	0.01	-0.56	-15.02	13.64	9.12	7.99	9.92	13.22	10.30	21.71
	Monongahela	0.94	0.93	0.95	0.92	0.86	0.94	0.00	-0.01	0.01	0.06	-9.20	9.68	11.71	11.02	12.33	13.73	11.73	17.29
	Wayne	0.96	0.90	0.96	0.90	0.68	0.94	0.00	-0.01	0.02	-0.49	-14.01	13.95	8.63	7.53	12.64	13.13	9.96	21.05
	Great Smoky Mountains	0.97	0.96	0.97	0.91	0.72	0.95	-0.01	-0.02	0.00	-0.10	-17.93	17.29	8.67	7.62	9.51	13.93	10.09	23.35
	Shenandoah	0.66	0.36	0.95	0.61	-0.47	0.90	1.90	-2.14	7.99	1.61	-24.07	27.77	23.79	8.63	33.96	26.39	12.21	40.14
GAMM	*Chattahoochee-Oconee*	0.96	0.96	0.97	0.95	0.93	0.96	0.00	0.00	0.00	-0.12	-1.77	1.49	8.22	7.71	8.64	9.87	8.65	11.22
	Cherokee	0.96	0.96	0.97	0.95	0.93	0.96	0.00	0.00	0.00	0.10	-2.07	2.13	9.00	8.46	9.48	10.81	9.49	12.22
	Nantahala	0.96	0.95	0.98	0.91	0.86	0.95	0.00	0.00	0.00	-0.07	-4.80	4.40	8.43	6.24	9.69	13.42	10.42	16.71
	Pisgah	0.98	0.97	0.98	0.94	0.89	0.97	0.00	0.00	0.00	0.00	-4.53	5.01	8.36	6.52	9.63	13.87	10.24	17.87
	George Washington-Jefferson	0.96	0.96	0.97	0.96	0.94	0.96	0.00	0.00	0.00	0.02	-1.37	1.26	7.65	7.29	7.95	8.73	7.93	9.52
	Daniel Boone	0.96	0.95	0.97	0.91	0.86	0.94	0.00	0.00	0.00	0.28	-3.79	4.55	8.66	7.39	9.68	13.52	10.89	16.62
	Monongahela	0.95	0.94	0.96	0.93	0.91	0.95	0.00	0.00	0.00	0.03	-2.34	2.30	10.47	9.86	11.04	12.46	10.92	14.38
	Wayne	0.95	0.94	0.97	0.87	0.78	0.92	0.00	0.00	0.00	-0.12	-4.77	4.93	8.17	6.86	9.35	14.64	11.78	18.27
	Great Smoky Mountains	0.96	0.95	0.97	0.90	0.83	0.94	0.00	0.00	0.00	-0.07	-5.62	4.71	8.41	6.92	9.90	14.92	11.67	18.52
	Shenandoah	0.99	0.99	1.00	0.81	0.10	0.93	0.00	0.00	0.00	0.45	-10.76	14.55	3.08	2.07	4.14	17.55	11.53	34.29

Name	Obs.	Name	Obs.
Shenandoah	169	*Daniel Boone*	576
Wayne	384	*Monongahela*	1244
Great Smoky Mountains	392	*Cherokee*	1335
Pisgah	469	*Chattahoochee-Oconee*	1681
Nantahala	515	*George Washington-Jefferson*	2491





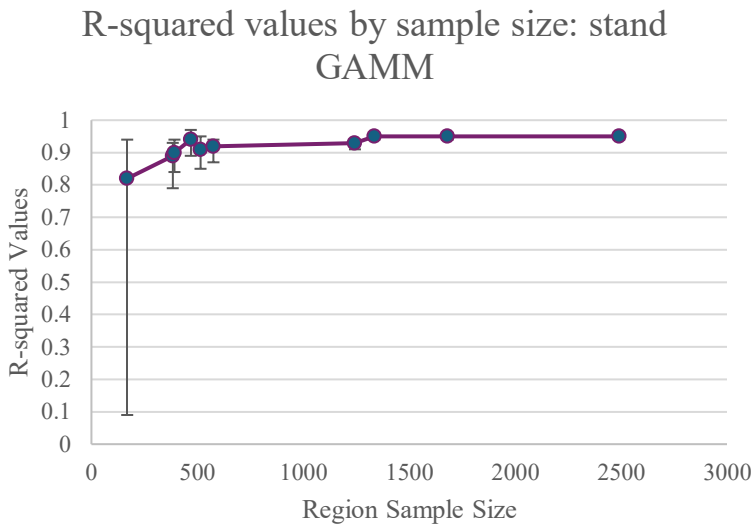
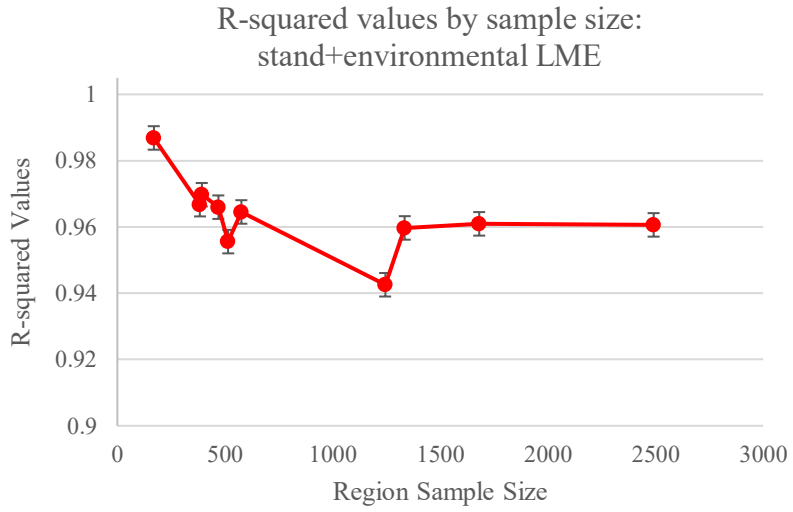


Figure 8: Median R-squared vs. sample size graph from LME, LASSO and GAMM models fit with stand-level variables the lower 2.5% and 97.5% confidence interval bars. Number of observations per region is provided in the side table.

Name	Obs.	Name	Obs.
Shenandoah	169	*Daniel Boone*	576
Wayne	384	*Monongahela*	1244
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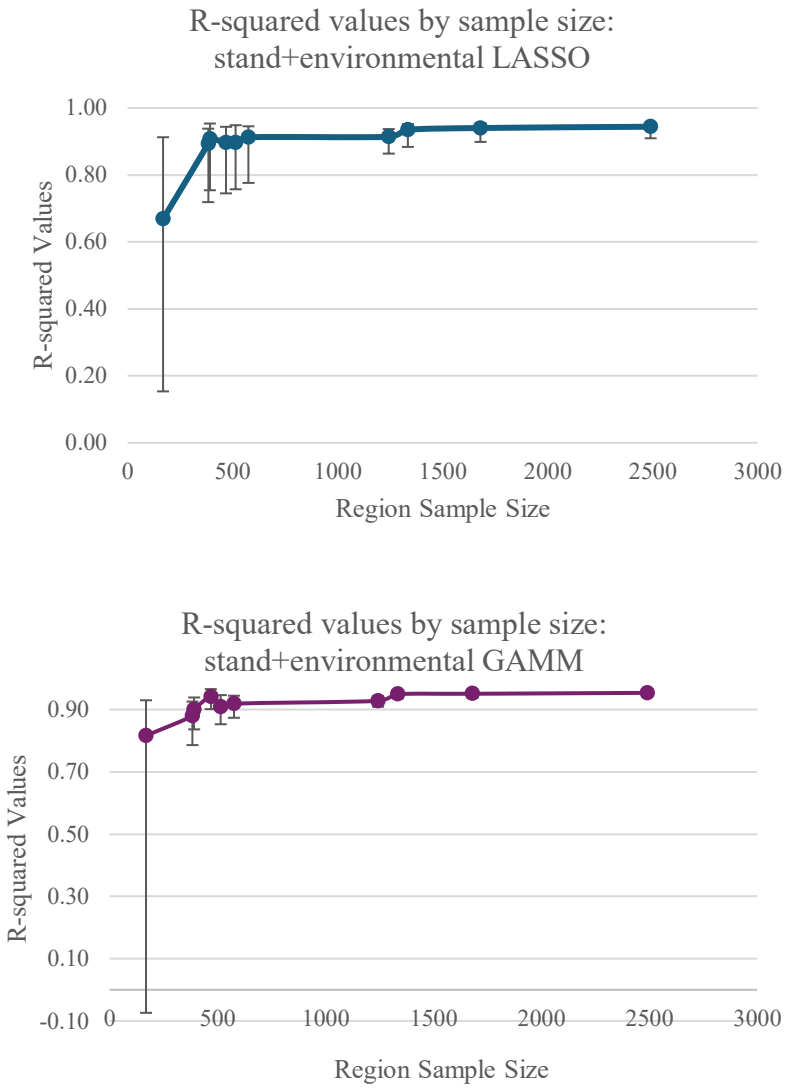
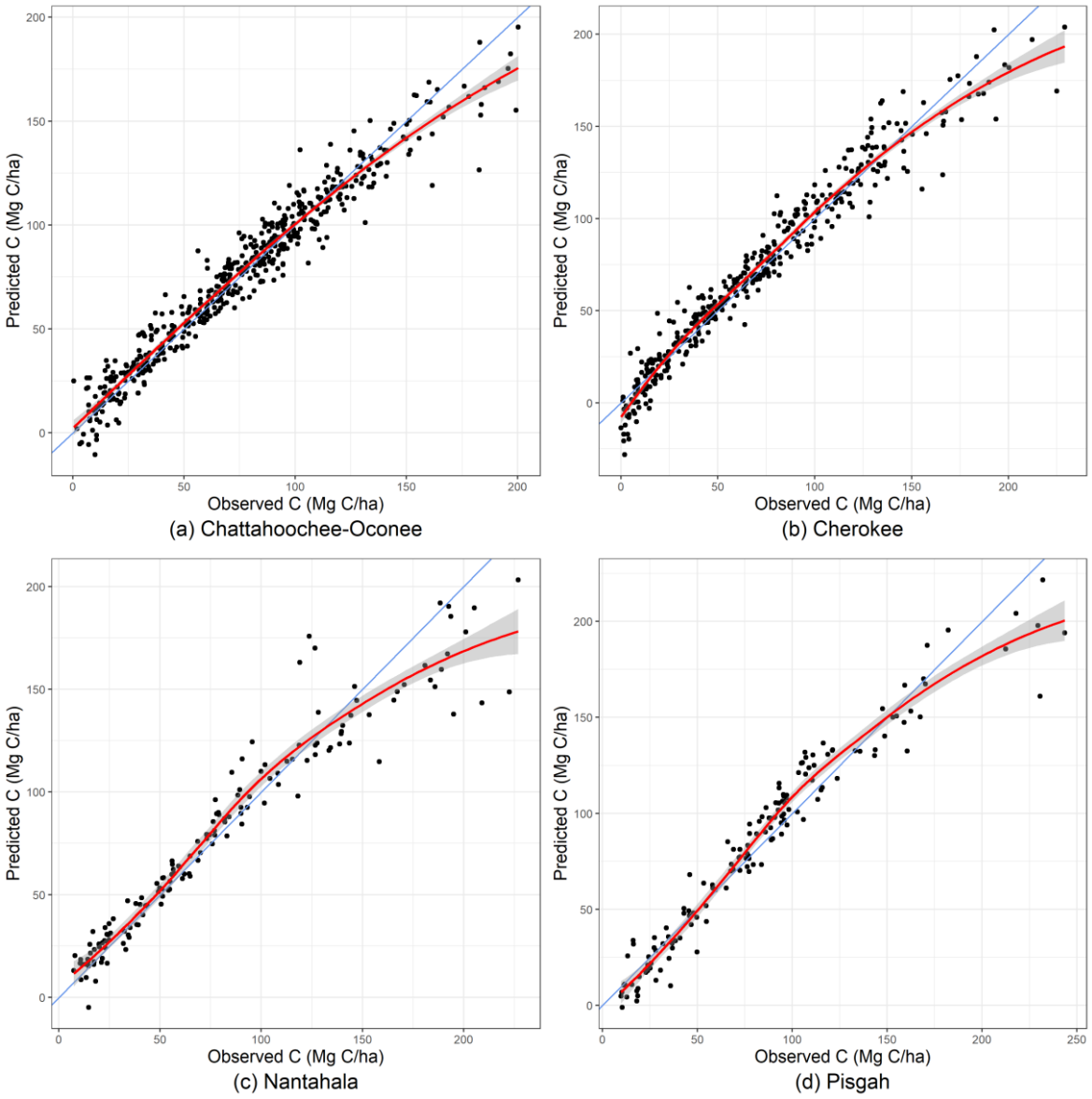
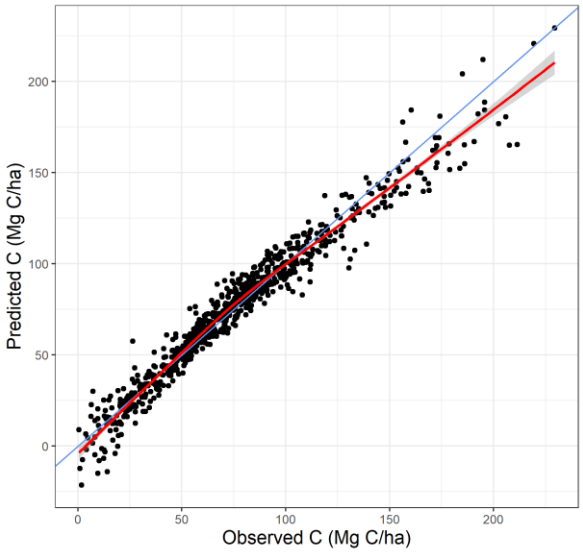
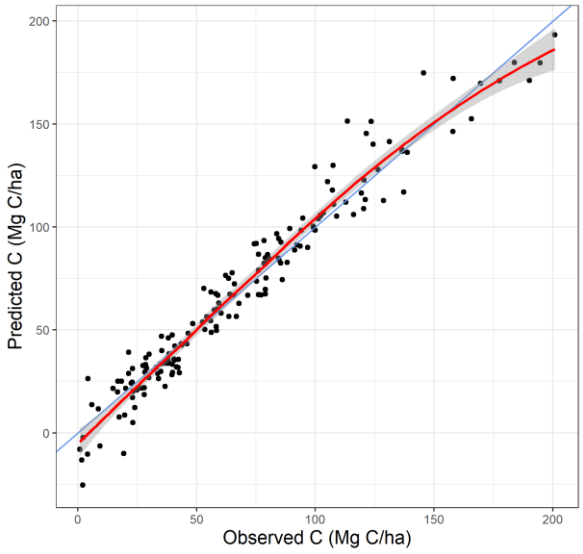


Figure 9: Median R-squared vs. sample size graph from LME, LASSO and GAMM models fit with stand+environmental variables the lower 2.5% and 97.5% confidence interval bars. Number of observations per region is provided in the side table.

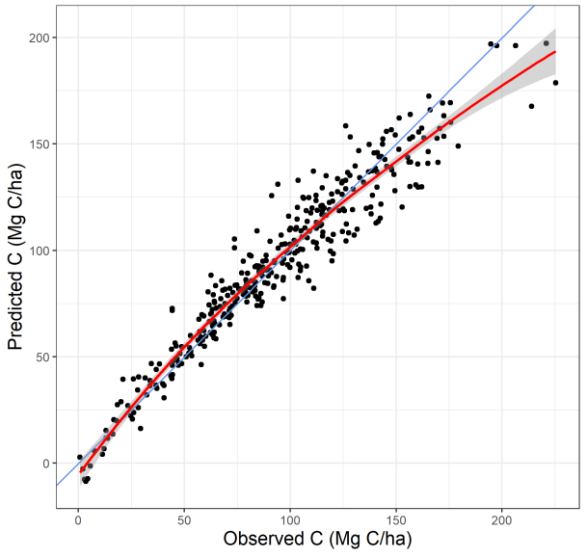

Predicted C (Mg C/ha)
Observed C (Mg C/ha)
(a) Chattahoochee-Oconee
(b) Cherokee
(c) Nantahala
(d) Pisgah



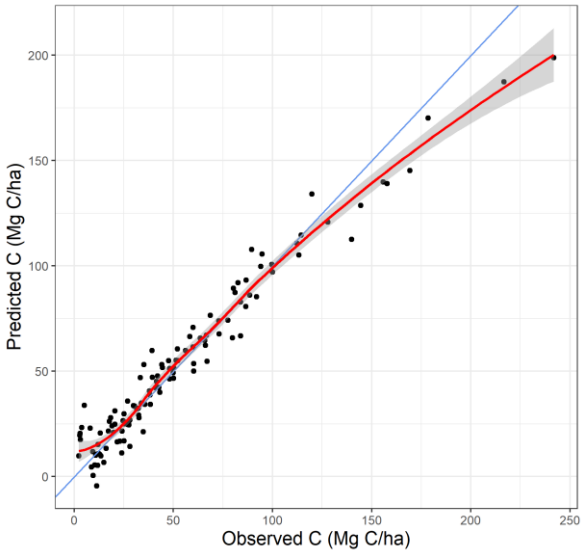
(e) George Washington-Jefferson



(f) Daniel Boone



(g) Monongahela



(h) Wayne

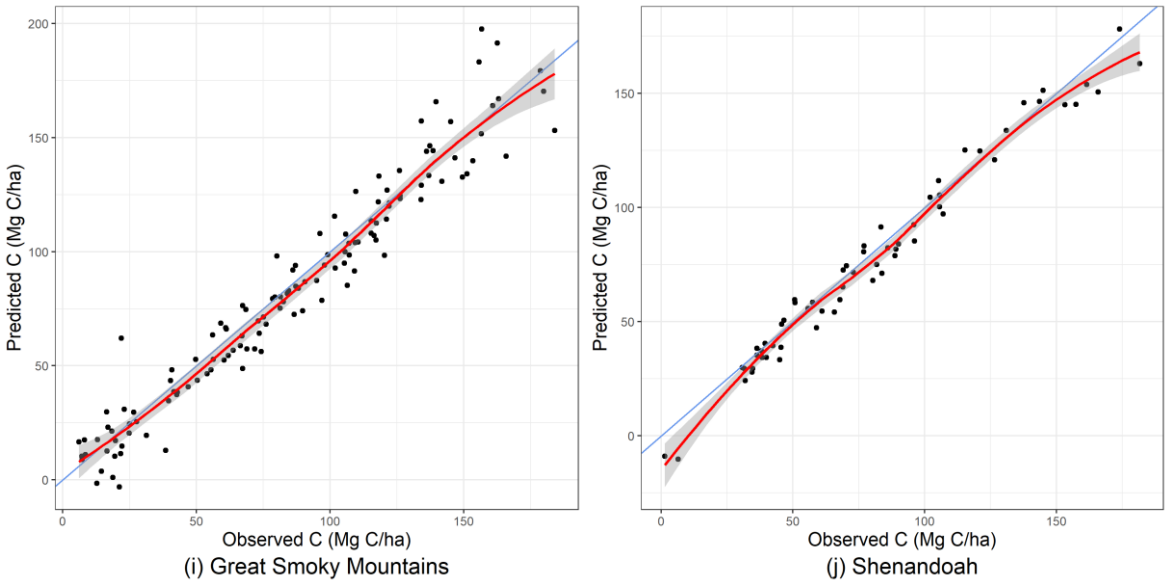
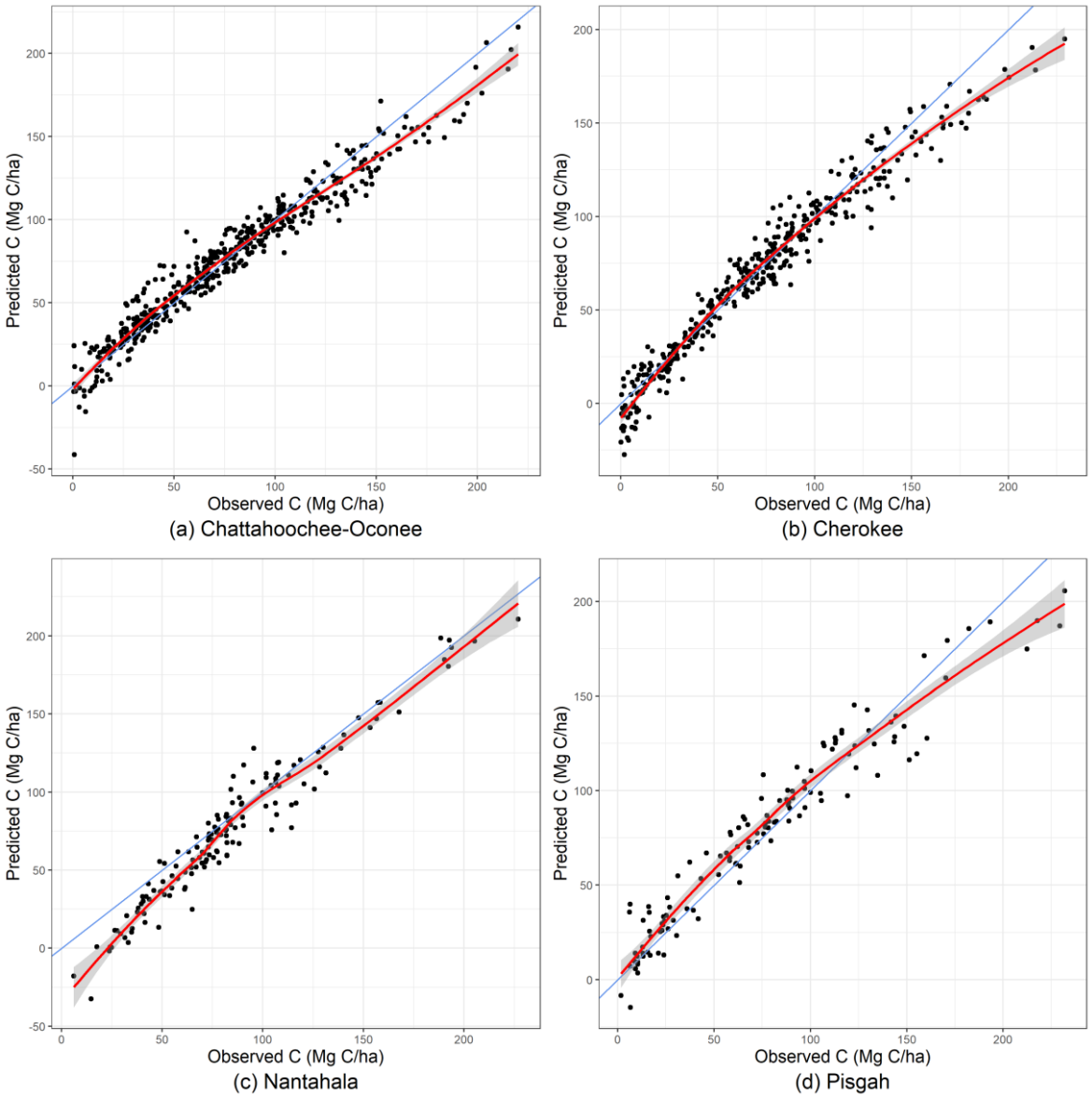
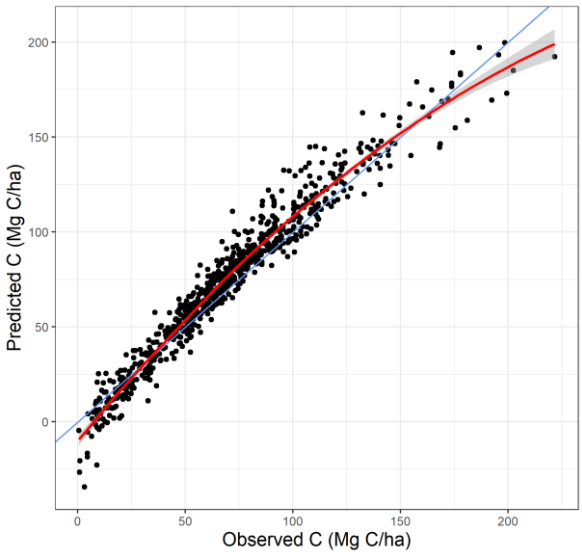
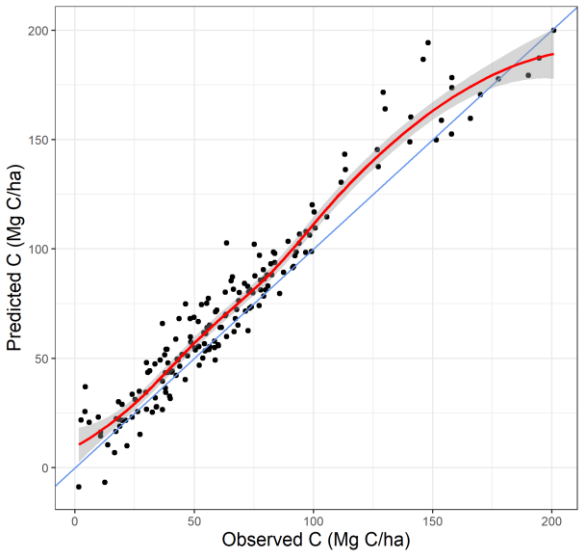


Figure 10: Predicted and observed carbon per hectare (Mg C/ha) from out-of-bag samples for linear mixed-effects using stand-level variables.

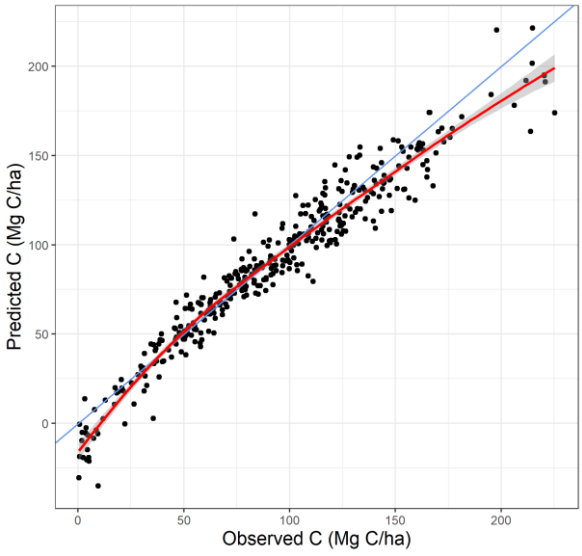

Predicted C (Mg C/ha)
Observed C (Mg C/ha)
(a) Chattahoochee-Oconee
(b) Cherokee
(c) Nantahala
(d) Pisgah



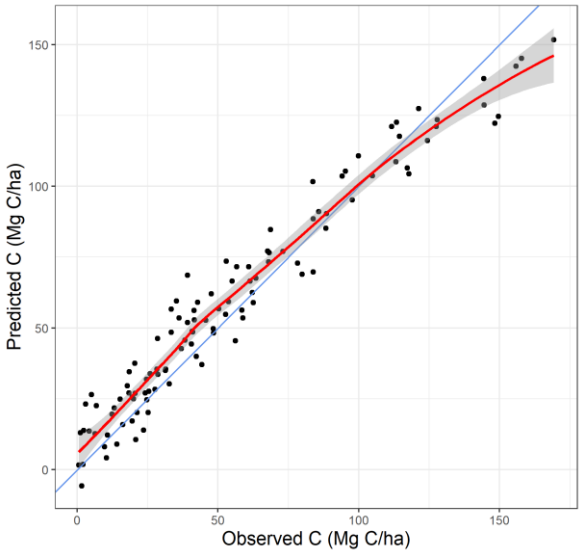
(e) George Washington-Jefferson



(f) Daniel Boone



(g) Monongahela



(h) Wayne

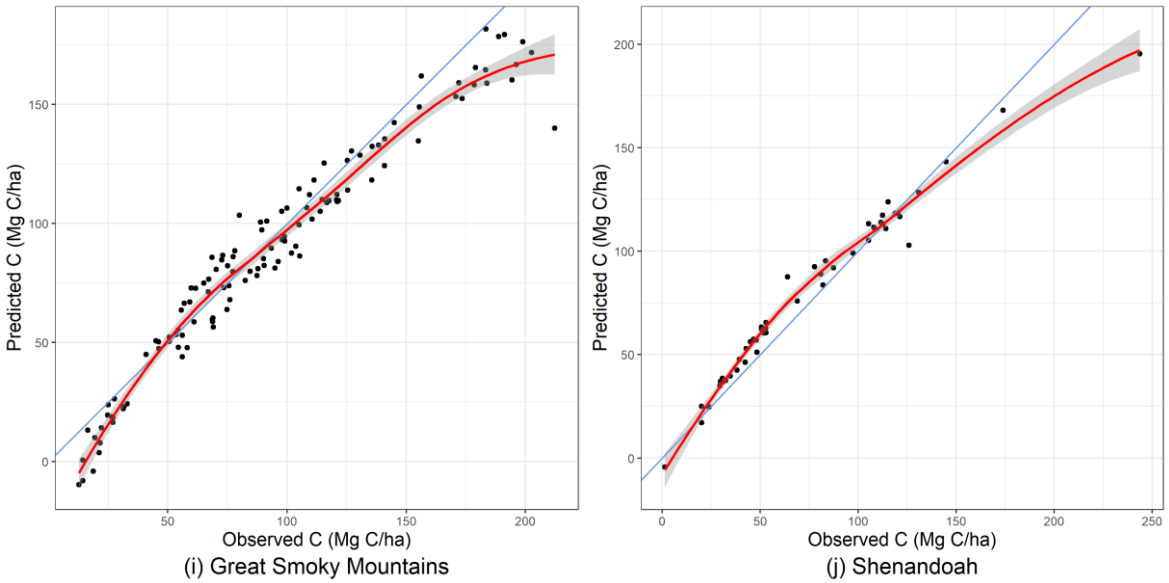
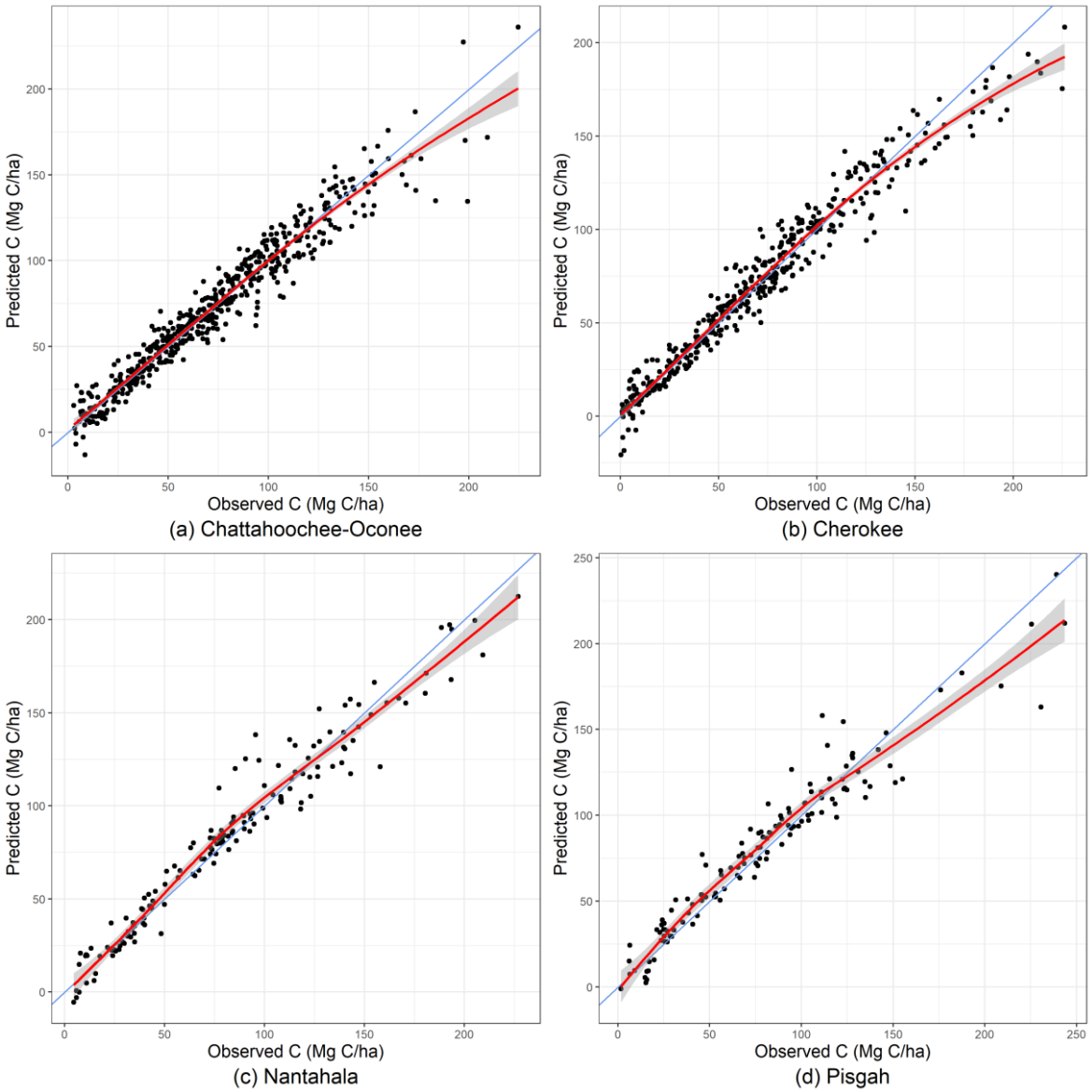
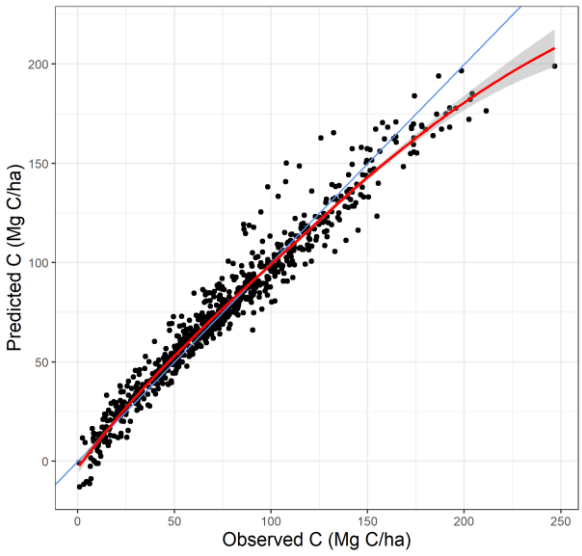
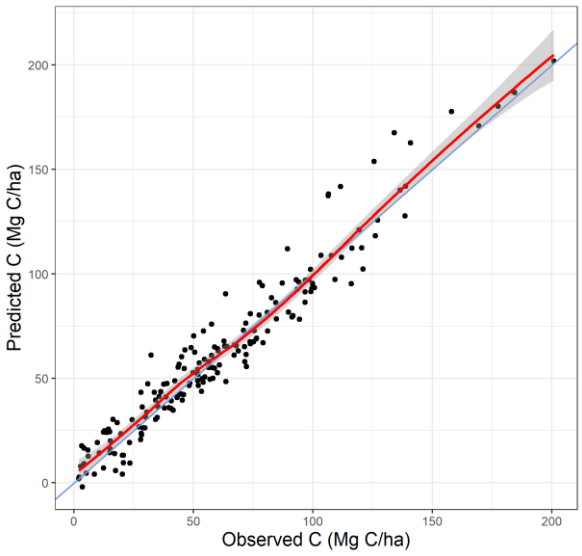


Figure 11: Predicted and observed carbon per hectare (Mg C/ha) from out-of-bag samples for LASSO models using stand-level variables.

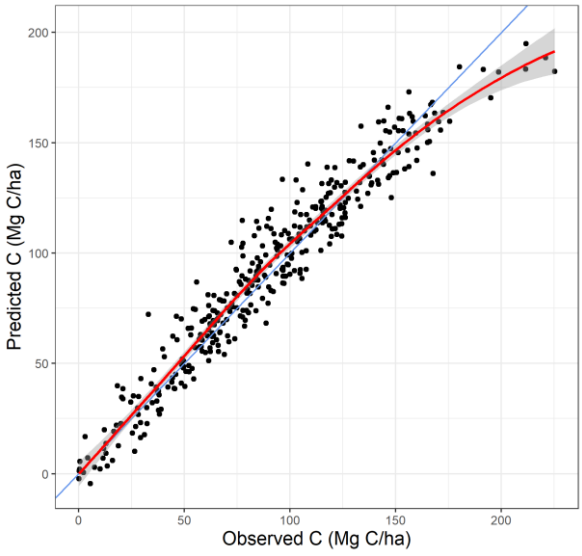

Predicted C (Mg C/ha)
Observed C (Mg C/ha)
(a) Chattahoochee-Oconee
(b) Cherokee
(c) Nantahala
(d) Pisgah



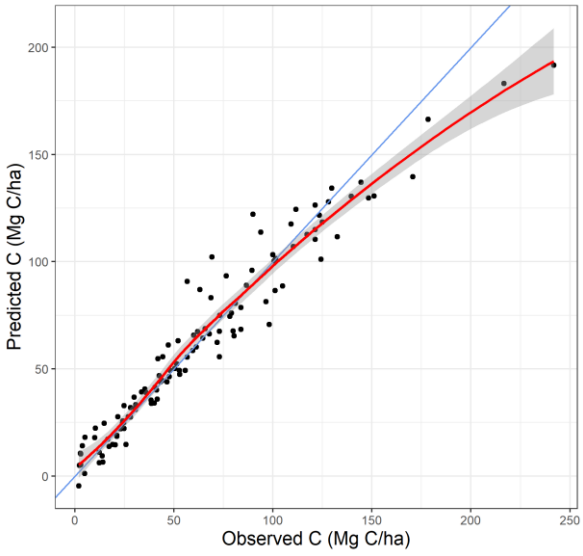
(e) George Washington-Jefferson



(f) Daniel Boone



(g) Monongahela



(h) Wayne

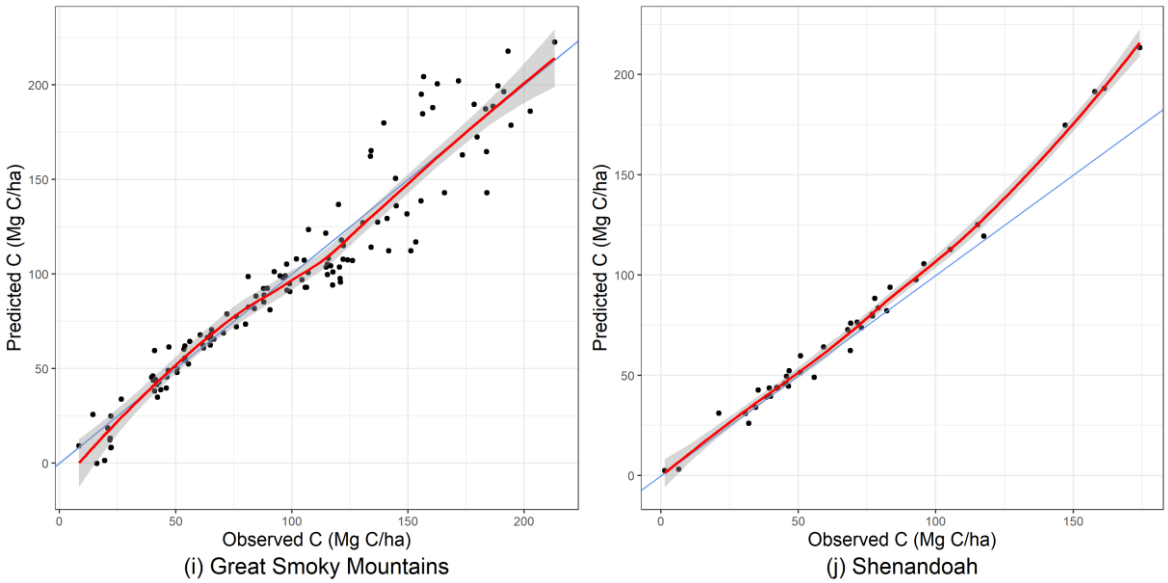
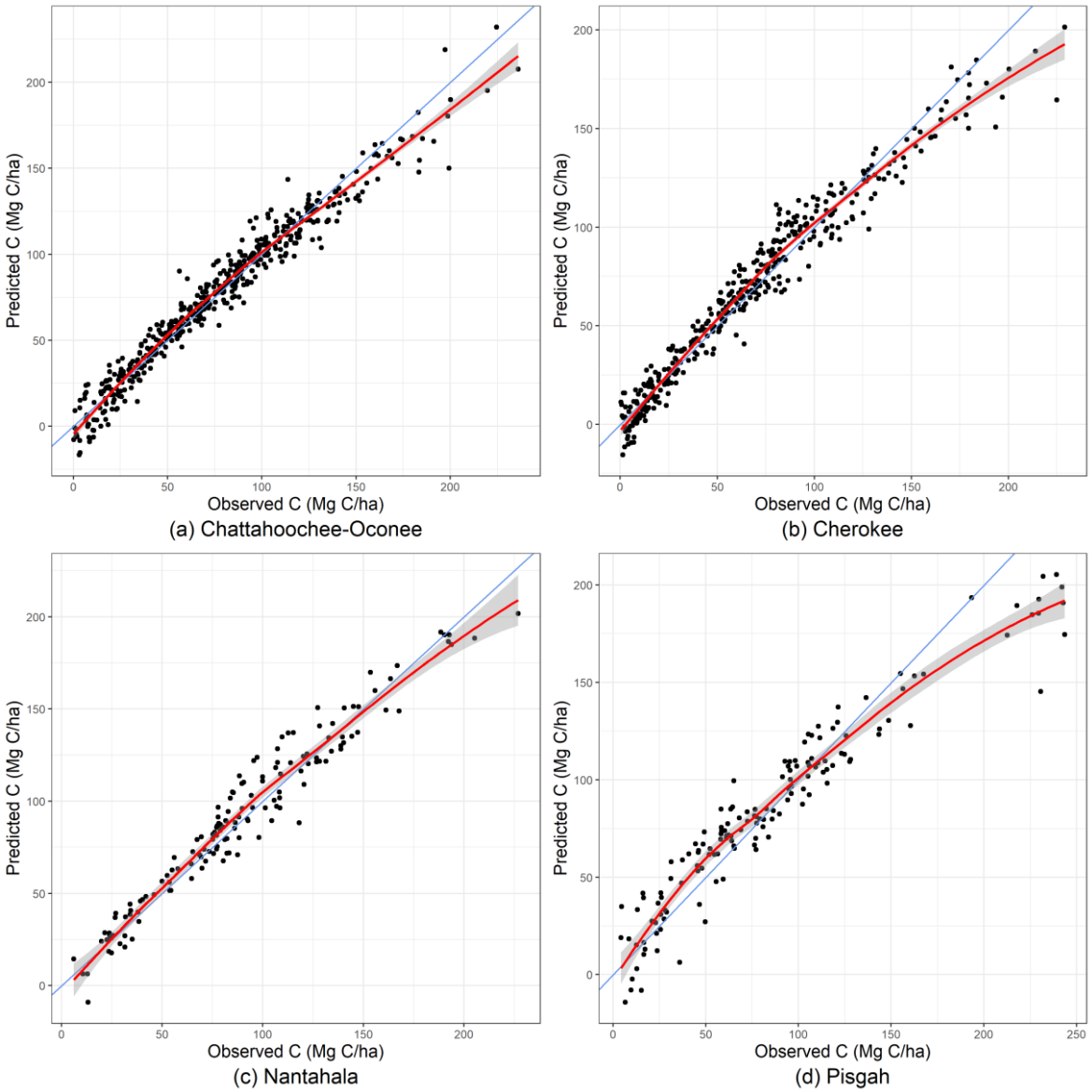
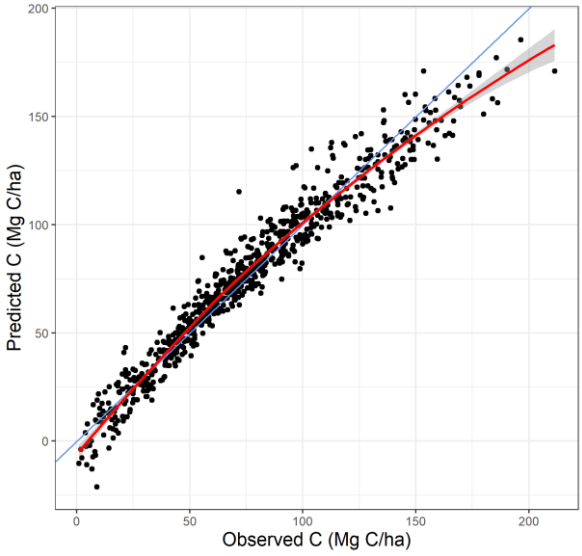
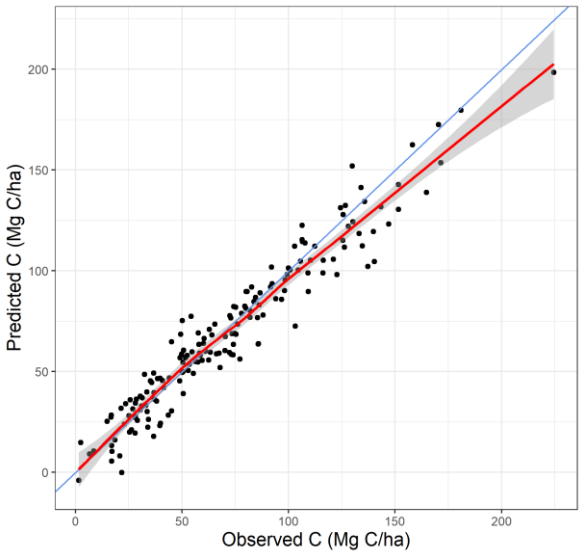


Figure 12: Predicted and observed carbon per hectare (Mg C/ha) from out-of-bag samples for GAMM models using stand-level variables.

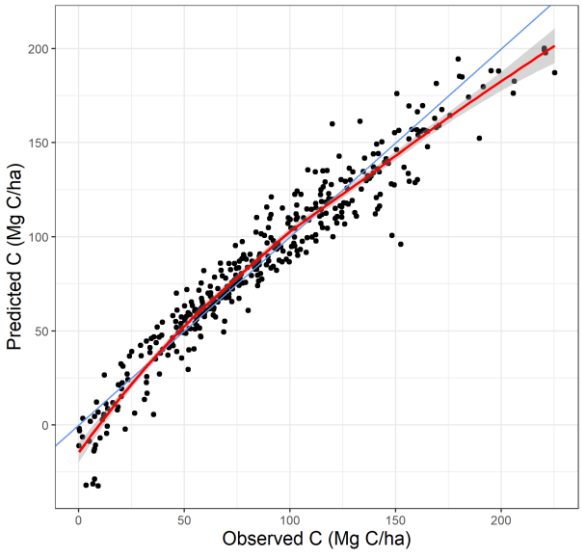

Predicted C (Mg C/ha)
Observed C (Mg C/ha)
(a) Chattahoochee-Oconee
(b) Cherokee
(c) Nantahala
(d) Pisgah



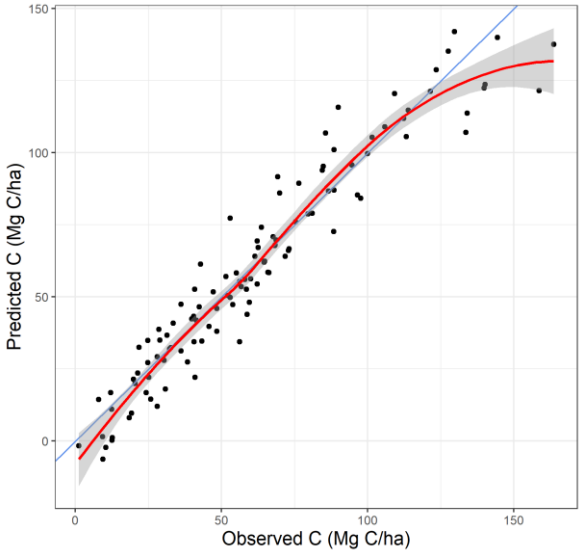
(e) George Washington-Jefferson



(f) Daniel Boone



(g) Monongahela



(h) Wayne

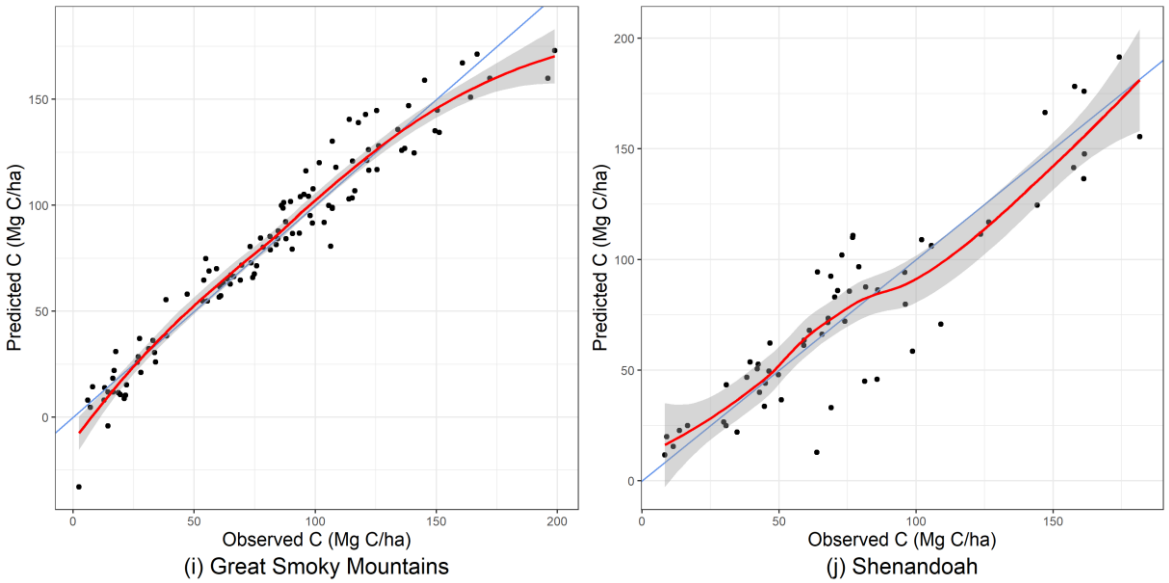
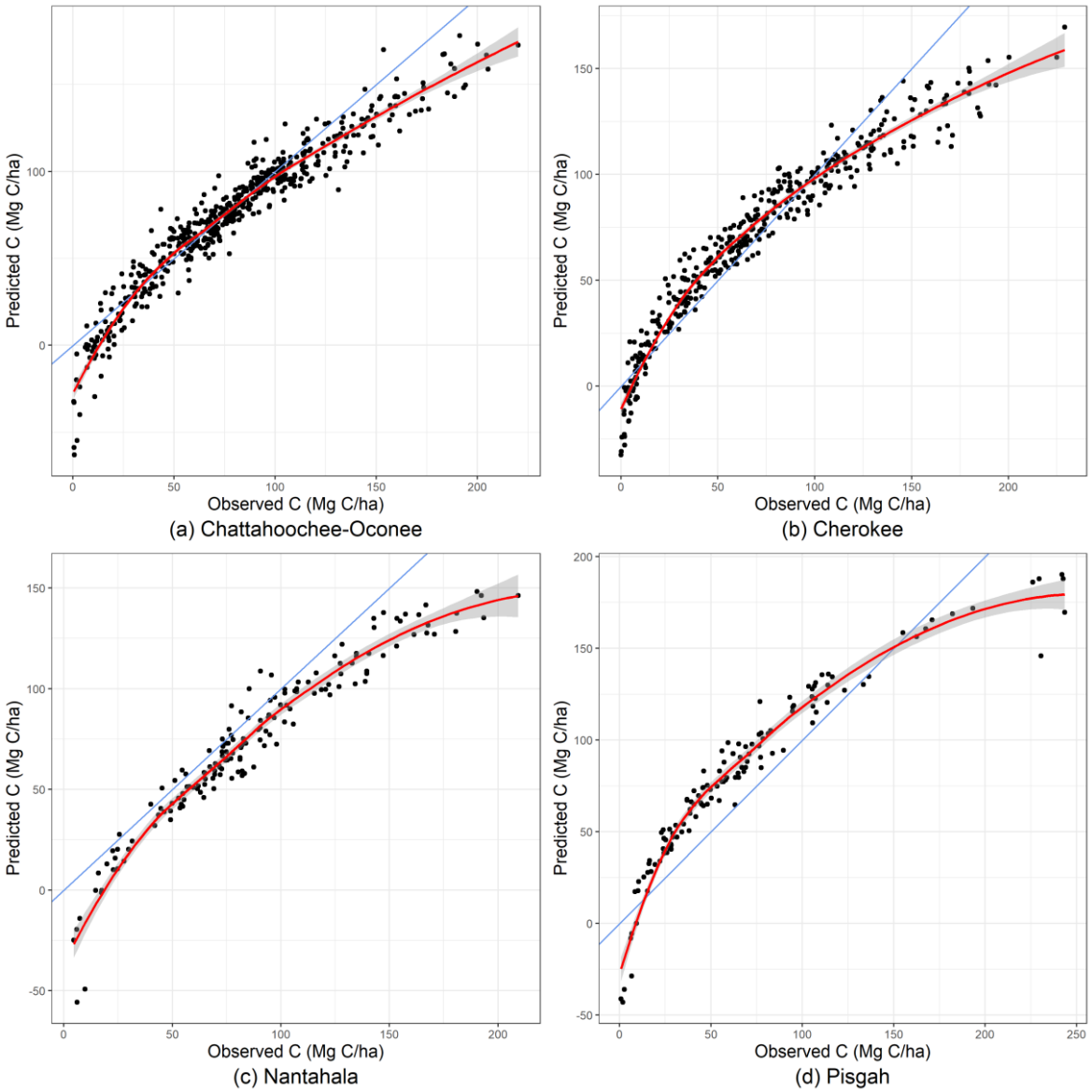
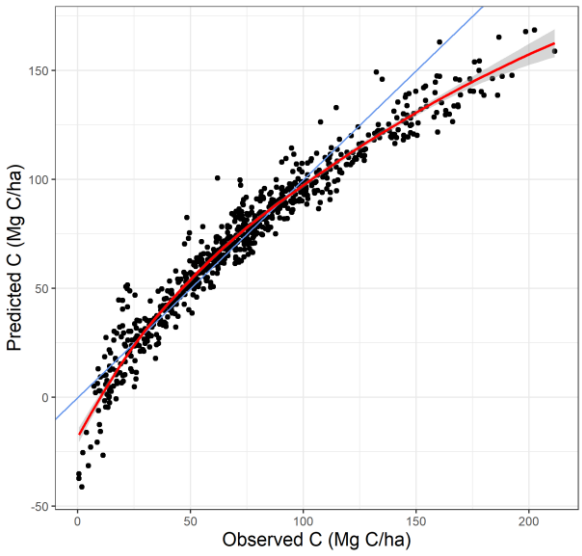
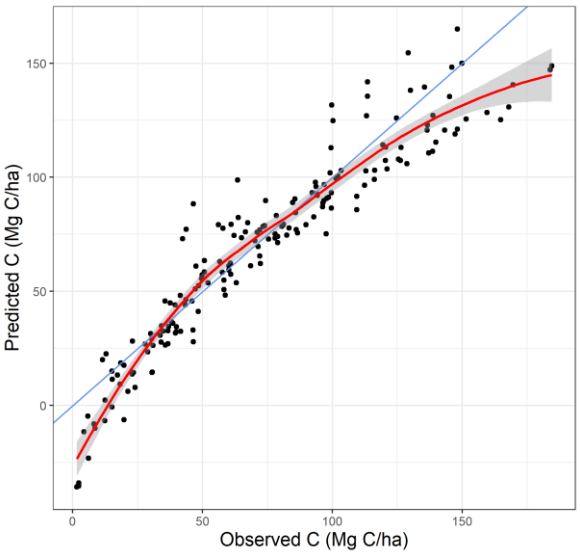


Figure 13: Predicted and observed carbon per hectare (Mg C/ha) produced from linear mixed-effect models using stand-level and WorldClim BIO environmental variables.

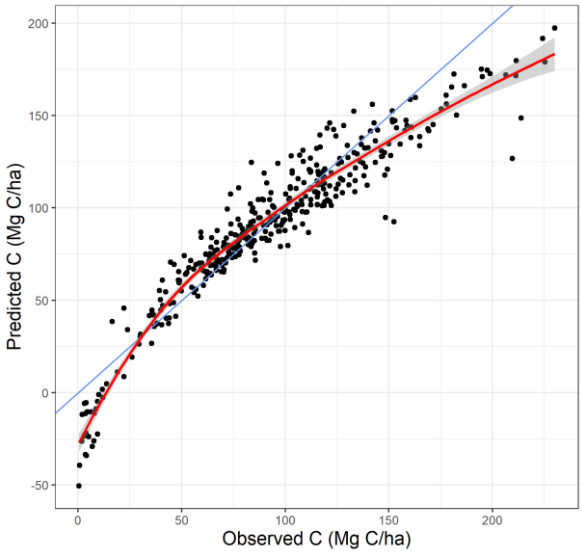

Predicted C (Mg C/ha)
Observed C (Mg C/ha)
(a) Chattahoochee-Oconee
(b) Cherokee
(c) Nantahala
(d) Pisgah



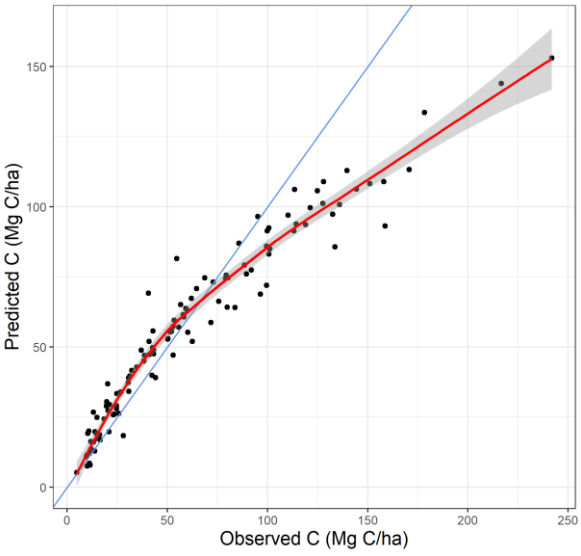
(e) George Washington-Jefferson



(f) Daniel Boone



(g) Monongahela



(h) Wayne

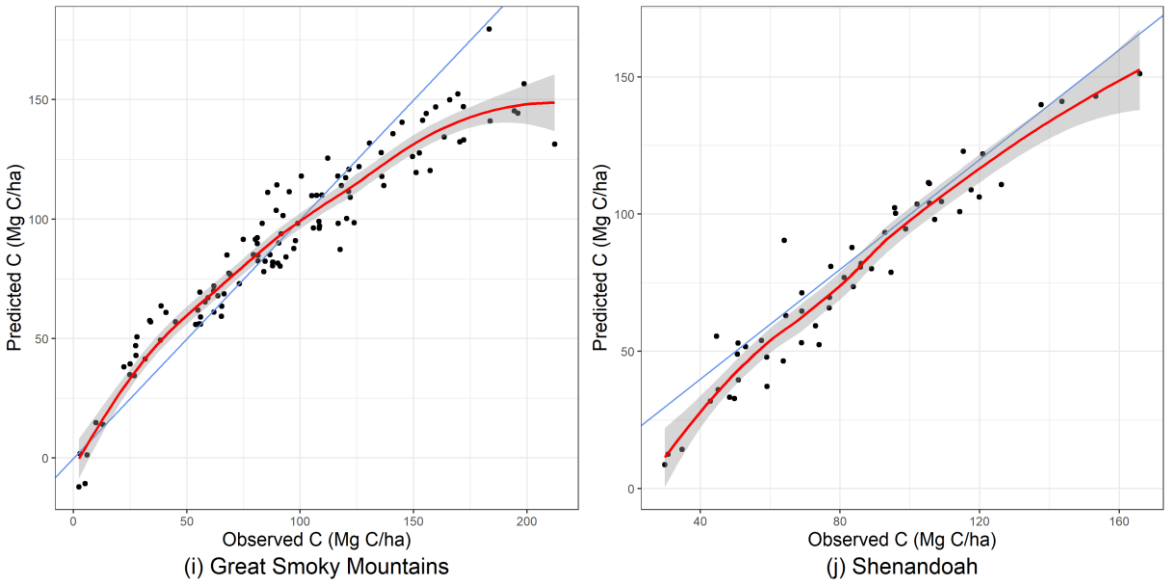
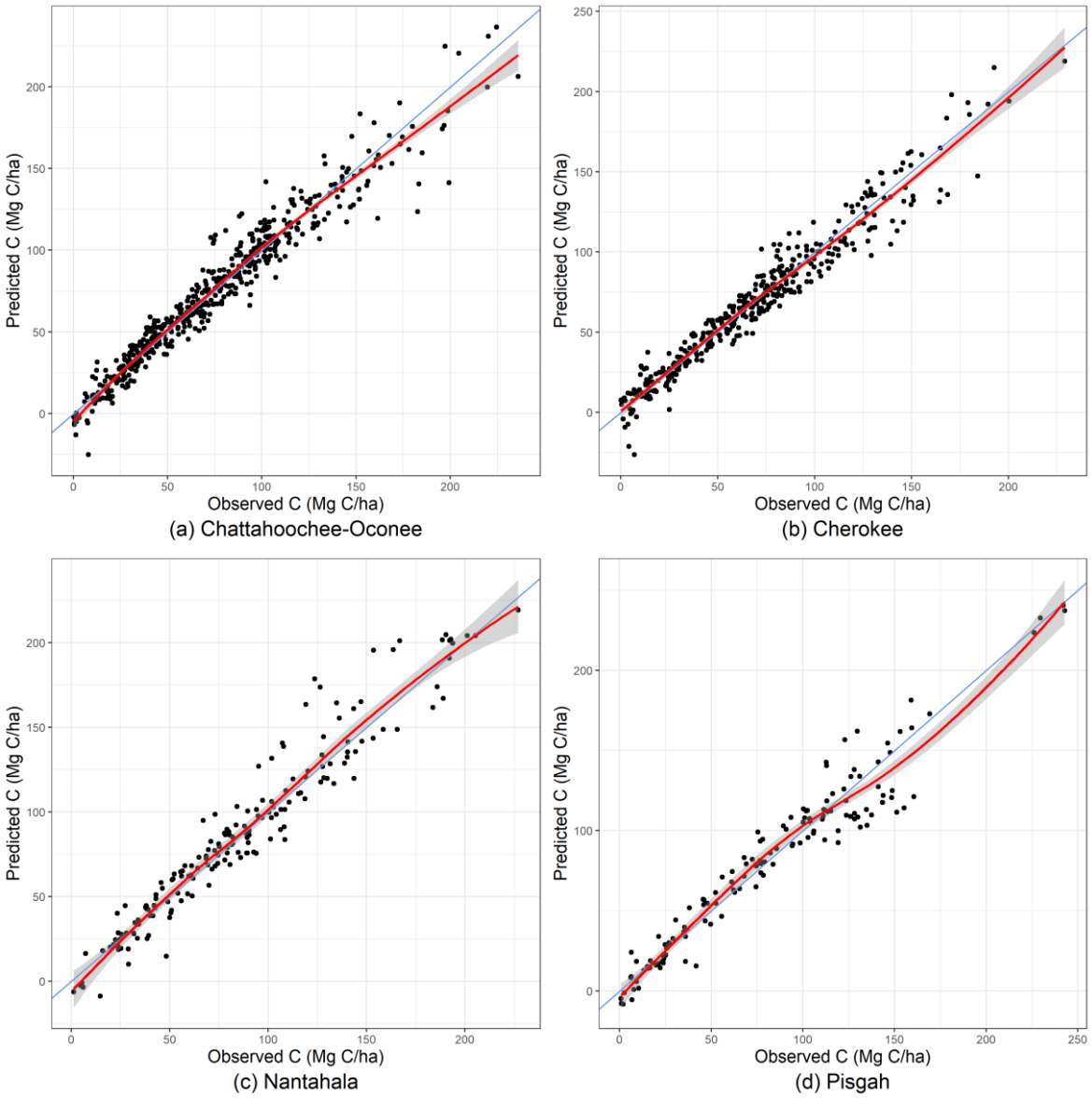
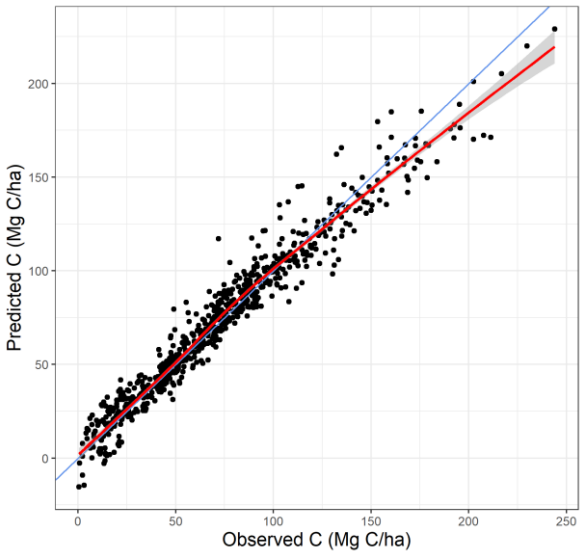
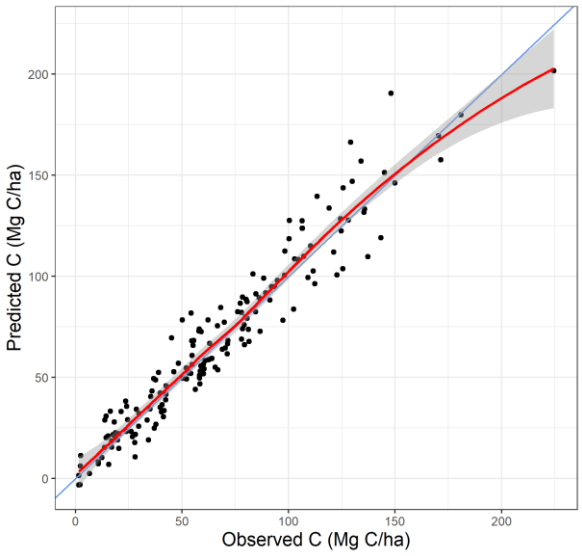


Figure 14: Predicted and observed carbon per hectare (Mg C/ha) produced from LASSO models using stand-level and WorldClim BIO environmental variables.

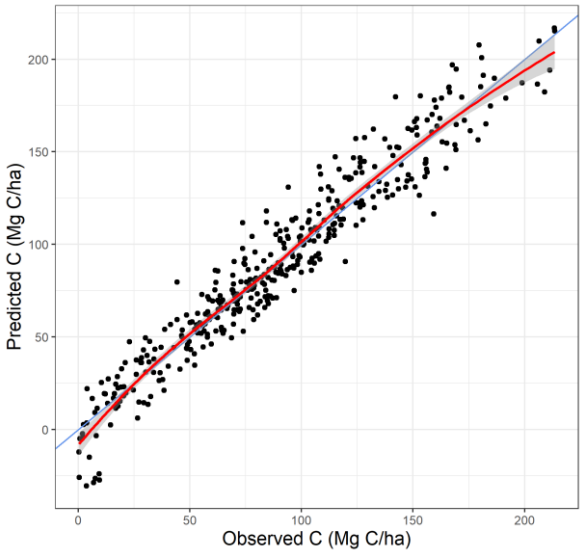

Predicted C (Mg C/ha)
Observed C (Mg C/ha)
(a) Chattahoochee-Oconee
(b) Cherokee
(c) Nantahala
(d) Pisgah



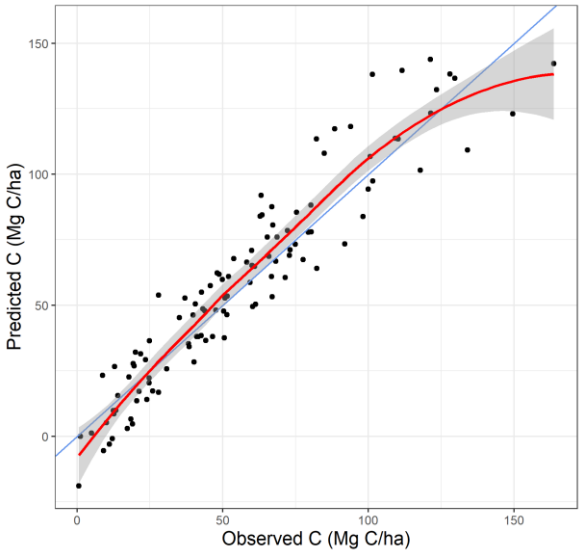
(e) George Washington-Jefferson



(f) Daniel Boone



(g) Monongahela



(h) Wayne

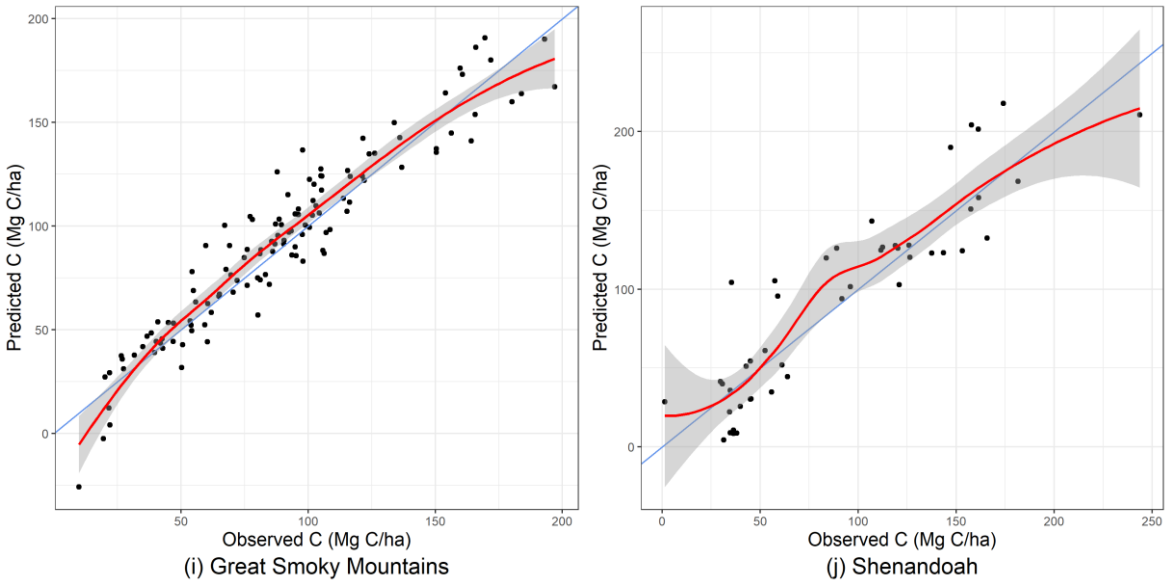


Figure 15: Predicted and observed carbon per hectare (Mg C/ha) produced from GAMM models using stand-level and WorldClim BIO environmental variables.

11 *Table 10: Median coefficients of predictor variables produced from the 1000 bootstrap samples for linear mixed-effect models by region.*

STAND VARIABLES		*D*	*H*	*H-D*	*QMD*	*BA*	*TPH*	*SDI*	*Aspect*	*Slope*	*Intercept*	σ_{fort}
	Chattahoochee-Oconee	3.26	-1.86	0.78	-0.18	4.26	-0.01	0.00	0.00	-0.02	-104.83	5.50
	Cherokee	2.48	-1.06	0.52	-0.26	4.50	-0.01	0.00	0.00	0.04	-73.27	5.41
	Nantahala	1.95	-1.20	0.71	-0.20	4.64	-0.01	0.00	-0.01	0.01	-76.92	5.02
	Pisgah	4.38	-3.19	0.86	-0.52	5.72	-0.01	0.00	-0.01	0.02	-102.21	8.06
	George Washington-Jefferson	2.41	-0.87	0.73	-0.15	4.12	-0.01	0.00	0.00	-0.01	-95.01	6.05
	Daniel Boone	2.64	-1.45	0.67	-0.28	5.37	-0.01	0.00	-0.01	-0.06	-87.41	7.81
	Monongahela	2.99	-1.60	0.94	-0.45	4.88	-0.01	0.00	-0.01	-0.01	-93.39	8.78
	Wayne	3.17	-2.20	0.74	-0.38	5.15	-0.02	0.00	0.00	-0.07	-82.19	7.94
	Great Smoky Mountains	2.04	-0.64	0.71	0.12	3.01	-0.01	0.00	0.01	0.01	-108.29	7.09
	Shenandoah	2.83	-2.02	0.90	-0.33	4.68	-0.01	0.00	0.00	0.00	-86.51	3.33
STAND + ENVIRONMENTAL VARIABLES		*D*	*H*	*H-D*	*QMD*	*BA*	*TPH*	*SDI*	*Aspect*	*Slope*	*bio01*	*bio02*
	Chattahoochee-Oconee	-8.66	-0.98	6.06	19.31	40.80	-7.45	0.86	0.27	-0.28	13.45	1.42
	Cherokee	-12.11	-2.14	5.65	22.61	43.42	-6.95	-2.21	-0.34	-0.09	-6.00	-0.59
	Nantahala	-1.92	-1.81	6.80	12.06	43.79	-8.12	-1.10	-0.22	-1.18	1.88	-3.55
	Pisgah	-11.03	1.31	6.72	19.42	36.10	-5.65	-1.14	-0.15	-0.50	8.01	-5.65
	George Washington-Jefferson	-21.62	9.36	0.58	22.49	42.99	-8.28	3.16	1.46	-0.52	69.94	-14.62
	Daniel Boone	-9.87	0.94	6.57	18.85	38.90	-8.94	5.06	-0.46	-0.14	-11.34	5.02
	Monongahela	-21.97	3.66	4.54	24.23	42.97	-5.62	-4.54	-0.60	0.39	10.62	-15.53
	Wayne	-16.91	-3.48	8.92	33.70	49.08	-6.56	-2.39	-0.25	-1.55	9.22	5.11
	Great Smoky Mountains	-16.80	0.20	3.53	30.28	32.84	-2.52	-3.26	-1.19	0.23	15.27	15.45
	Shenandoah	17.85	-8.50	8.76	-0.57	33.69	-9.58	6.03	0.74	-1.12	9.25	-4.57
		bio03	*bio04*	*bio05*	*bio06*	*bio08*	*bio09*	*bio10*	*bio11*	*bio12*	*bio13*	*bio14*
	Chattahoochee-Oconee	0.47	7.11	-15.55	-0.69	-5.07	-0.65	2.39	0.46	-85.91	18.60	9.34
	Cherokee	2.50	-0.65	4.13	0.70	2.58	0.22	4.43	-3.34	5.71	1.11	6.16
	Nantahala	3.02	-17.78	9.58	-1.21	0.11	10.89	9.04	-30.75	25.78	-1.71	-1.63
	Pisgah	0.18	-7.02	20.18	-6.43	0.67	0.91	-7.00	-9.78	-4.06	5.75	-0.44
	George Washington-Jefferson	7.38	-7.02	-36.31	-23.57	3.91	3.16	10.15	-10.73	-68.66	57.73	10.69
	Daniel Boone	0.85	3.05	-11.56	5.59	2.98	0.56	-5.19	6.92	-13.04	11.74	-9.50
	Monongahela	8.20	-2.50	14.82	-3.92	-0.26	0.24	-8.28	-9.27	-167.53	58.48	79.81
	Wayne	5.78	-3.63	15.28	8.65	-4.49	-5.68	2.36	-33.71	-114.92	44.26	63.40
	Great Smoky Mountains	2.47	12.46	36.86	40.20	-0.27	10.15	-114.41	0.00	-4.66	17.36	-36.67
	Shenandoah	1.10	-11.62	14.26	-3.67	-6.47	0.87	2.77	-22.84	-14.59	-6.71	21.20
		bio15	*bio16*	*bio17*	*bio18*	*bio19*	*Intercept*	σ_{fort}				
	Chattahoochee-Oconee	-2.19	-17.69	10.82	22.25	42.78	74.56	5.27				
	Cherokee	0.19	6.05	-19.95	1.37	1.42	70.29	5.72				
	Nantahala	5.66	-10.84	-2.87	8.09	-14.47	68.17	7.68				
	Pisgah	0.48	-2.60	6.72	-0.94	-3.13	71.18	6.12				
	George Washington-Jefferson	-1.44	-7.32	-33.83	37.30	-5.45	88.57	7.84				
	Daniel Boone	-2.40	-28.57	0.00	23.58	10.14	91.07	8.37				
	Monongahela	-6.16	15.01	-74.01	61.31	17.19	75.48	4.53				
	Wayne	-7.88	10.48	-114.40	73.01	35.29	80.74	7.01				

	Great Smoky Mountains	-3.05	-8.58	9.29	15.88	0.00	78.95	7.24					
	Shenandoah	4.93	6.13	-61.96	-3.65	61.11	56.12	7.29					

12 *Table 11: Median coefficients of predictor variables produced from the 1000 bootstrap samples for LASSO models by region.*

STAND VARIABLES		*D*	*H*	*H-D*	*QMD*	*BA*	*TPH*	*SDI*	*Aspect*	*Slope*	*Intercept*	σ_{fort}	λ
	Chattahoochee-Oconee	-1.88	-2.42	7.35	14.43	40.10	-7.98	1.96	0.39	-0.58	73.65	5.30	0.00
	Cherokee	-2.00	-3.23	7.04	14.25	42.30	-7.70	-0.21	-0.33	0.23	69.37	5.79	0.00
	Nantahala	-1.60	-2.05	8.13	11.13	41.54	-6.52	-2.70	-1.11	-0.02	74.61	6.08	0.00
	Pisgah	-0.61	-9.01	11.97	23.45	49.17	-8.37	-0.10	0.08	-0.88	80.18	11.87	5.00
	George Washington-Jefferson	-6.98	0.94	7.84	15.82	35.89	-6.40	-0.24	-0.15	-0.40	69.79	6.29	0.00
	Daniel Boone	0.00	-1.77	6.91	10.06	43.98	-8.10	-0.75	-0.46	-1.57	67.18	7.89	5.00
	Monongahela	-1.64	-0.82	8.08	12.89	39.14	-9.65	6.04	-0.54	-0.43	89.49	9.08	5.00
	Wayne	0.25	-0.19	4.34	9.03	33.64	-6.83	3.39	-0.02	-1.50	54.92	8.24	25.00
	Great Smoky Mountains	-1.05	2.72	5.32	8.32	42.37	-8.22	4.78	1.35	-0.10	88.65	7.58	5.00
	Shenandoah	0.19	-0.01	4.27	11.07	30.77	-1.46	-1.50	0.10	-0.47	78.99	6.13	10.00
STAND + ENVIRONMENTAL VARIABLES		*D*	*H*	*H-D*	*QMD*	*BA*	*TPH*	*SDI*	*Aspect*	*Slope*	*bio01*	*bio02*	*bio03*
	Chattahoochee-Oconee	-0.51	-3.17	7.34	13.70	40.37	-8.18	2.18	0.28	-0.31	0.36	0.40	0.13
	Cherokee	-1.55	-4.26	6.68	14.19	42.78	-7.87	-0.19	-0.45	-0.12	0.02	-0.09	2.08
	Nantahala	-0.51	-0.29	7.55	8.32	41.44	-6.40	-1.85	-0.72	0.10	-0.02	-0.20	1.54
	Pisgah	-0.53	-8.44	11.67	21.92	49.19	-8.65	0.00	0.17	-1.49	-0.14	-1.10	3.40
	George Washington-Jefferson	-3.36	-0.54	8.21	13.22	35.98	-6.86	0.40	-0.19	-0.59	0.07	0.74	-0.35
	Daniel Boone	0.00	-1.71	6.72	10.11	43.35	-7.99	-0.11	-0.69	-1.28	-0.26	-0.92	1.41
	Monongahela	-1.16	-1.00	8.10	12.31	39.09	-9.79	6.33	-0.48	-0.29	0.01	-0.03	0.56
	Wayne	0.18	-0.95	4.11	10.22	33.34	-6.57	3.32	0.00	-1.10	0.00	-0.31	-0.04
	Great Smoky Mountains	-0.75	2.24	5.06	8.38	41.98	-6.97	4.03	1.72	-0.56	0.24	-0.30	1.53
	Shenandoah	6.82	6.95	0.00	6.75	11.46	-0.79	0.92	1.15	-0.10	0.00	0.00	0.00
		bio04	*bio05*	*bio06*	*bio08*	*bio09*	*bio10*	*bio11*	*bio12*	*bio13*	*bio14*	*bio15*	*bio16*
	Chattahoochee-Oconee	0.29	-0.02	-0.02	-0.01	-0.57	0.04	0.00	-0.09	0.02	0.00	-0.12	0.06
	Cherokee	1.04	0.01	-0.02	0.61	-1.24	0.03	0.00	0.00	0.02	0.17	0.87	0.40
	Nantahala	0.00	-0.01	0.02	-0.65	-1.61	0.00	-0.01	0.00	0.16	0.00	-0.11	0.23
	Pisgah	1.18	-0.06	0.00	-1.54	-1.84	-0.02	-0.87	-1.18	0.02	1.29	1.24	-0.23
	George Washington-Jefferson	0.19	0.25	-0.23	0.29	0.72	-0.01	-0.01	-0.88	1.61	0.06	-0.08	-0.07
	Daniel Boone	0.20	1.53	0.02	-0.11	2.27	1.22	-1.17	0.26	-0.72	0.00	2.83	0.05
	Monongahela	0.36	-0.30	-0.11	0.00	-0.04	0.00	0.05	-0.06	0.65	-1.24	-0.32	0.03
	Wayne	-0.07	1.64	0.09	-0.01	0.95	0.01	0.00	-0.24	0.00	0.20	1.73	0.02
	Great Smoky Mountains	-0.56	0.00	1.04	2.15	1.40	0.08	0.25	0.03	0.37	0.26	0.03	0.20
	Shenandoah	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		bio17	*bio18*	*bio19*	*Intercept*	σ_{fort}	λ						
	Chattahoochee-Oconee	0.00	-0.77	0.07	74.30	8.07	0.00						
	Cherokee	-0.40	0.48	-0.07	70.13	8.85	0.00						
	Nantahala	-0.02	0.00	0.03	74.84	7.58	0.00						
	Pisgah	0.04	0.03	-0.01	79.64	10.97	0.00						
	George Washington-Jefferson	0.23	-0.40	-0.07	70.61	10.88	0.00						
	Daniel Boone	0.28	0.38	0.00	67.16	8.05	0.00						

Monongahela	-0.19	0.40	-0.31	88.99	10.61	0.00					
Wayne	0.07	-0.36	0.09	55.46	8.70	5.00					
Great Smoky Mountains	0.11	0.05	0.00	88.13	9.46	0.00					
Shenandoah	0.00	0.00	0.00	79.33	29.99	500.00					

13

14 Appendix

Chapter 3

Evaluation of expanding a regionally trained model compared against off-the-shelf carbon products and carbon inventory estimates: a case study in southern Appalachia.

Introduction

Forests occupy 31% of land area on planet Earth and, along with oceans, are able to store enormous amounts of carbon (Reay *et al.* 2008, Wei *et al.* 2025). To combat rising levels of carbon dioxide in the atmosphere, quantifying carbon and understanding temporal patterns has become a focal point. Quantifying and tracking the flows of carbon, like the live aboveground carbon stored in forests, has received increased attention as the global average temperature continues to increase. As a global issue, international efforts have been made to combat, or at the very least mitigate, the effects of the ever-increasing consumption of fossil fuels, as evidenced by the United Nations Framework Convention on Climate Change (UNFCCC) (1994), the updated Kyoto Protocol to the United Nations Framework Convention on Climate Change (1997), and the most recent Paris Agreement (2015). As a way to offset ever increasing emissions from ever increasing fossil-fuel usage and achieve the UN's goal of net-zero emissions, carbon credit programs and cap-and-trade marketplaces have proliferated and incentivized private landowners' enrollment (Charnley, Diaz, and Gosnell 2010, White *et al.*, n.d.). Because of its importance in understanding global carbon patterns and monetary potential, many carbon and biomass estimation products have been developed to meet the needs of various users.

The gold standard of carbon and biomass estimates are produced through a rigorous and time-intensive process, using tree-level forest inventory data combined with species-specific biomass equations (carbon is generally accepted as the following: $c \approx 0.5 * biomass$). Because carbon estimations are performed at the individual tree-level, these estimates are highly accurate and considered the "true" value. Despite the uncertainty still present from various sources like measurement and sampling error, this methodology is still considered best practice by forest monitoring programs like the United Nations Reducing Emissions from Deforestation and Forest Degradation (REDD) program and Verra (Verra, n.d.). However, given how rigorous the methodology is, completing inventories to this full extent can strain resources, particularly at large landscape levels. Alternative methodologies like maps and geospatial datasets have been developed to generate estimates more efficiently at large scales, but at the cost of error being introduced. The amount of error introduced varies, and previous research has focused on quantifying and comparing error present among various products (Hudak *et al.* 2026). Understanding systematic differences between estimates and ground truthed values. While these models will not be able to replace the detail and precision of boots-on-the-ground inventories, generating estimates can allow for easier monitoring between inventories. Comparing the estimates against a certified carbon program can help us quantify how much error exists at different spatial scales to make more informed conclusions. Previous research has investigated such questions by comparing multitudes of models. However, given how many products are available, many models remain untested.

As alternatives to this rigorous procedure, aboveground biomass density (AGBD) maps and products provide large-scale, such as national or global, estimates of biomass at varying resolutions. The number of products available continues to grow, varying by input and validation

data and estimation approach. Mission purpose also differs, with some maps being the product of international and national efforts, like the U.S.'s National Aeronautics and Space Administration (NASA), the U.S. Forest Service Forest Inventory and Analysis (FIA), or the European Space Agency (ESA). While validation data normally uses forest inventory data, some products like the ESA's Climate Change Initiative (CCI) biomass data is produced through a methodology consisting solely of remotely sensed data. A biomass map derived from both remotely sensed and ground inventory with a proprietary methodology has been released by CTrees, a non-profit organization. Both are available at 100 m resolutions, these products are efficient alternatives to best practices, not only having global scale data, but can also be updated more frequently than global estimates produced via national forest inventories (NFIs). However, there is a trade-off with increased error, and previous studies have quantified error present across several carbon products that differed in input data and methodology. Error also varied within products based on the spatial aggregation level, like county, stand, and plot level.

An intermediate alternative to the best practices and off-the-shelf global carbon products may lie in using a model trained on a nearby federal region, like national parks and forests. Training models on federal regions can capitalize on federal regions' large contiguous areas, and they may even share management objectives with privately owned forests. Historical records can be accessed through NFIs to increase observations and study temporal changes. One methodology is described in Yang *et al.* (2025) and Chapter 2 that delivers highly accurate carbon estimates. Having the ability to expand these federally trained models beyond the training area has not yet been tested but may yield promising results. In southern Appalachia, which is the focus area of the two aforementioned studies, forests do not always follow human-created property boundaries, and nearby forests can be similar to those within the property line. These forest characteristics include but are not limited to stand age, composition, topography, and climate. The amount of diversity in southern Appalachia was previously an obstacle, as mixed-species models faced particular challenges (Yang *et al.* 2022b, Yang, Brandeis, and Skiba 2025). But, when using a linear mixed-effects model approach with a random effect on forest type, high diversity can be an advantage to share information between forest type levels. Furthermore, this global model can utilize the increased combined number of observations to provide a more stable model. This approach was shown to be successful in Yang *et al.* (2022b, 2025), but its use beyond the training region has not been tested.

Our hypothesis is that this regionally trained model will perform better than the global AGBD maps and worse than best practice estimates. As a model trained specifically for a nearby region, we would expect it How much better or worse is unknown. Using the maps and best practice as reference points helps rank this new methodology in terms of predictive power. The ability to use this regional model outside of the trained region may be more time-intensive to produce that using the AGBD products, but improved predictions can be more desired in some cases to warrant the extra effort needed. As the scale of ~1000 acre properties, this study's areas of interest are smaller than conducted in other studies. Using a working research forest

maintained by the University of Tennessee, the properties are at a similar scale to those of private landowners, demonstrating how the extension of this model can be applied beyond this study by other landowners and managers.

Although the three products described above all aim to quantify aboveground live carbon, estimates can vary greatly. For carbon monitoring programs, this uncertainty between products can have consequences, like failing to detect changes or determining appropriate management strategies. As the assumed "true" quantity of carbon is estimated using a combination of ground-based inventories with species-specific biomass equations, the University of Tennessee Forest Resources AgResearch and Education Center's (FRECC) existing forest carbon inventory program serves as ground truthed validation data at spatial scales on par with private landowners. Quantifying how much error is present among and between the model and products tested at this aggregated spatial scale provides an updated understanding of which approach may be suitable as an efficient alternative to conducting full inventories. Comparing estimates from the various carbon products against a validated carbon inventory programs sheds light on how close estimates are to the accepted true value of the plot.

Materials and Methods

Study Area and Validation Data

The areas of interest for this study are the regions of the University of Tennessee Forest Resources AgResearch and Education Center (FRREC) Forest Properties enrolled in a verified carbon monitoring program. These regions are spread across multiple stratum in middle and east Tennessee, as shown in Figure 16, Figure *17*, Figure *18*, and Figure *19*. The stratum considered for this study include two regions located in the Cumberland Mountains, one region within the Oak Ridge, Tennessee Arboretum, and two regions in the Highland Rim (shown in *Table 12*). The stratum are further divided by forest composition, either as hardwood, softwood, or a mix of the two. Geographically, the Highland Rim stratum are located the furthest west near Tullahoma, at an elevation of approximately 1000 ft. Similarly, the Oak Ridge Arboretum is at an elevation of 1000 feet, but it is located further east and situate in a valley surrounded by more mountainous regions. Both Cumberland Forest stratum are located in the Cumberland Mountains west of the Appalachia mountains and are at higher elevations than the other stratum (approximately 2000 ft.).

To meet best practice standards as defined by the United Nations Reducing Emissions from Deforestation and Forest Degradation (REDD) program and protocol set by the American Carbon Registry (ACR), the sampling design was created to follow ACR methodology guidelines (The ACR Standard, n.d.). "[target] 0% uncertainty as assessed by the ACR methodology, corresponding to a 90% confidence interval less than or equal to +/-10% of mean forest carbon stock" (FINAL UTIA SOP revOct 2020 – how to cite?) The inventory process consists of 1/10 acre plots, with live and dead trees ≥ 5" diameter at breast height (dbh)

measured. A nested 1/100th acre plot around the center record all live trees within 1-4.9' dbh recorded. Species were recorded for all trees, and heights were measured for trees within the eastern hemisphere of each plot. The collected inventory information serves as a real-world scenario and the validation data used for this study. Individual tree carbon was calculated using Jenkins (2003) carbon and biomass equations and aggregated to produce stand-level estimates. A histogram of stand-level carbon is shown in

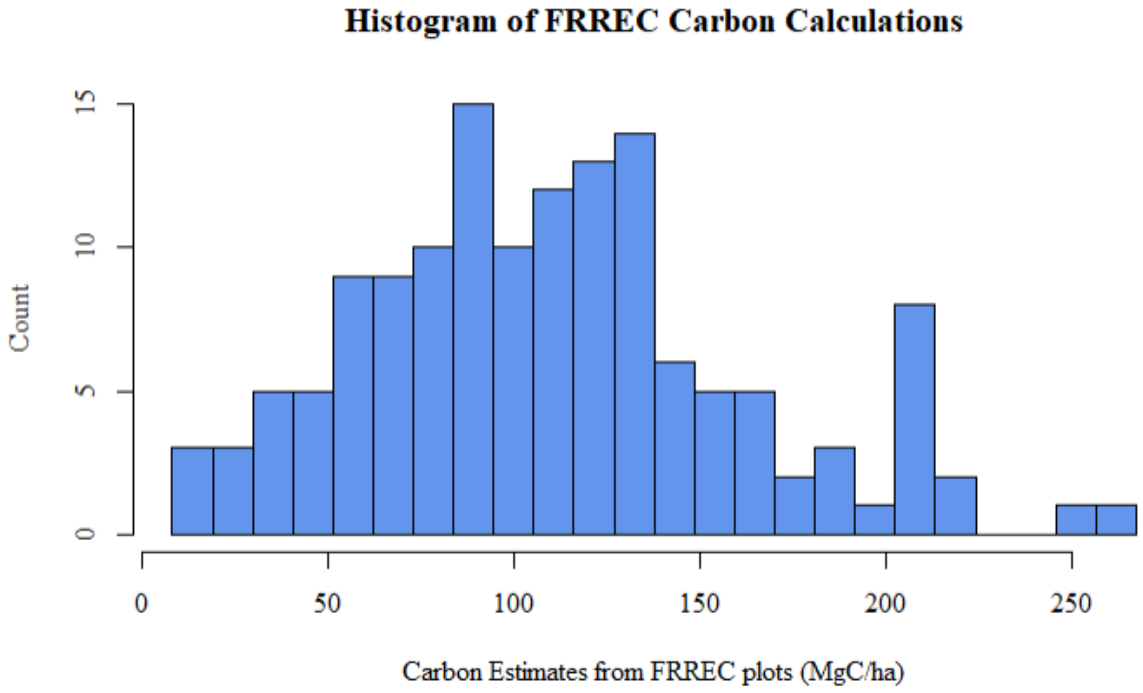


Figure 20. A total of 142 plots located across the regions described were used for this study.

Table 12: FRECC's 5 stratum used within the study. Additional stratum exist that were excluded.

Stratum (general forest type)	Abbreviation	No. of obs.	Acres
Cumberland Forest North (hardwood)	CN	56	4403.9
Cumberland Forest South (hardwood)	CS	40	3996.0
Highland Rim (hardwood)	HR_HW	8	514.0
Highland Rim (pine)	HR_PI	9	255.4
Oak Ridge Arboretum (hardwood and pine)	OR	37	2145.5

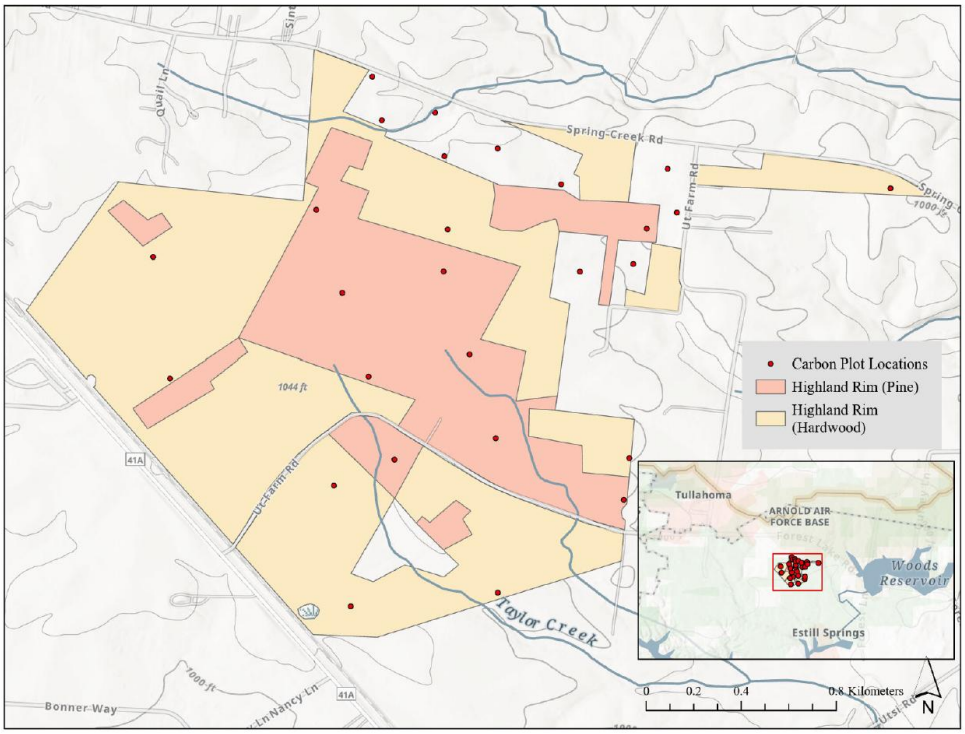


Figure 16: Map of UTK FRECC's Highland Rim region south of Tullahoma, TN with stratum and carbon plot locations.

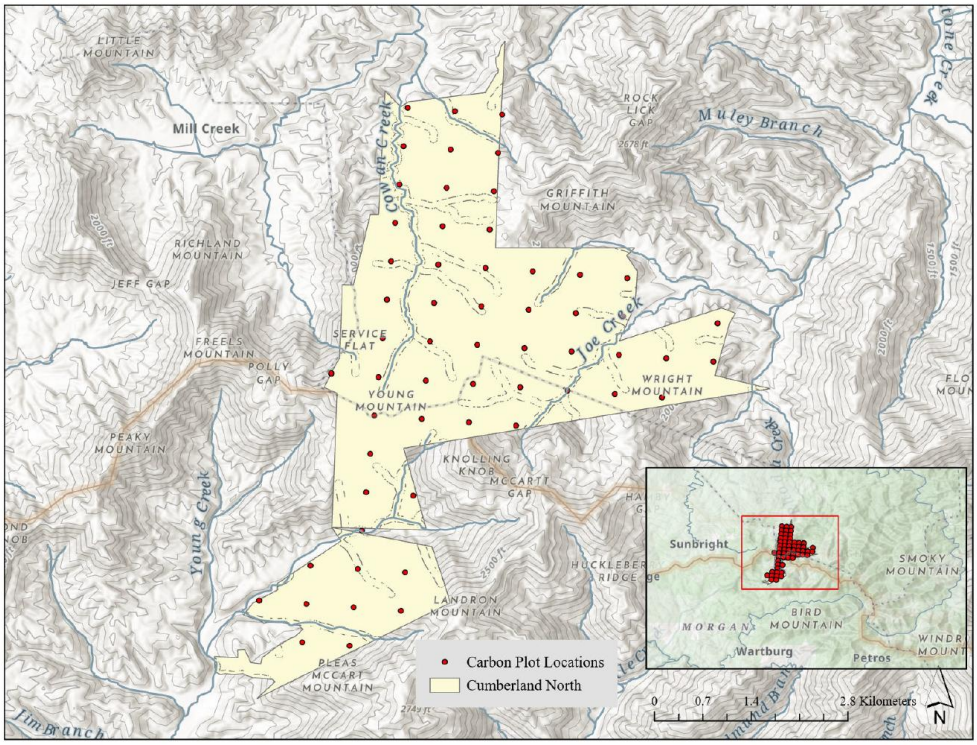


Figure 17: Map of UTK FRECC's Cumberland Forest North region with stratum and carbon plot locations.

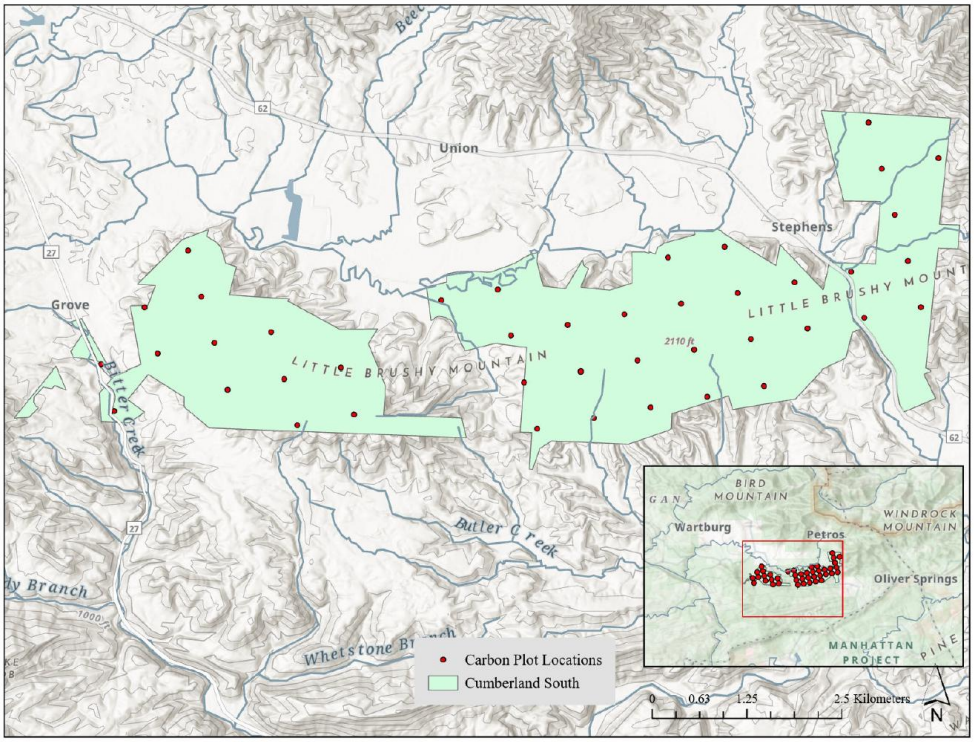


Figure 18: Map of UTK FRECC's Cumberland Forest South region with stratum and carbon plot locations.

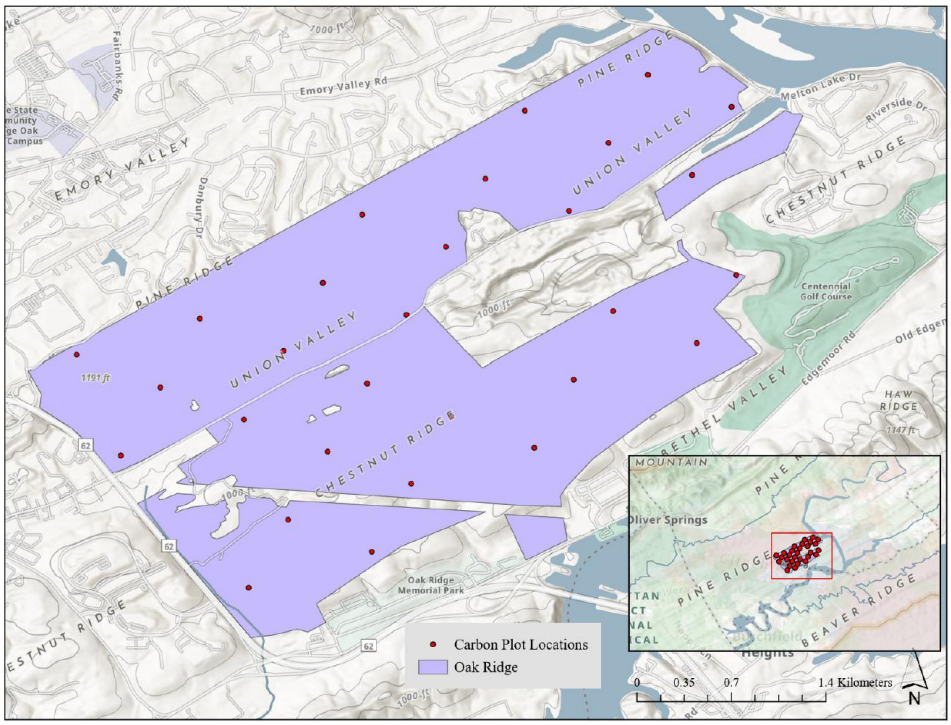


Figure 19: Map of UTK FRECC's Oak Ridge region with stratum and carbon plot locations.

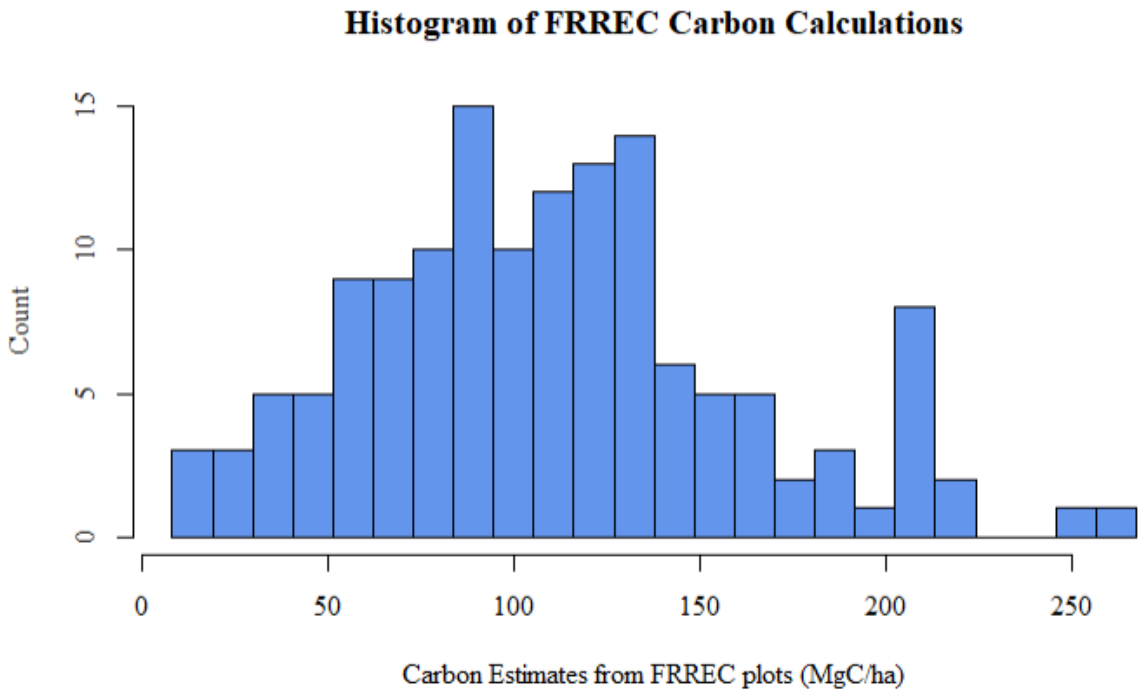


Figure 20: Histogram of calculated stand-level carbon estimates from FRREC carbon inventory plots.

To supplement slope and aspect variables into the FRECC dataset, we used the slope and aspect spatial analysis tools in ESRI's ArcGIS software (version 3.6.0, https://www.esri.com/en-us/arcgis/geospatial-platform/overview) to calculate the two metrics from USGS's 3DEP 1 m resolution data (Figure out how to cite).

ESA CCI

One AGBD map selected for this study was the European Space Agency's (ESA) Climate Carbon Initiative (CCI) global biomass estimates at a 1 km resolution. ESA's CCI aboveground live biomass map is produced through the combination of multiple remotely sensed products including but not limited to Sentinel-1, ALOS-2 PALSAR-2, Spaceborne LiDAR, Digital Elevation Models, and MODIS (Santoro and Cartus 2025) It is a popular dataset that has been widely studied, available through many popular geospatial repositories like ESRI's Living Atlas and Google Earth Engine (GEE). Its data availability spans from 2005-2012 and 2015-2024 and is updated annually. This dataset was selected because of its use of solely remotely sensed input data and being a dataset produced by international government agencies, as well as its overlapping time availability with our validation data (from 2020).

The ESA CCI aboveground biomass dataset was accessed via ESRI's Living Atlas repository, filtered for 2020 data. Estimates were extracted using the zonal statistics function in ArcGIS Pro for individual plots and for entire stratum.

CTrees

Another off-the-shelf carbon map selected for this study was produced by CTrees, a nonprofit to "monitor, report, and verify (MRV) carbon stocks and land-use activities for every ecosystem on land" (CTrees, n.d.). Along with differing objectives from ESA's CCI mission, CTrees also produces estimates differently from ESA's CCI biomass algorithm. Although CTrees's estimation methodology is proprietary, they report that both ground-based forest inventories and remotely sensed are used to create estimates. This differs from ESA solely using remotely sensed data to generate their estimates. CTrees's Land Carbon Map contains annual estimations from 2000 to 2025 at a 100 m resolution.

CTrees's Land Carbon Map is available through EarthMover, which we filtered for only 2020 data. Extraction was produced following CTrees's provided example code to handle and extract estimates.

Regional model

The regionally trained model was fit using FIA data and WorldClim BIO variables (V1) from Daniel Boone National Forest from Chapter 2. This region was selected due to as a national forest that is near the study area and has similar characteristics and management objectives. National forests typically have more active best management practices compared to national parks. FRREC forest landscapes often change as the results of active management, teaching lessons, and research studies conducted throughout the properties, which more closely matches the management plans of national forests. The regional model consists of a linear mixed-effect (LME) model with the following form:

$$C = \sum x\beta + \mathrm{b} + \epsilon$$

$\sum x\beta$ is the linear combination of predictors x and their coefficients β, b is a forest-type-specific random effect, and error term ϵ. Predictors used are listed in *Table 13* (all variables except for *cph*, the response variable). Model fitting was performed in R statistical software (Version 4.5.3, https://www.r-project.org/). To fit the model on FRREC data, some variables needed to be supplemented via height-diameter models, forest type assignment, and use of geospatial datasets.

As the FRREC data only measured tree heights on the eastern hemisphere of a plot, a non-linear model with a species-level random effect applied was used to predict missing heights. The Pearl and Reed population growth was selected as it has previously been used to predict tree height, like oaks (Pearl and Reed 1920, Misik *et al.* 2016). This model takes the following form:

$$h = \frac{\beta_0}{1 + \beta_1 e^{(\beta_2 DBH)}}$$

h is height in meters, DBH is diameter at breast height in cm, and β_0, β_1, and β_2 are coefficients to be estimated. To fit models to the many species present, a species-specific random effect on β_0

was applied to allow flexibility for different species' height-diameter relationships. Start values for the coefficient estimations were as follows: $\beta_0 = 109, \beta_1 = 3.71, \beta_2 = -0.18$. Coefficients were chosen with consideration to estimates provided by Jha *et al.* (2023).

To assign a FIA forest type, plurality of basal area was calculated using trees greater than 5 in. DBH. Species with highest basal area(s) were used to match the plot's composition with a FIA forest type. This allowed for better classification for the mixed-species forests found in this study. However, misclassification may still occur and could be further reduced by following the full FIA classification methodology (Burrill *et al.* 2024). Distributions of the forest types present within the Daniel Boone National Forest training dataset and the assigned forest types for the FRREC data are presented in Figure 22.

Table 13: Summary statistics of predictors from the FRREC study areas Mean and standard deviation (in parenthesis) are provided.

	CPH	*TPH*	*BAPH*	*Mean D (cm)*	*Mean H (m)*	*Mean H:D*	*QMD*	*SDI*	*Slope*
CN	125.69 (51.21)	1264.21 (1094.29)	33.2 (10.41)	31.13 (9.09)	21.86 (4.79)	78.92 (12.78)	33.9 (9.72)	1792.48 (1452.12)	65.24 (16.69)
CS	115.84 (55.42)	1292.98 (1201.85)	31.93 (11.71)	28.5 (6.51)	21.66 (4.07)	83.02 (10.23)	30.97 (7.71)	1593.29 (1077.13)	70.71 (15.43)
HR_HW	85.53 (34.69)	1284.95 (653.11)	23.87 (8.73)	26.73 (8.44)	20.44 (5.44)	79.64 (8.46)	27.81 (8.59)	1372.84 (694.37)	55.61 (21.9)
HR_PI	91.3 (24.48)	2223.95 (1151.5)	42.15 (10.48)	23.82 (6.22)	18.24 (4.72)	79.97 (5.98)	24.57 (6.65)	1910.77 (704.12)	65.55 (23.05)
OR	86.96 (44.51)	2405.44 (2102.44)	27.37 (9.05)	25.67 (8.38)	19.35 (4.51)	83.02 (13.14)	27.45 (9.86)	2140.04 (1420.33)	65.17 (16.16)

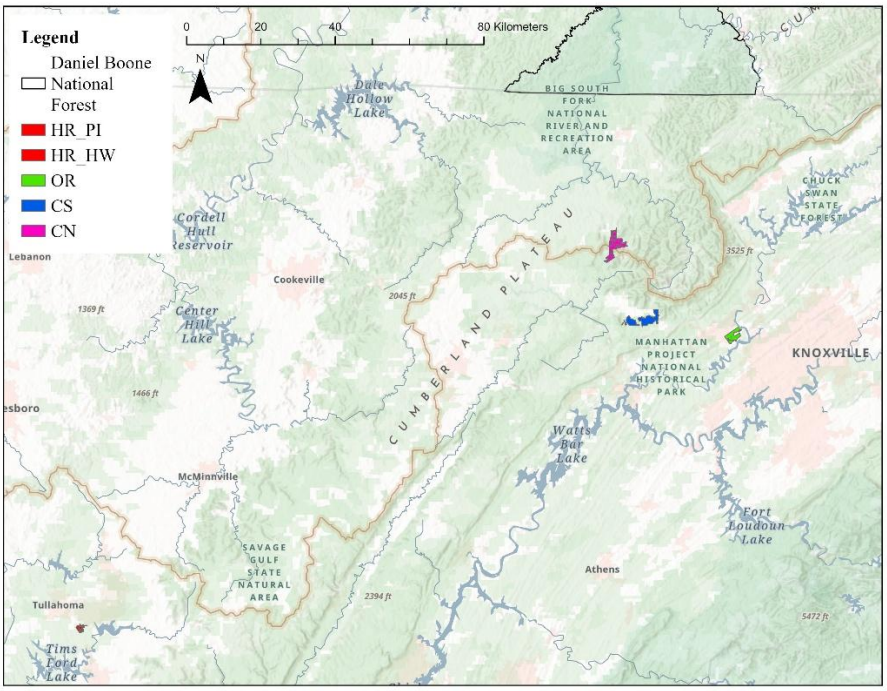


Figure 21: Map of FRREC study regions with respect to Daniel Boone National Forest, which the regional model was trained on.

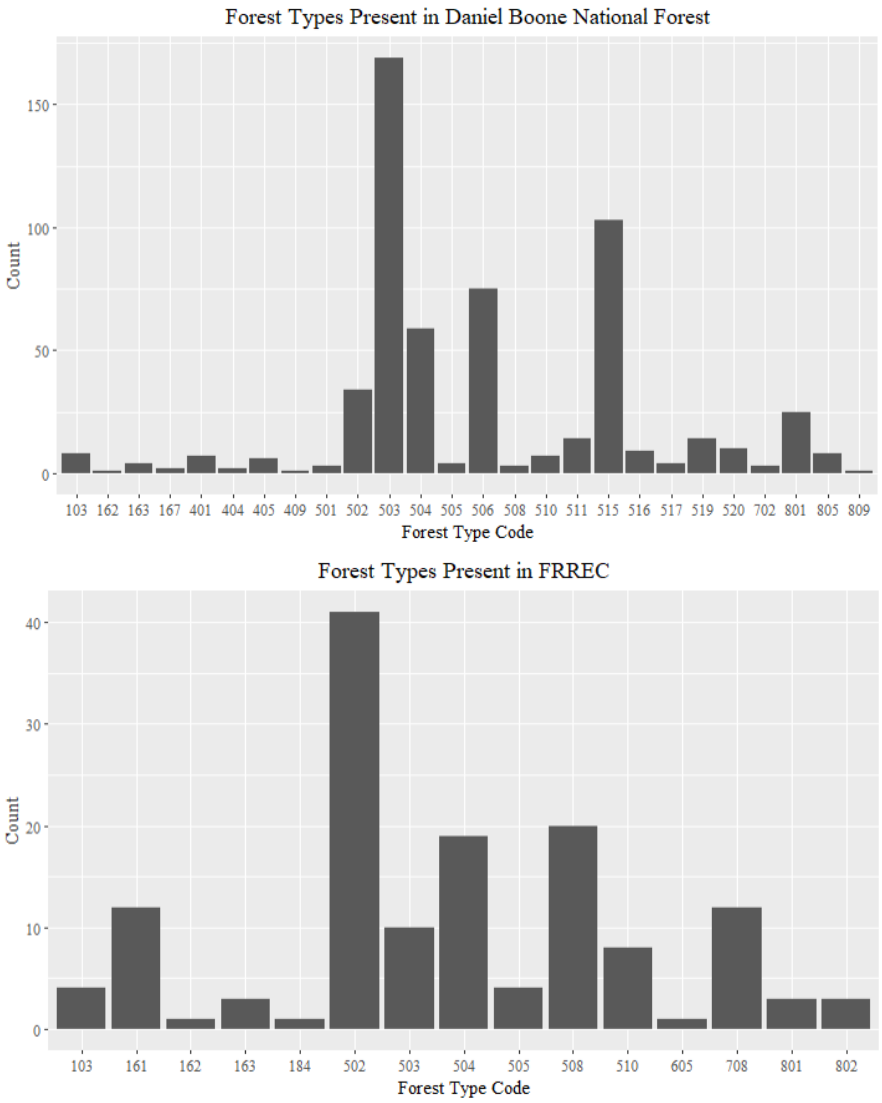


Figure 22:Distributions of forest types present in Daniel Boone National Forest (as assigned by the FIA) and in FRREC (manually assigned using plurality of basal area).

Analysis

All analysis was completed in R statistical software (Version 4.5.3, https://www.r-project.org/). For the analysis, we compared estimates at plot and stratum level to study performance at these two scales. The plot-level estimates are consistent with the regional model, as the regional model produces carbon estimates at the plot-level, while stratum-level estimates are more compatible when utilizing geospatial data where resolution can be too course for plot-level estimates. As the regional model produced plot-level estimates, these estimates are aggregated to produce stratum-level estimates. For the AGBD maps that produced biomass estimates, a ½ factor was applied to convert to carbon. For plot-level comparisons, R-squared, mean bias (MB) and root mean square error (RMSE) were calculated. At the stratum level,

differences (Δ) were calculated. Fitted values were plotted against observed values with a 1:1 diagonal line using *ggplot2* R package (Wickham 2016).

Comparing plot-level estimates against the FRECC inventory estimates showed severe differences in estimates, both over and underestimating carbon values.

To produce stratum-level estimates, the FRECC data was summarized at the stratum level, the ESA data was produced by averaging extracted pixel values from the stratums' polygons, the CTrees data was…, the regional model was fit at the plot-level and then aggregated to the stratum-level.

Results

ESA

ESA's performance varied by spatial aggregation level. At the plot level, more than half of all predictions were smaller than observed values, with all of Highland Rim (pine) and all but one observation of Highland Rim (hardwood) having estimates smaller than observed. When looking at the fitted versus observed plot (

Figure *23*), a distinct pattern can be observed between the stratum. Although observed values within each stratum spanned wide ranges, ESA estimates had narrow ranges within stratum. A stratum's estimate range appears to be related to acreage, with ranges increasing with study area size. Highland Rim (pine) is the smallest region (103.4 ha) and has the narrowest range (54.5 MgC/ha). The average returned estimates of each stratum appears to better capture the stratum-level estimates, compared to average *cph* values in Table 13. This is congruent with what we expect with the 1 km resolution of the ESA data used, with the courser resolution failing to capture detail needed at a plot level. However, regardless of if estimates are aggregated from plot-level estimates or returned for the whole stratum, ESA estimates were lower than observed. Using the stratum-level estimates, Figure 24 and Table 14 show ESA underestimates between 20 and 40 MgC/ha.

CTrees

Plot-level estimates of CTrees produced larger ranges and more varied estimates than the ESA dataset, which follows the pattern we expect from CTrees's finer resolution of 100 m. However, even though CTrees's has finer resolution, plot-level estimates were still not accurately captured, again with the majority of values being underestimates of the observed values. At a 100 m resolution, these pixels cover 1 ha, which is larger than the measured plots' area. At this scale,

non-forested areas can be included within a pixel and affect the aggregated estimate. While we don't observe the same layered stratum pattern as with ESA and some estimates are very

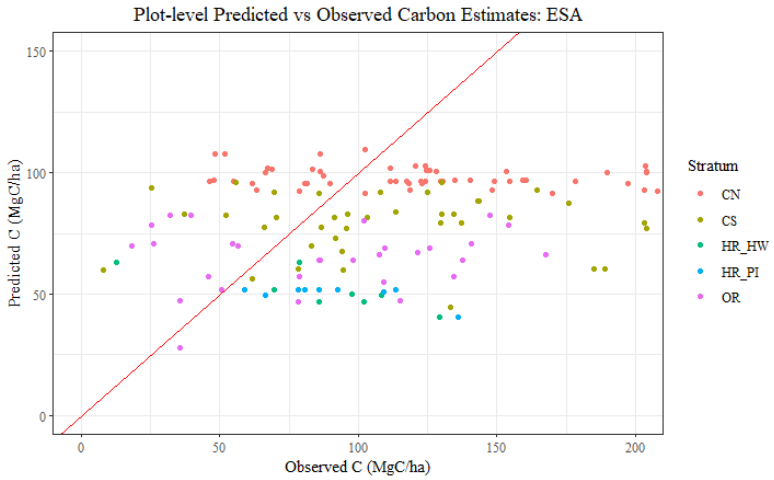


Figure 23: Plot-level returned versus observed estimates of ESA CCI.

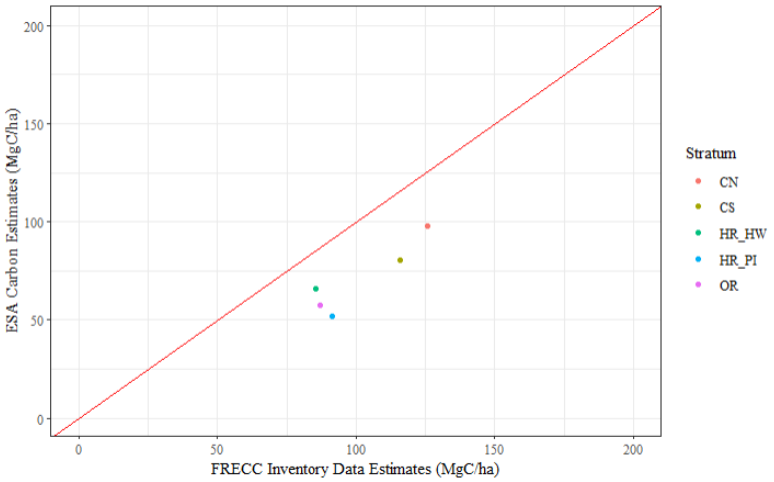


Figure 24: Stratum-level returned versus observed estimates of ESA CCI.

different from the observed values, CTrees was able to better capture observed values across the full range of carbon values present.

Plot-level returned values show that CTrees failed to capture any estimates above 125 MgC/ha except for one observation at 154.75 (which is an extreme overestimation of the observed value of 13.0 MgC/ha, *Plot-level fitted versus observed estimates of CTrees.*). At the stratum level, CTrees underestimated carbon values, even more than observed with ESA except for the Oak Ridge stratum. (Figure 26, Table 14).

Regional Model

The regional model produced the best estimates at both plot and strata levels. *Figure 27* show that plot-level estimates produced by the model closely follow the 1:1 diagonal line to

show model fit, with fewer underestimations than with the two AGBD maps. The regional model was best able to capture larger values of carbon as well, while the AGBD maps were not able to produce estimates at this scale.

The one exception in improved performance occurs in the Highland Rim (pine) stratum, which shows the largest differences in plot and stand-level fitted versus observed values and in difference Δ. It is also the only instance of an overprediction at strata-level.

Table 14: Stratum-level returned estimates and differences between returned estimates and observed values (FRREC).

Stratum	*FRREC (baseline)*	*ESA*	*ΔESA*	*CTrees*	*ΔCTrees*	*Regional Model*	*ΔRegional Model*
Cumberland Forest North	125.69	97.71	-27.99	77.19	-48.50	119.52	-6.17
Cumberland Forest South	115.84	80.32	-35.53	63.83	-52.02	111.51	-4.33
Highland Rim (hardwood)	85.53	66.19	-19.34	54.67	-30.86	74.19	-11.34
Highland Rim (pine)	91.30	52.00	-39.30	58.37	-32.94	136.49	45.18
Oak Ridge	86.96	57.50	-29.46	70.01	-16.95	79.11	-7.85

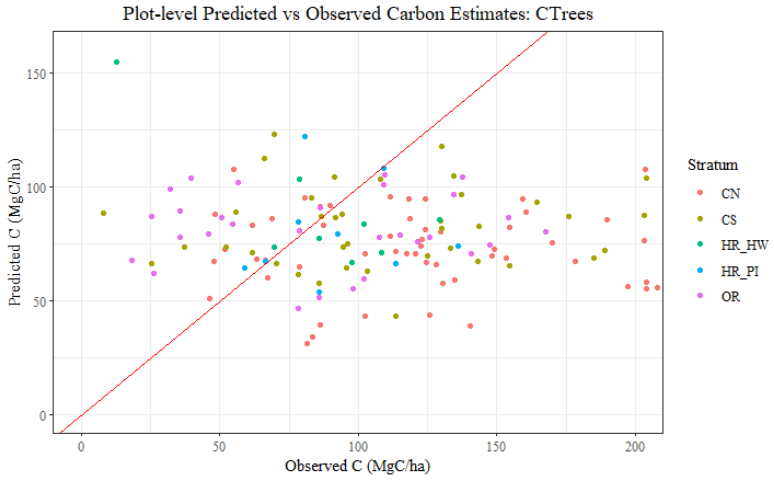


Figure 25: Plot-level fitted versus observed estimates of CTrees.

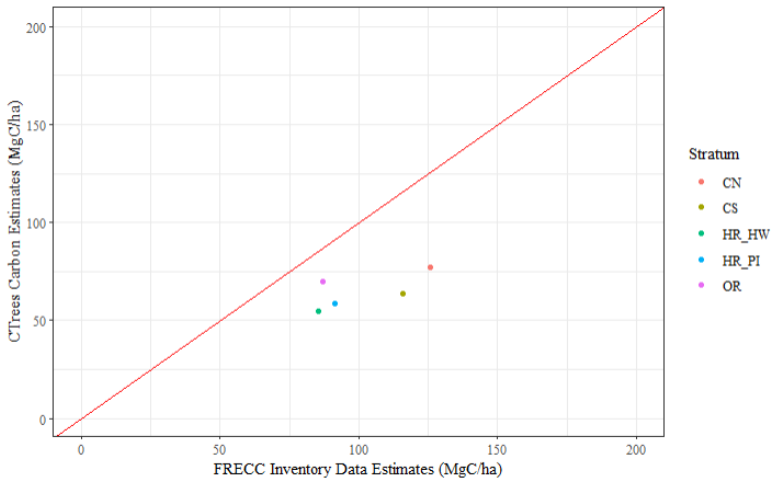


Figure 26: Stratum-level fitted versus observed estimates of CTrees.

Discussion

Overall, AGBD maps produced poorer estimates than the regional model, as expected. Specific limitations with AGBD maps being able to capture larger carbon values either due to the methodology or resolution scale can make them poor contenders to use at the scale of the properties used in this study. This applies more explicitly at the plot-level, whereas strata-level aggregations were closer to the observed values, although these estimates consistently underestimated true values.

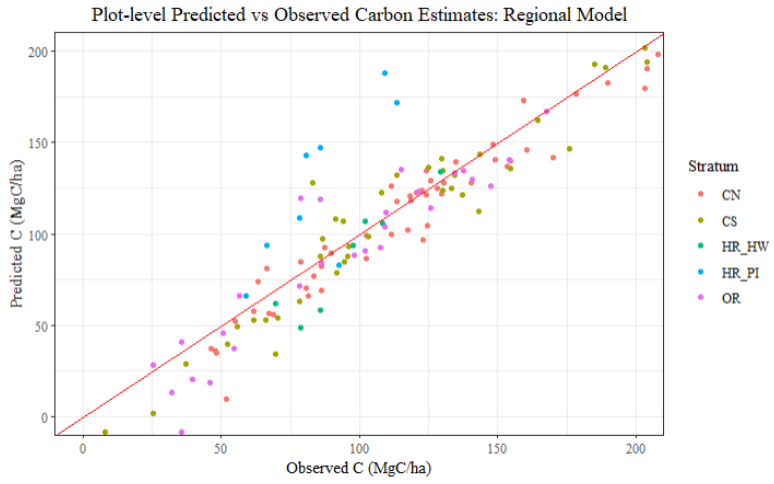


Figure 27: Plot-level fitted versus observed estimates of the regional model.

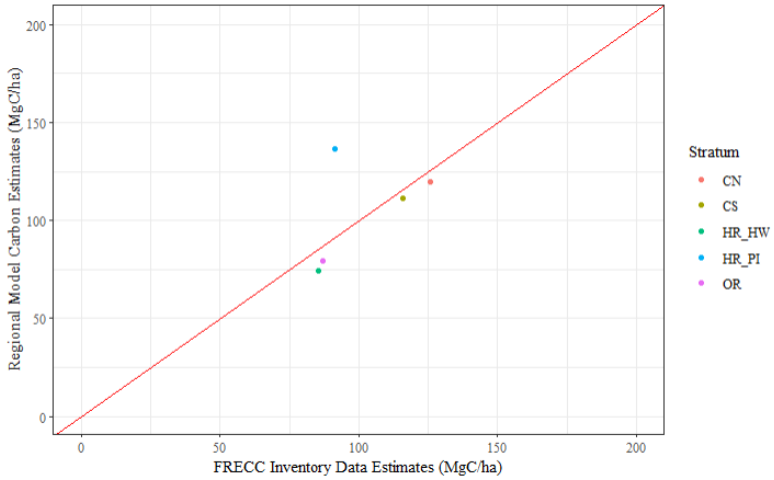


Figure 28: Strata-level fitted versus observed estimates of the regional model.

Conclusion

CONCLUSION

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